

X4Pro. Plotting via Plotly.

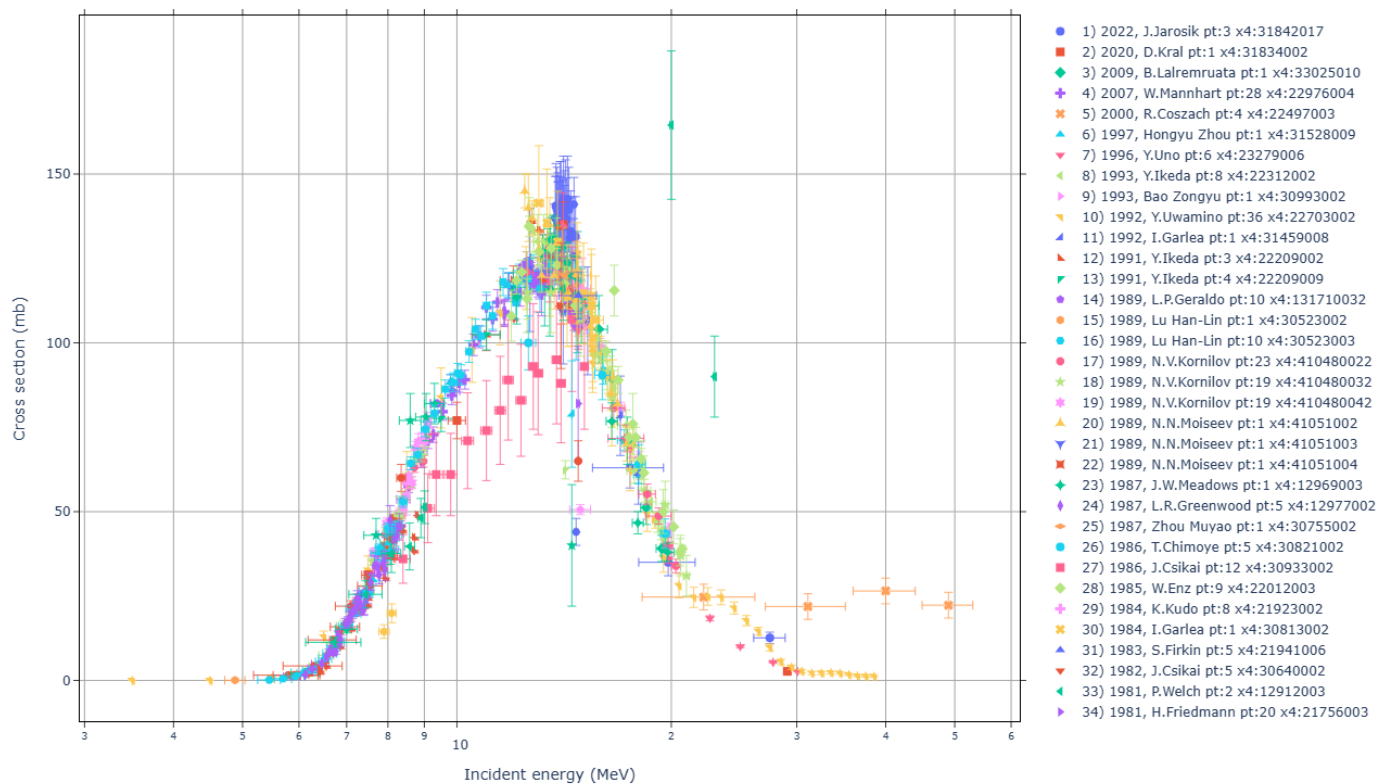
Generated: 2026-10-04,15:44:32

part1-1-sig/ Retrieve and plot EXFOR cross sections (modular Python-program)

1) part1-1-sig/out00/Al27na-1.html.png

EXFOR cross sections $\sigma(E)$: Al-27(n,a),sig

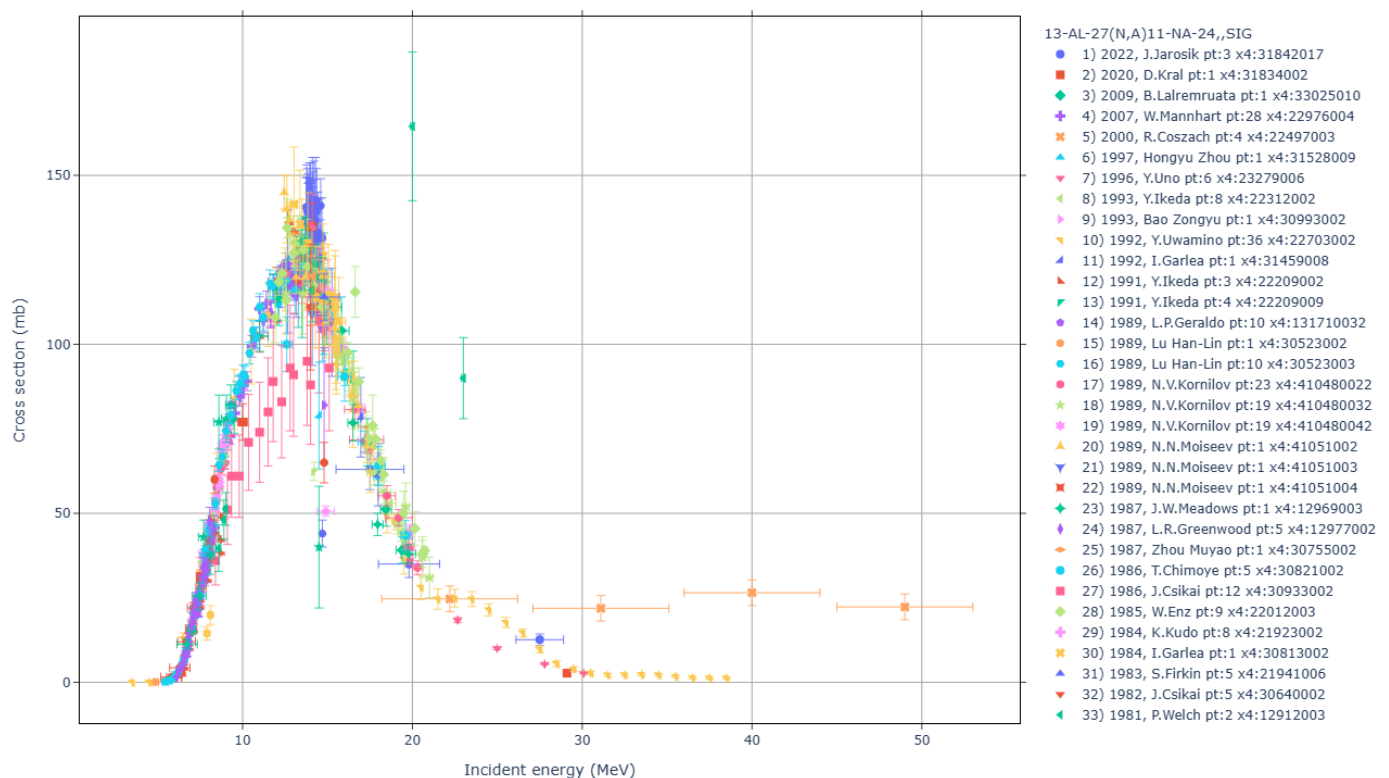
X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-10-02 13:51:53



2) part1-1-sig/out00/Al27na-2.html.png

EXFOR cross sections $\sigma(E)$: Al-27(n,a),sig

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-10-02 13:51:56

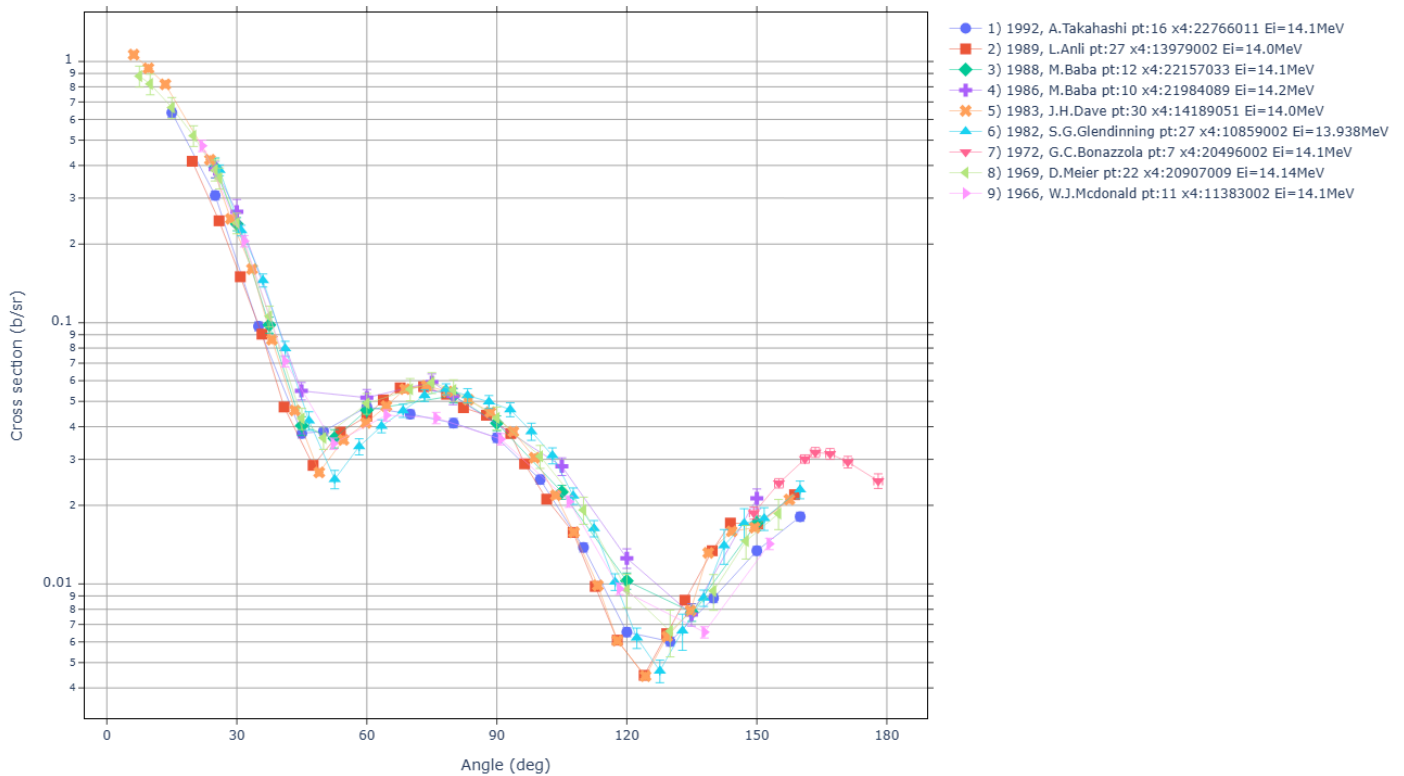


part1-2-da/ EXFOR angular distributions

3) part1-2-da/out00/da0an.html.png

Plot EXFOR angular distributions $d\sigma/d\Omega(E,\theta)$: O-16(n,e),da

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-12-20 //running:2026-10-02 13:51:59

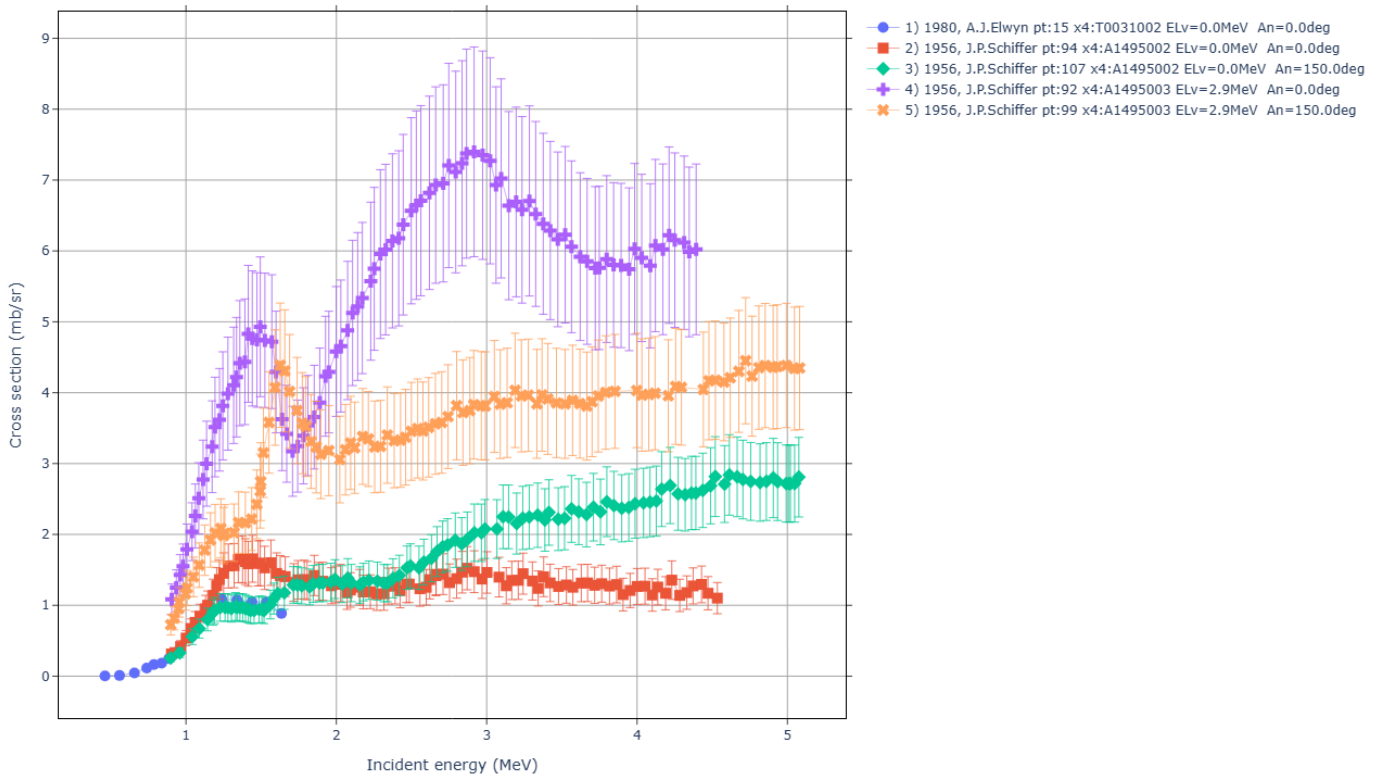


part1-3-dap/ EXFOR partial angular distributions

4) part1-3-dap/out00/dap0ei.html.png

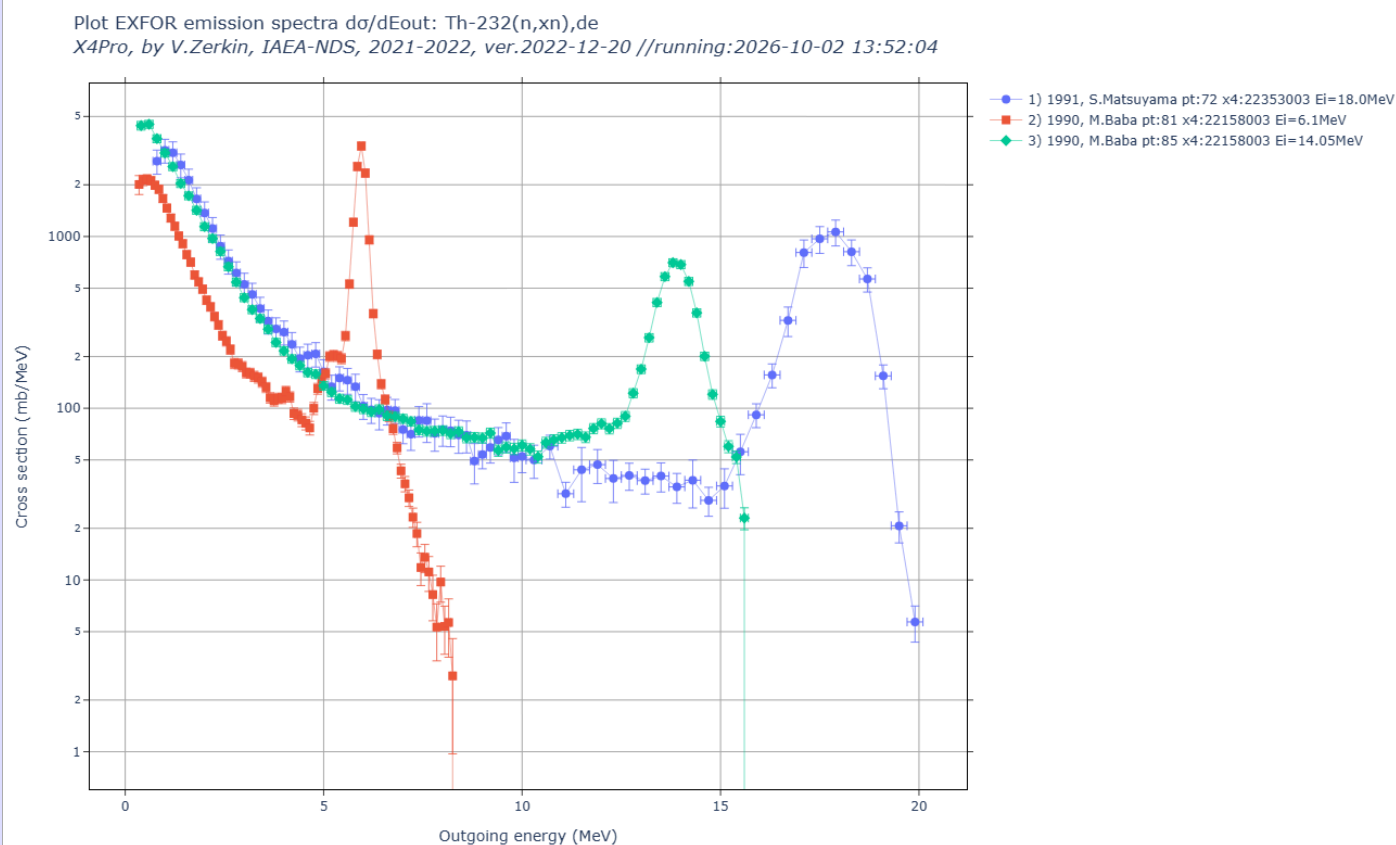
Plot EXFOR angular distributions $d\sigma/d\Omega(E,\theta)$: Li-6(he3,p)par,da

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-10-02 13:52:01



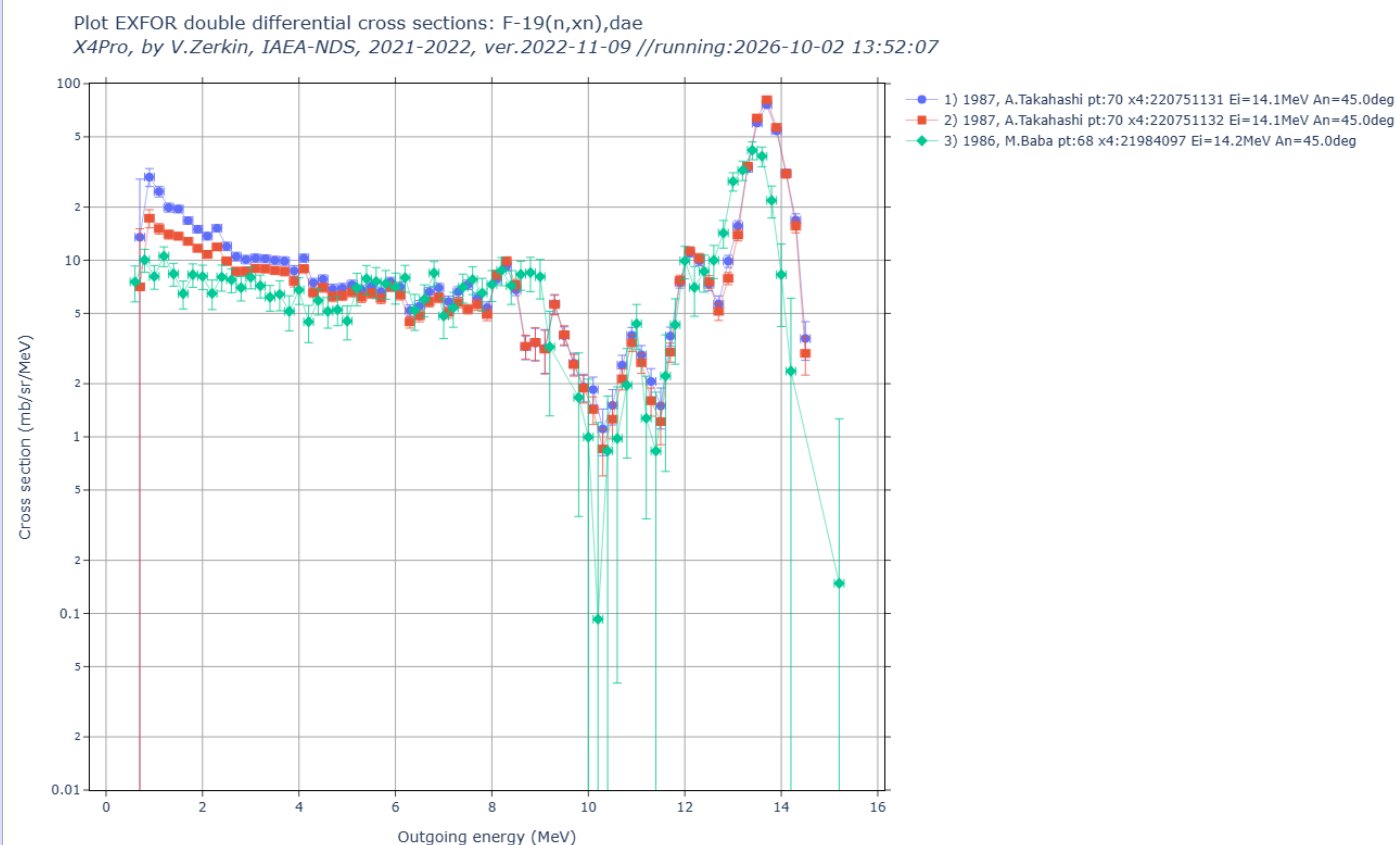
part1-4-de/ EXFOR energy spectra of outgoing particles

5) part1-4-de/out00/de0eo.html.png



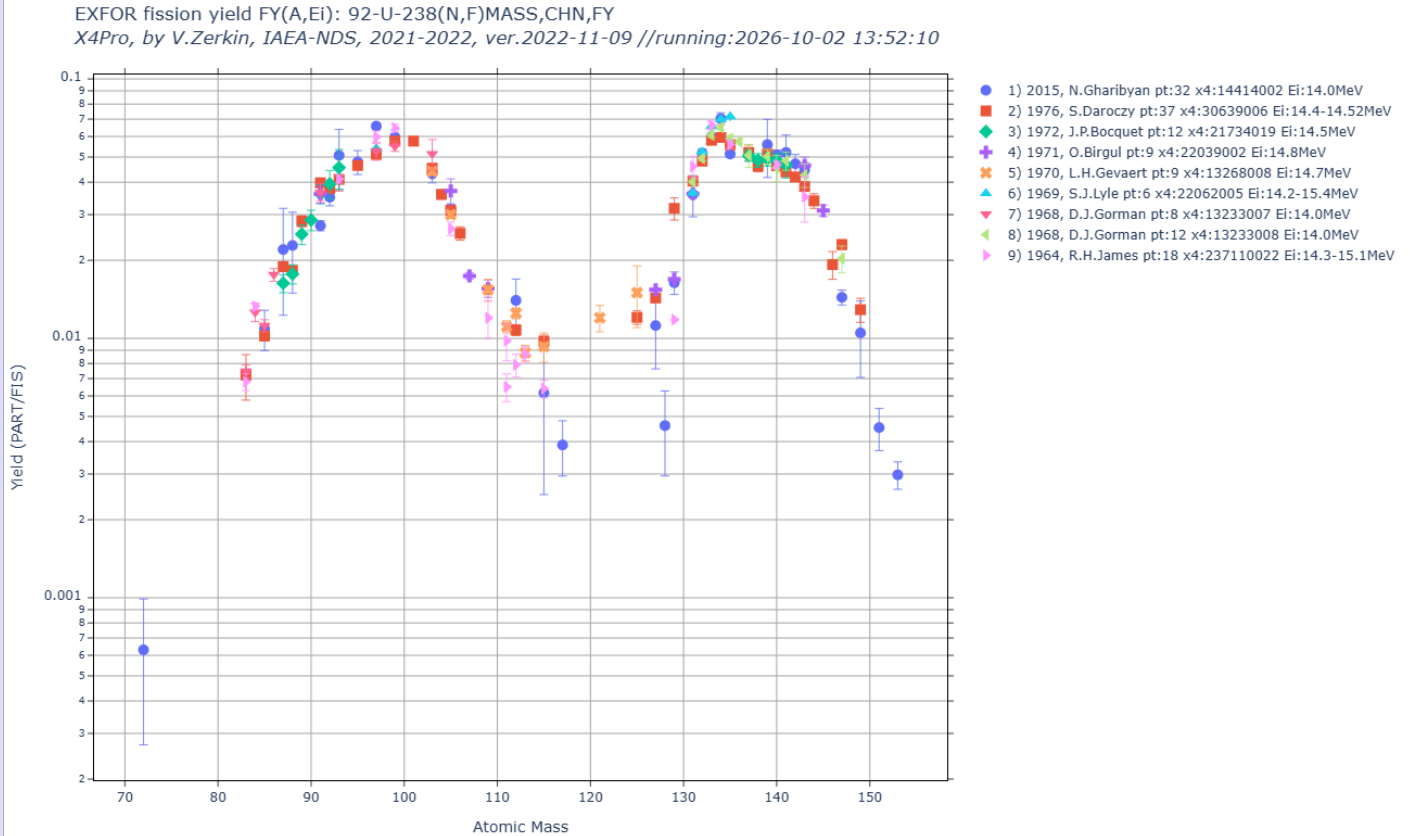
part1-5-dae/ EXFOR double differential cross sections

6) part1-5-dae/out00/dae0eo.html.png



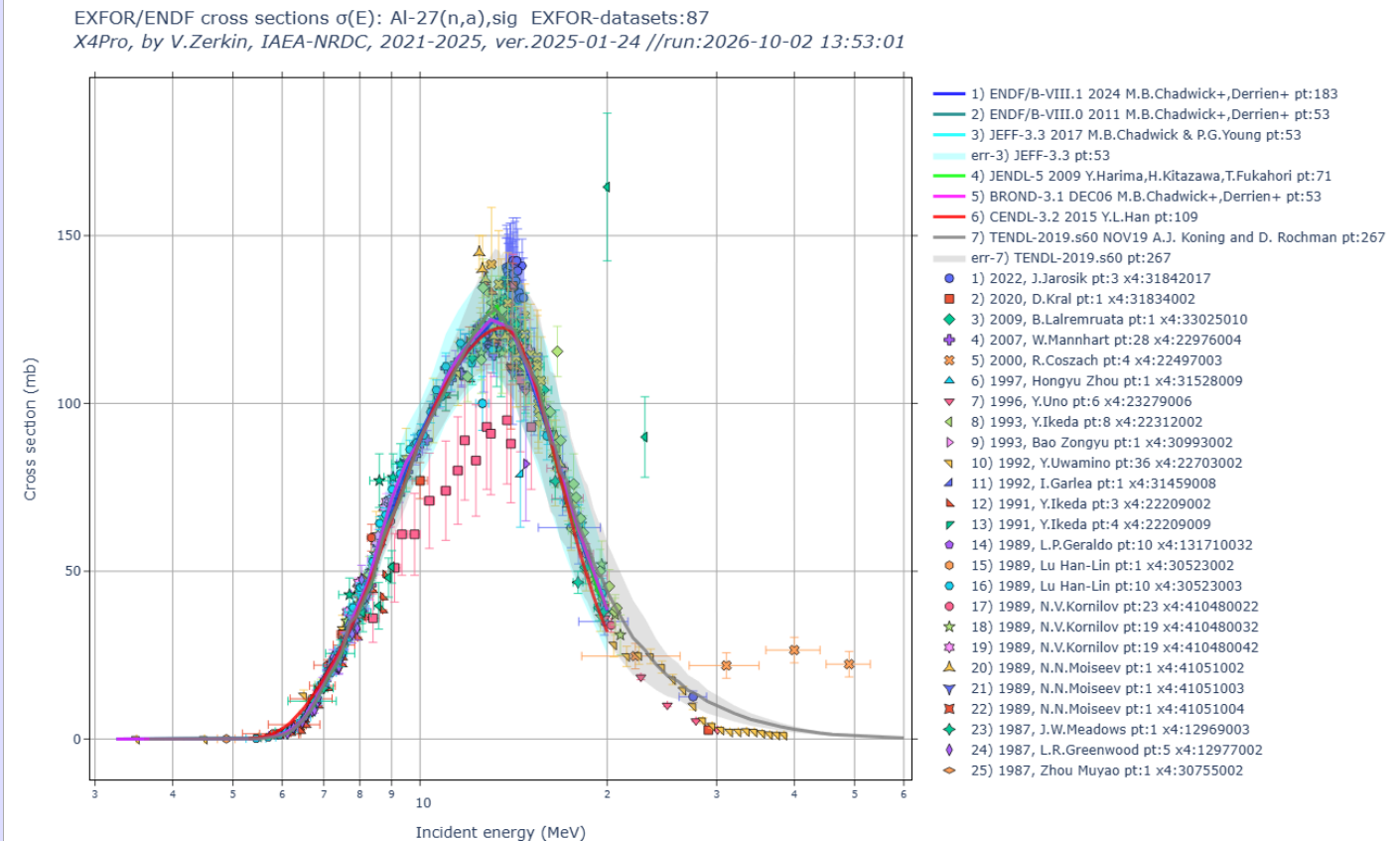
part1-6-fy/ EXFOR fission yield

7) part1-6-fy/out00/U238nf-fy.html.png



part2-1-sig1/ Retrieve and plot EXFOR/ENDF cross sections

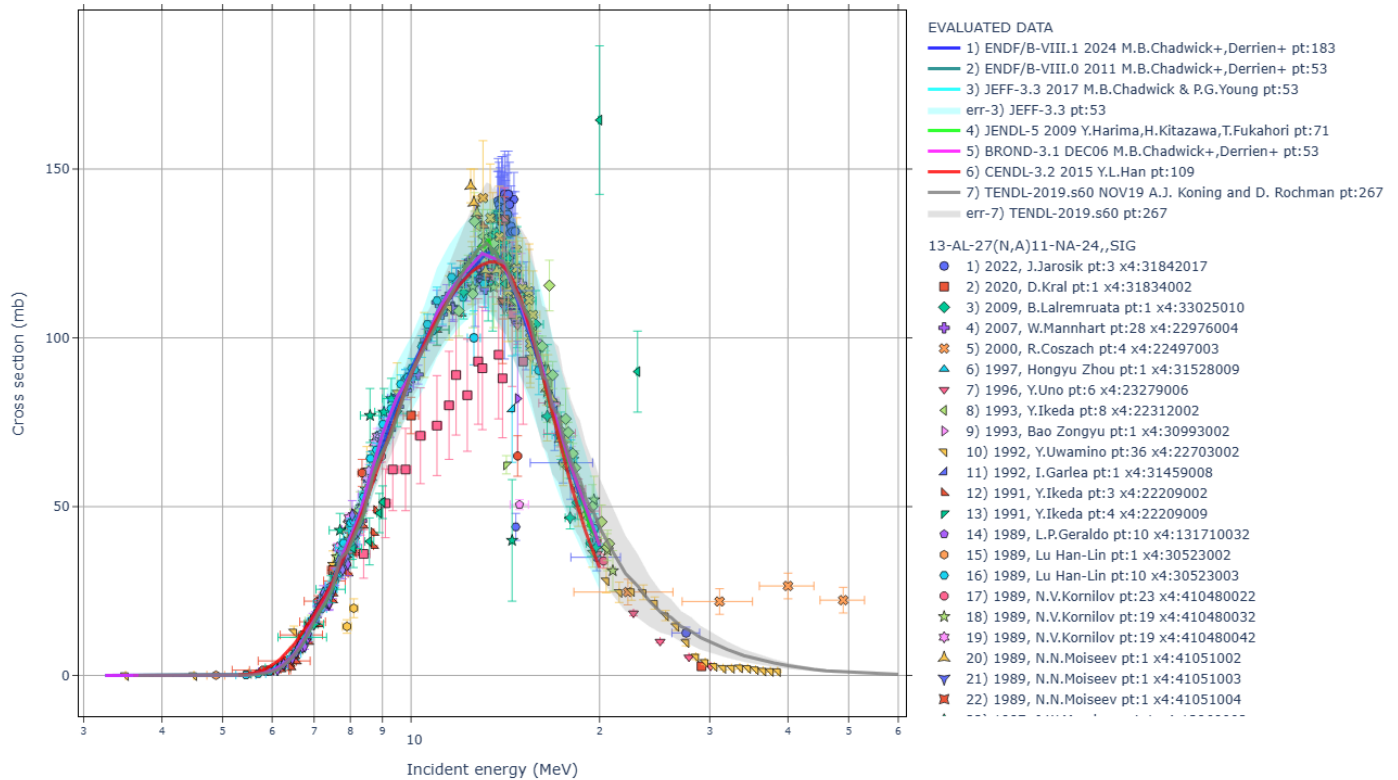
8) part2-1-sig1/out00/Al27na.html.png



9) part2-1-sig1/out00/Al27na2.html.png

EXFOR/ENDF cross sections $\sigma(E)$: Al-27(n,a),sig EXFOR-datasets:93

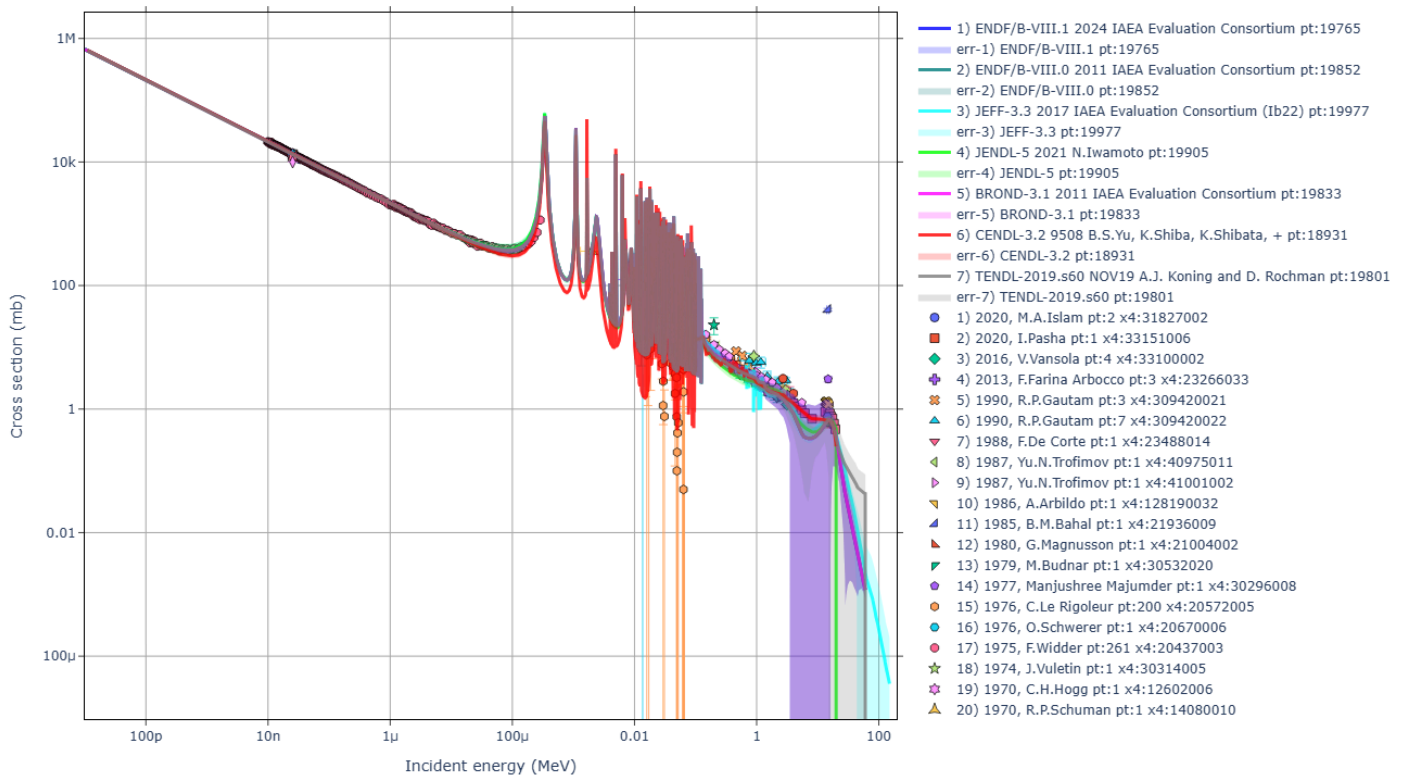
X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-01-24 //run:2026-10-02 13:53:04



10) part2-1-sig1/out00/mn55ng.html.png

EXFOR/ENDF cross sections $\sigma(E)$: mn-55(n,g),sig EXFOR-datasets:49

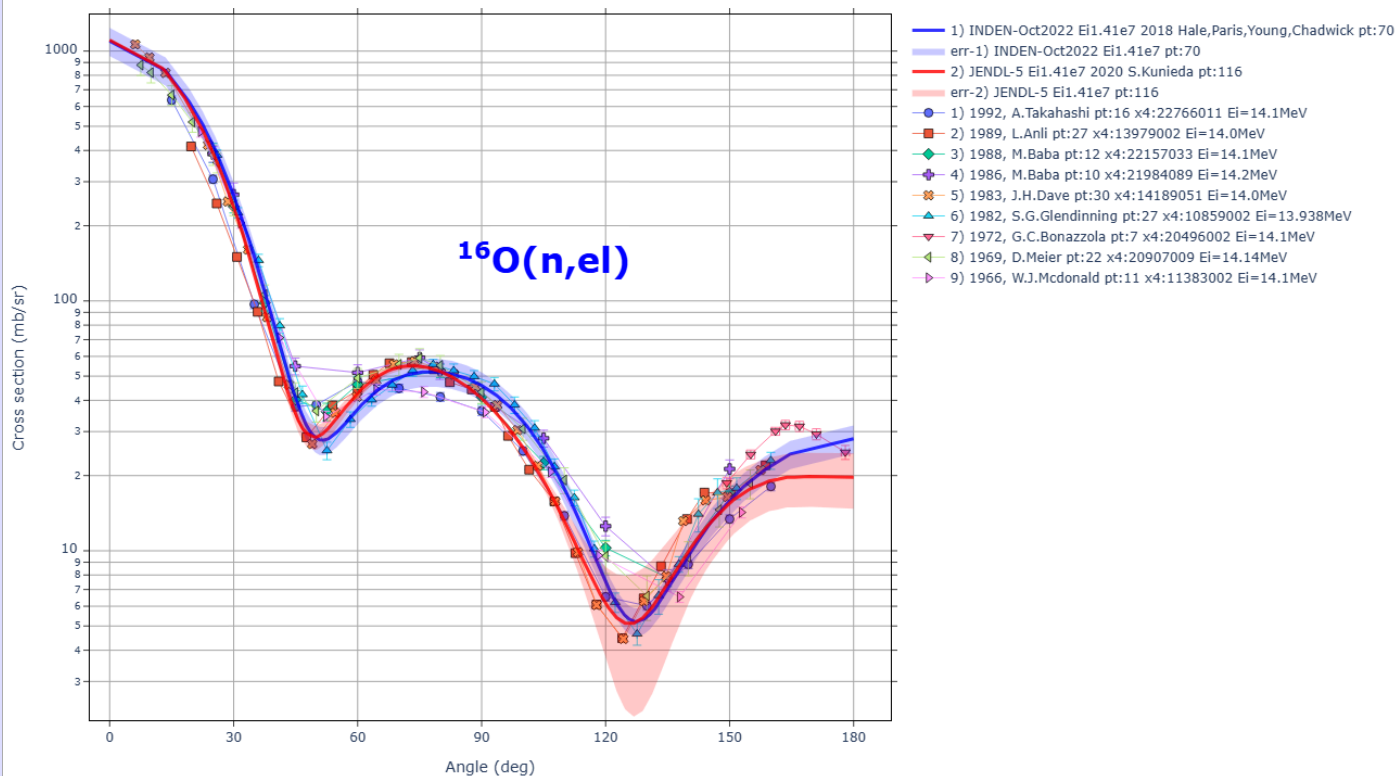
X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-01-24 //run:2026-10-02 13:53:07



part2-2-da1an/ Angular distributions with fixed incident energy

11) part2-2-da1an/out00/da1an.html.png

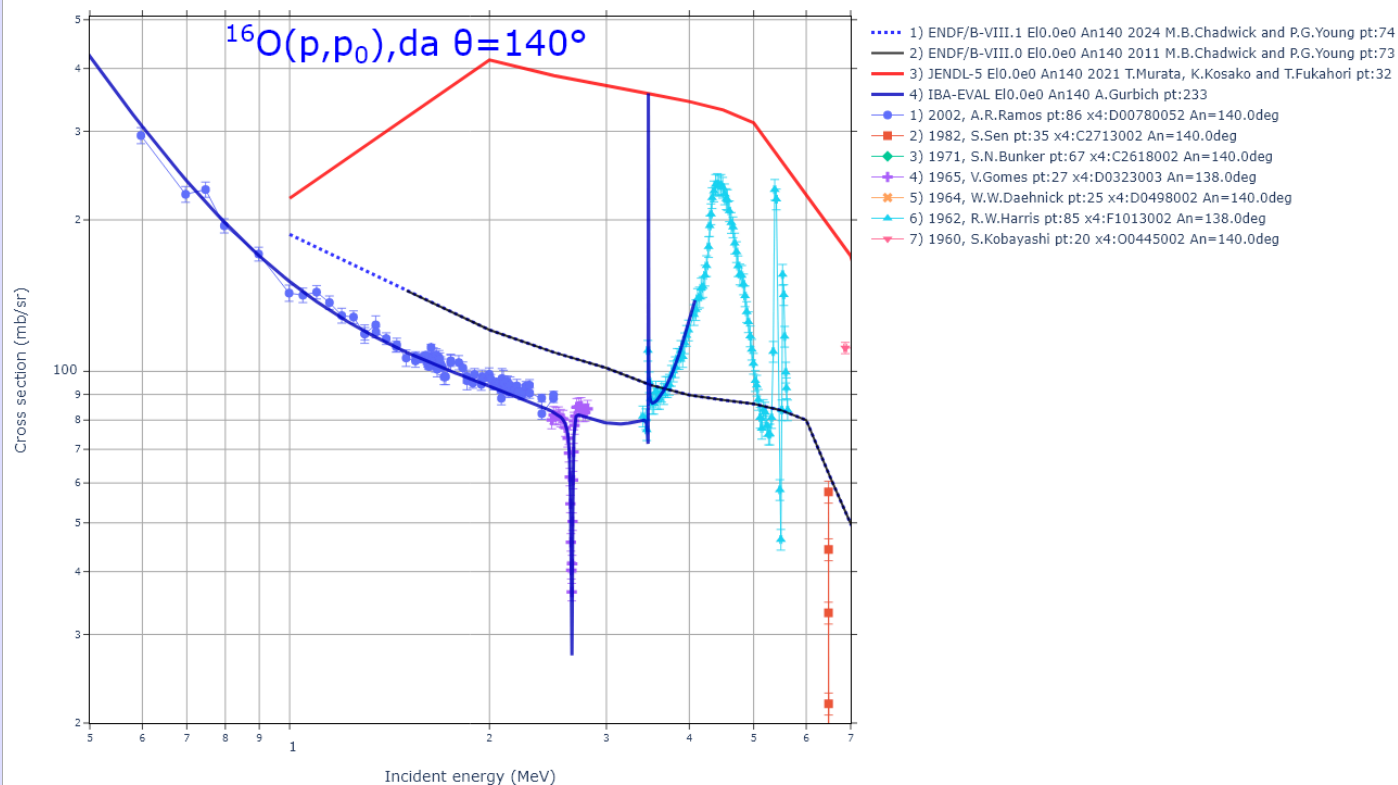
EXFOR/ENDF angular distributions $d\sigma/d\Omega(E,\theta)$: O-16(n,el),da EXFOR-datasets:9
 X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-01-24 //run:2026-10-02 13:53:14



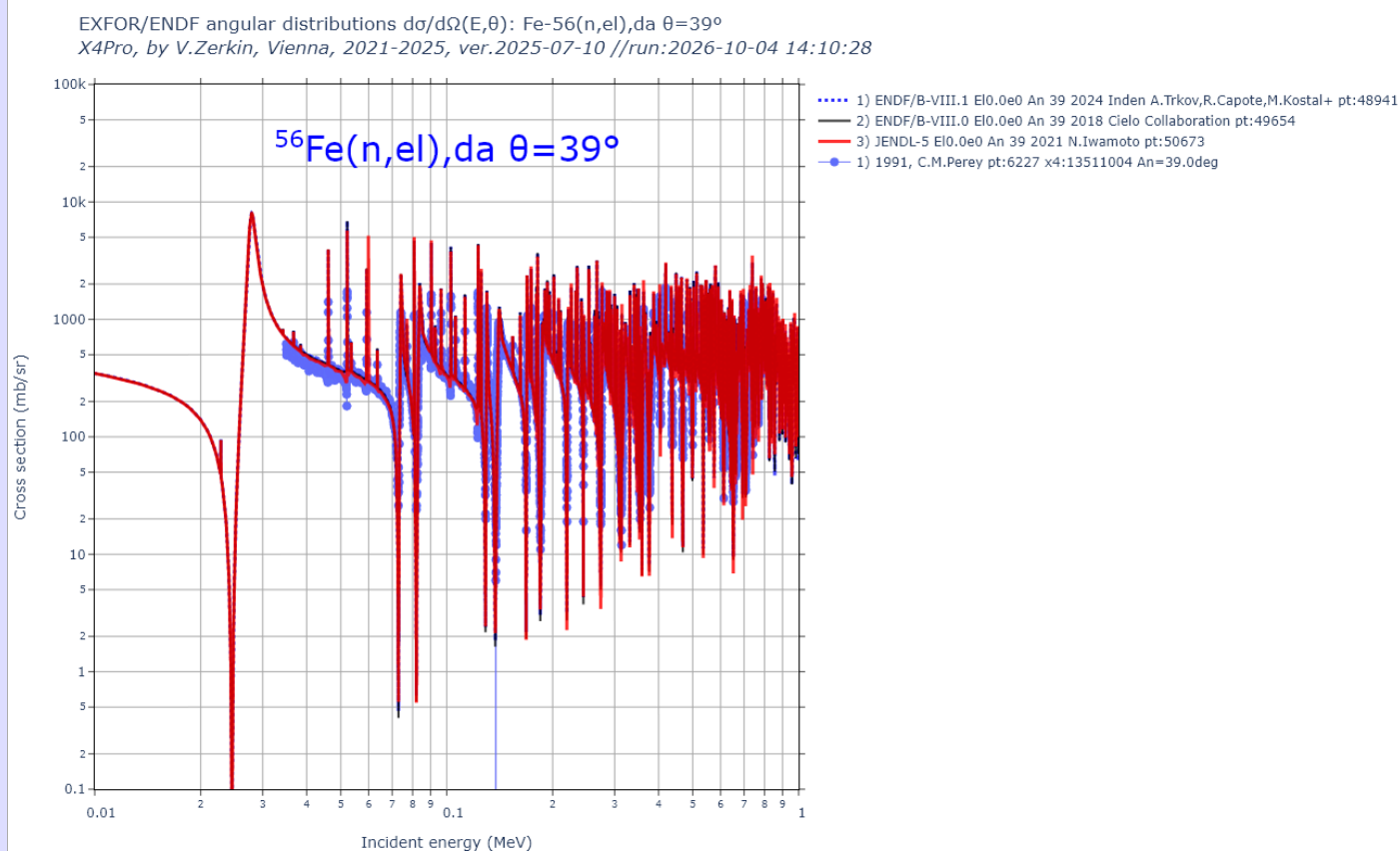
part2-3-da1ei/ Angular distributions with fixed angle

12) part2-3-da1ei/out00/da1ei-ex1.html.png

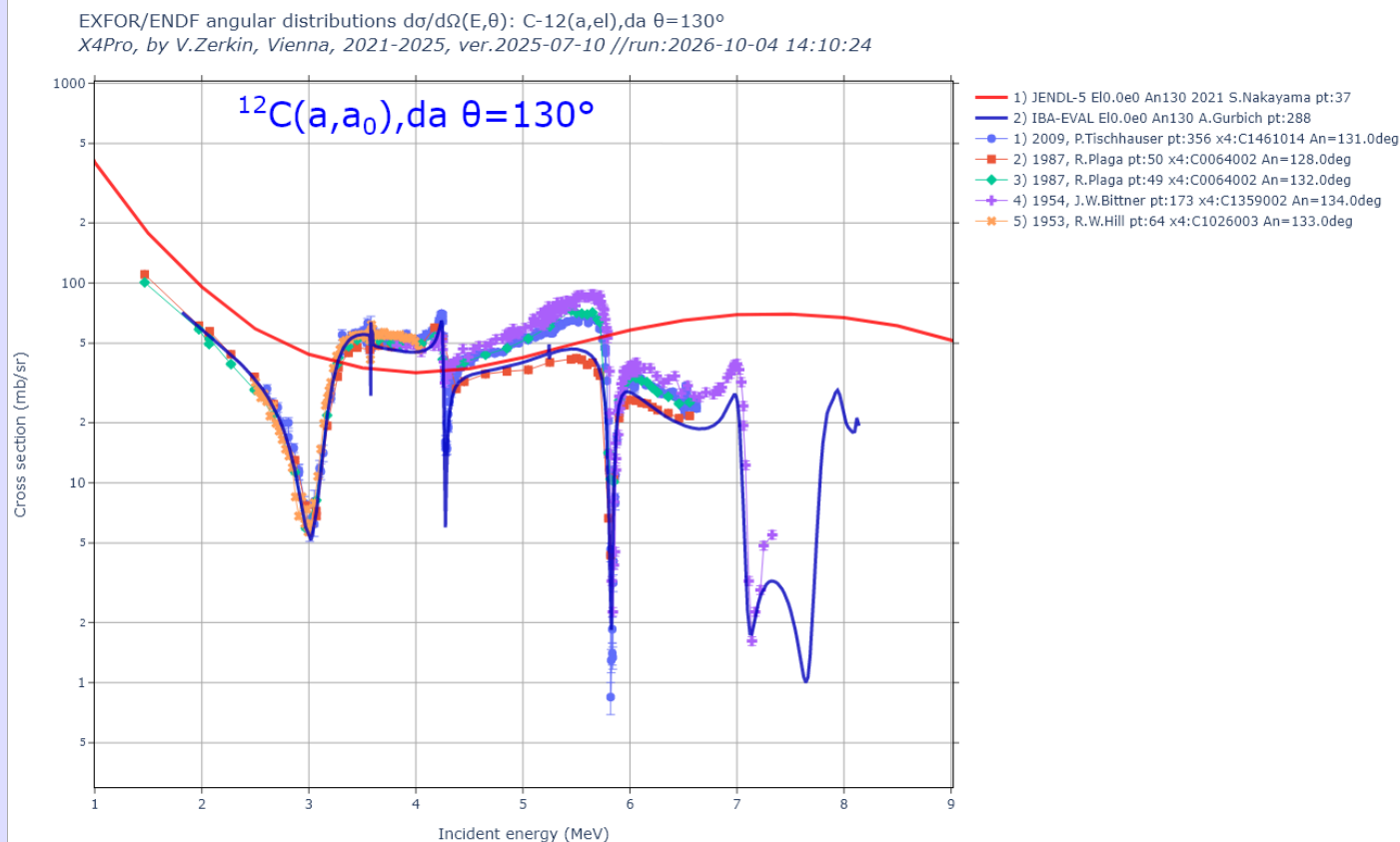
EXFOR/ENDF angular distributions $d\sigma/d\Omega(E,\theta)$: O-16(p,p₀),da $\theta=140^\circ$
 X4Pro, by V.Zerkin, Vienna, 2021-2025, ver.2025-07-10 //run:2026-10-04 14:10:26



13) part2-3-da1ei/out00/da1ei-ex2.html.png

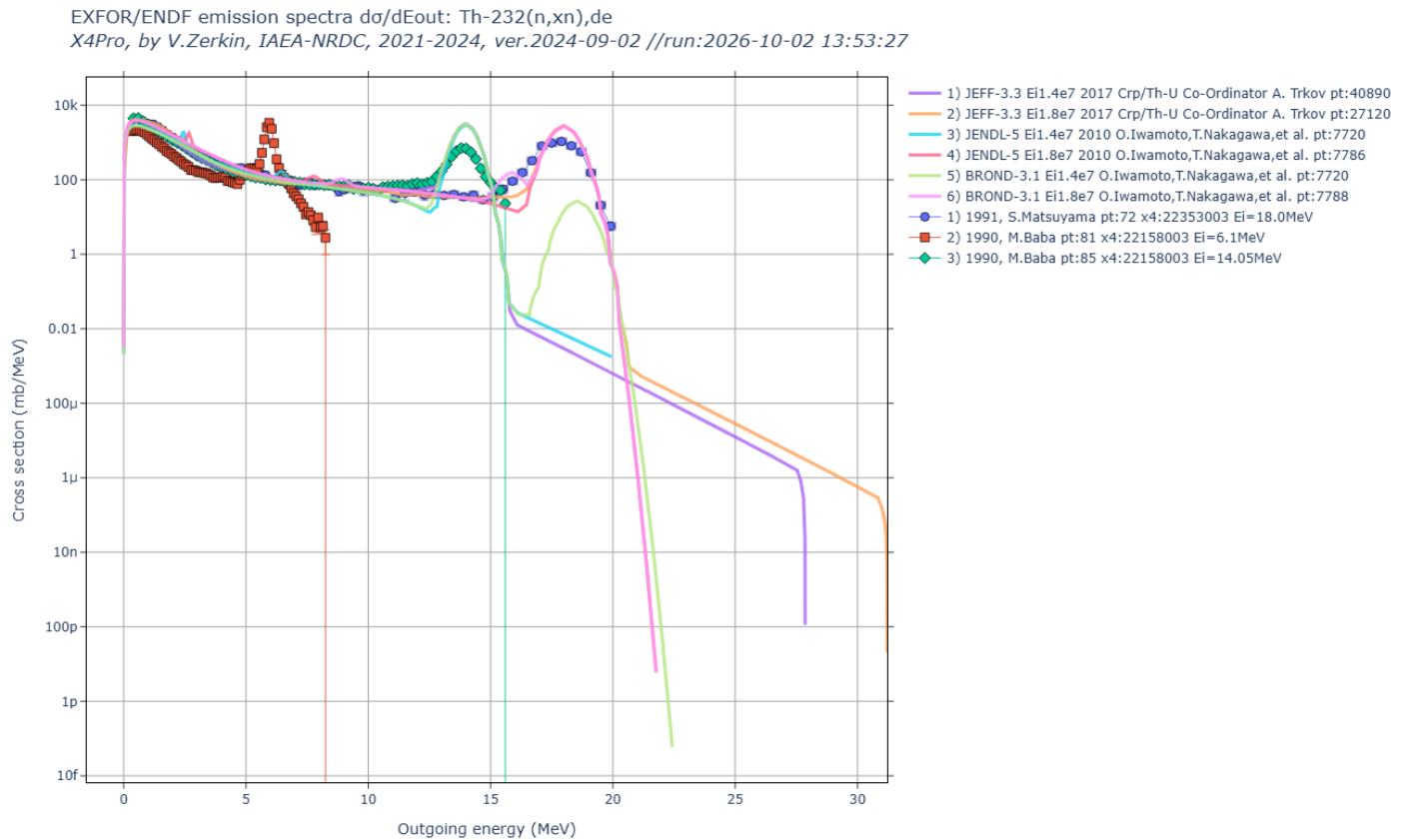


14) part2-3-da1ei/out00/da1ei.html.png



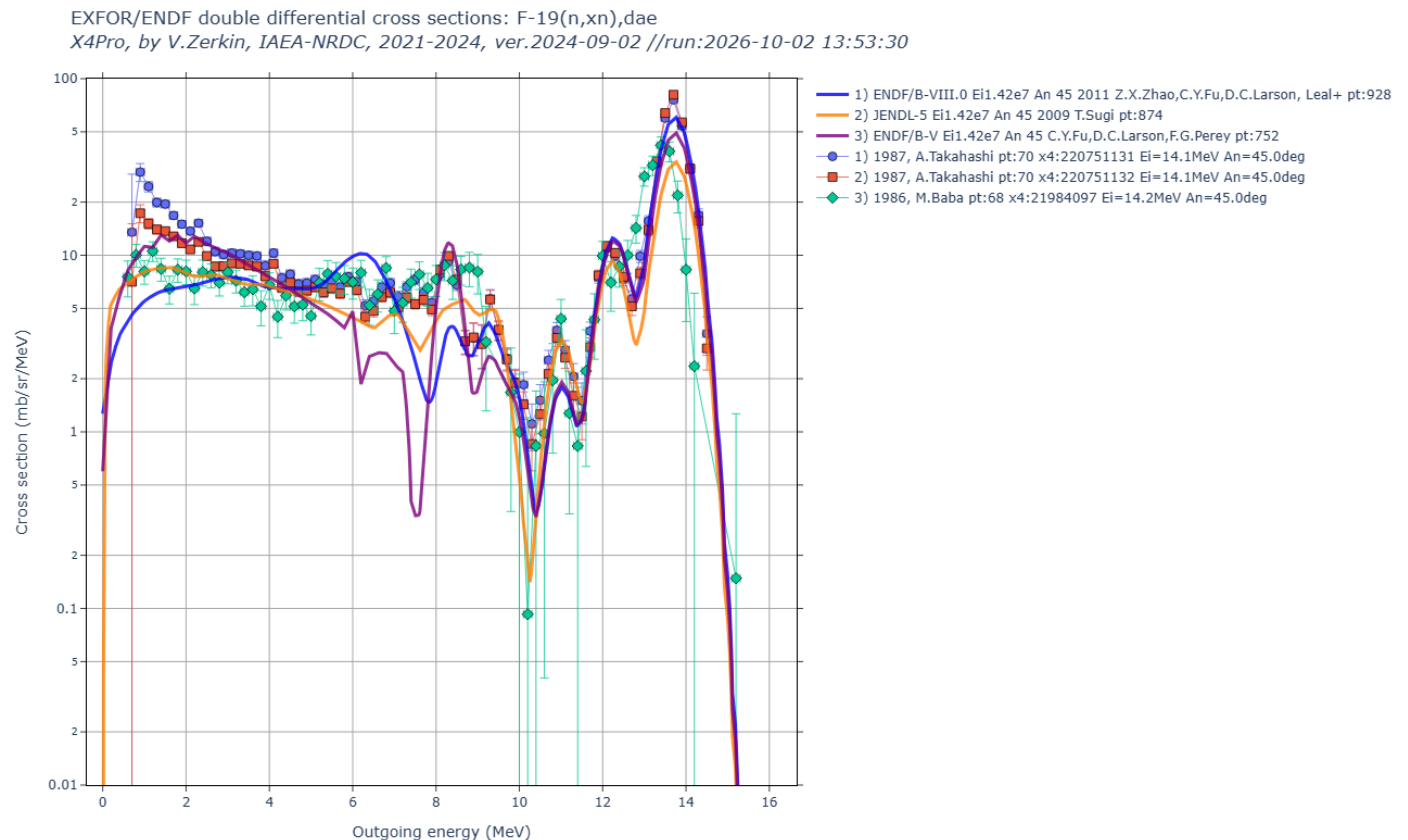
part2-4-de1eo/ Energy spectra of outgoing particles

15) part2-4-de1eo/out00/de1eo.html.png



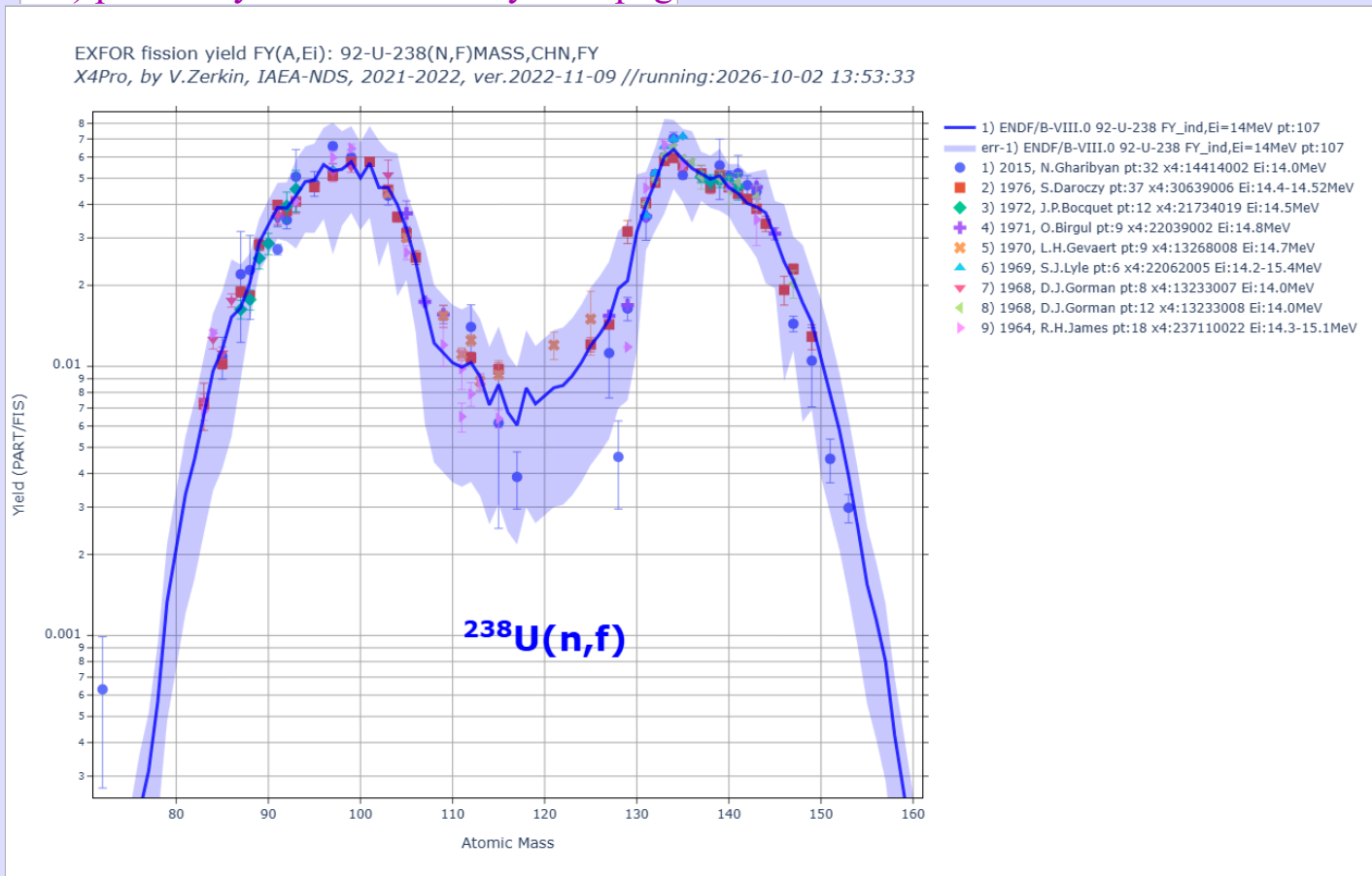
part2-5-dae1eo/ Double differential cross sections

16) part2-5-dae1eo/out00/dae1eo.html.png



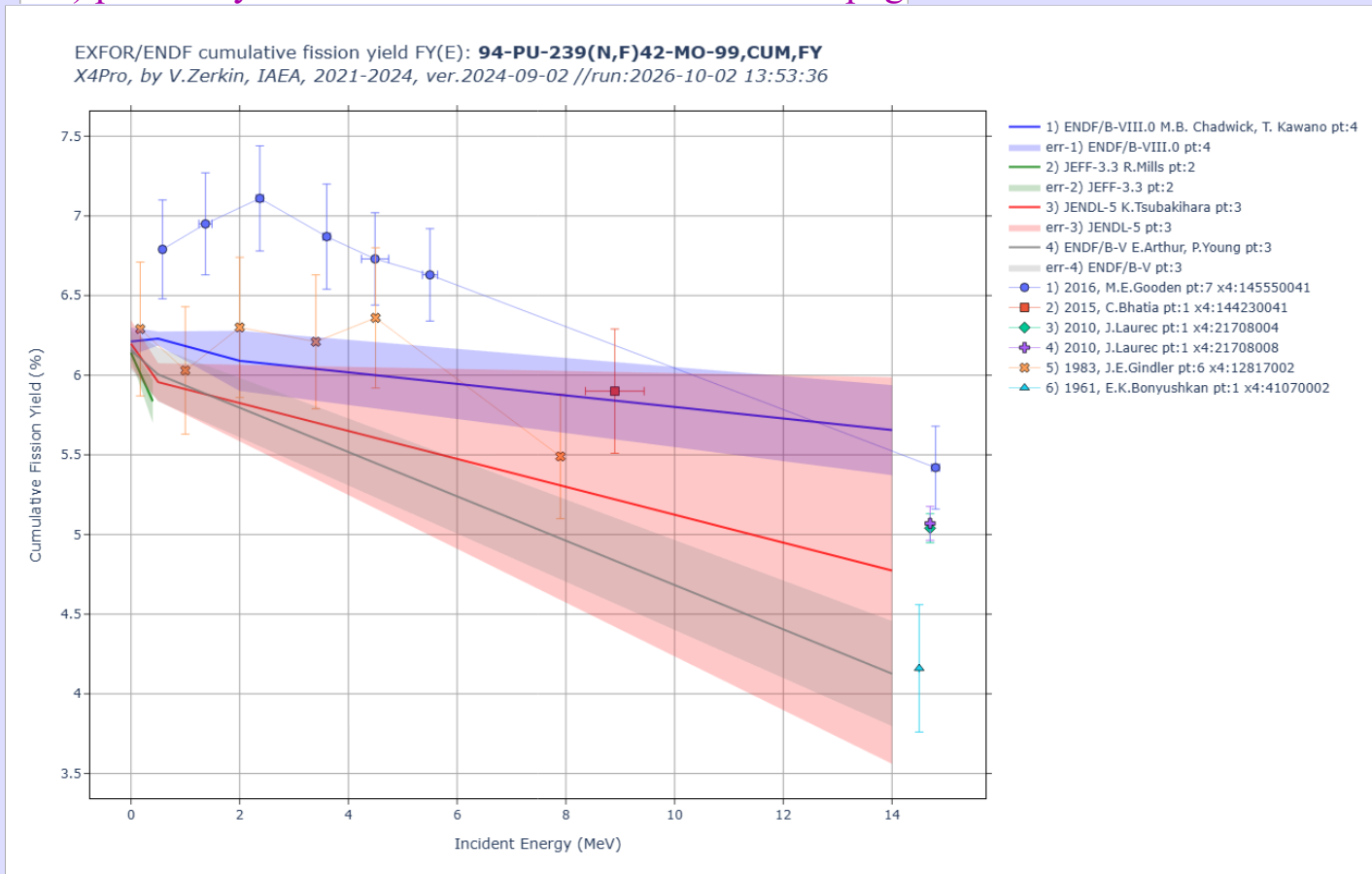
part2-6-fy/ Fission yeild (mass distribution)

17) part2-6-fy/out00/U238nf-fy.html.png

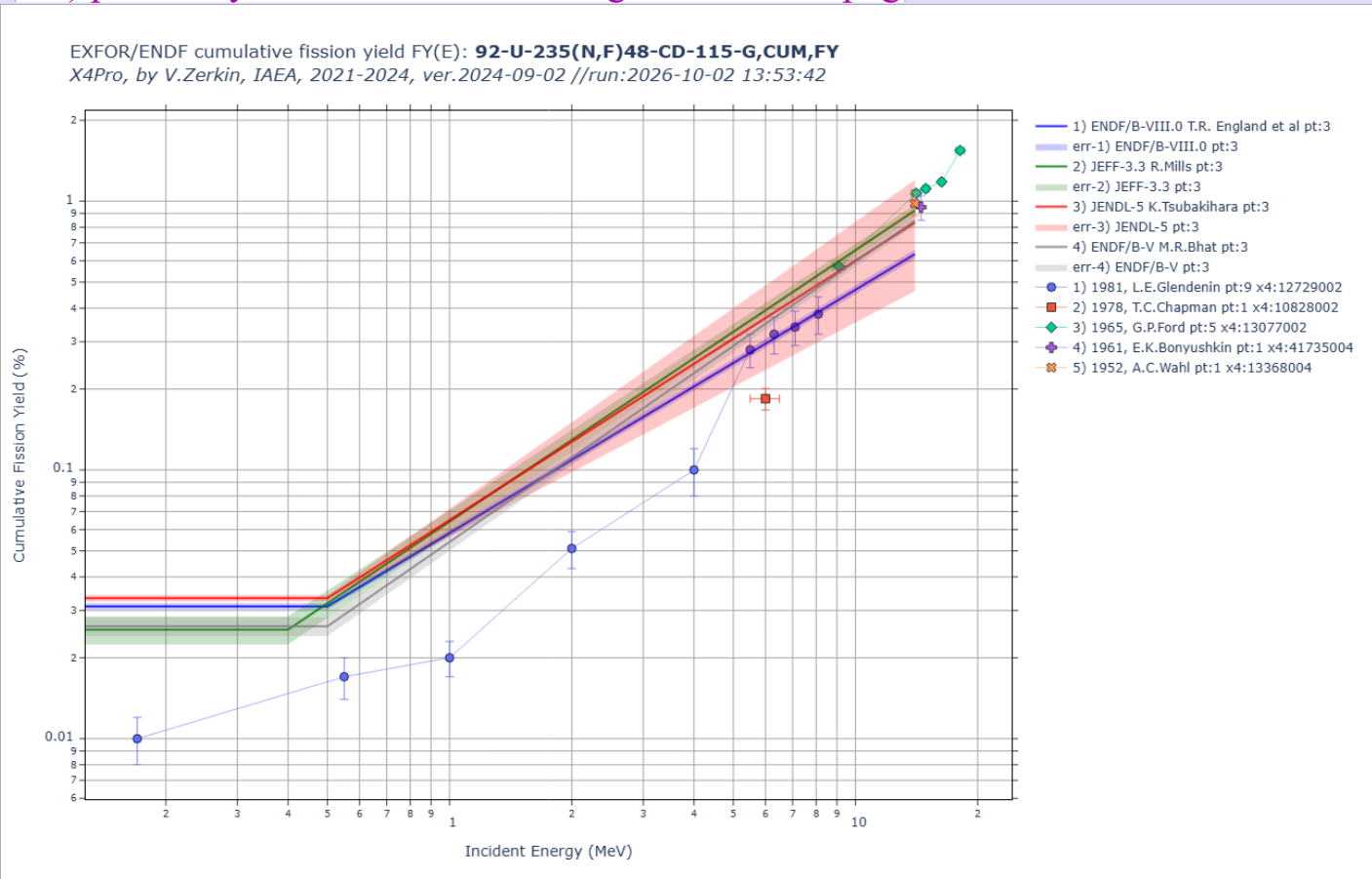


part2-7-fye/ Fission yeild (energy distribution)

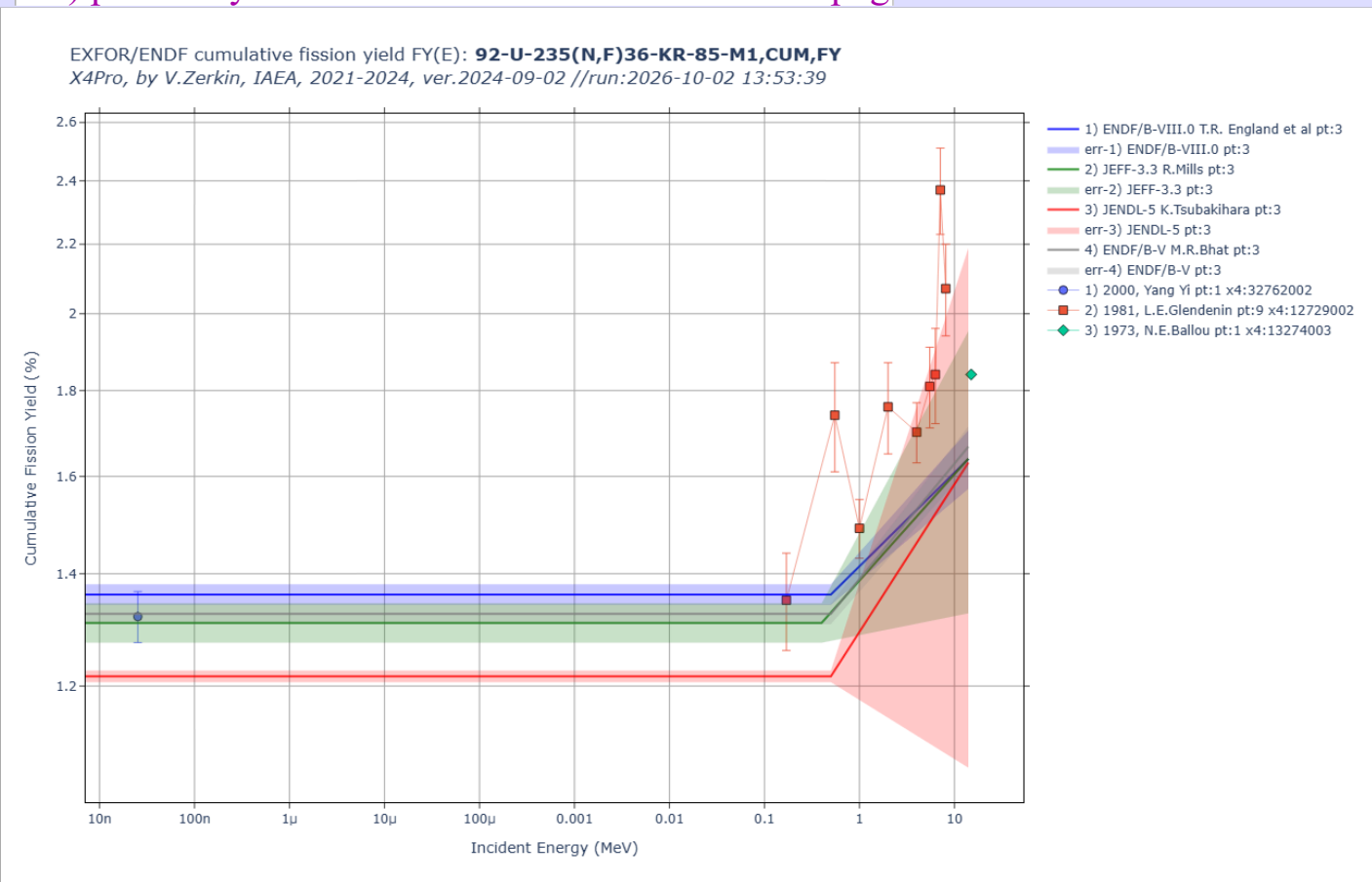
18) part2-7-fye/out00/Pu239-nf-Mo99-cumFY.html.png



19) part2-7-fye/out00/u235-nf-cd115g-cumFY.html.png



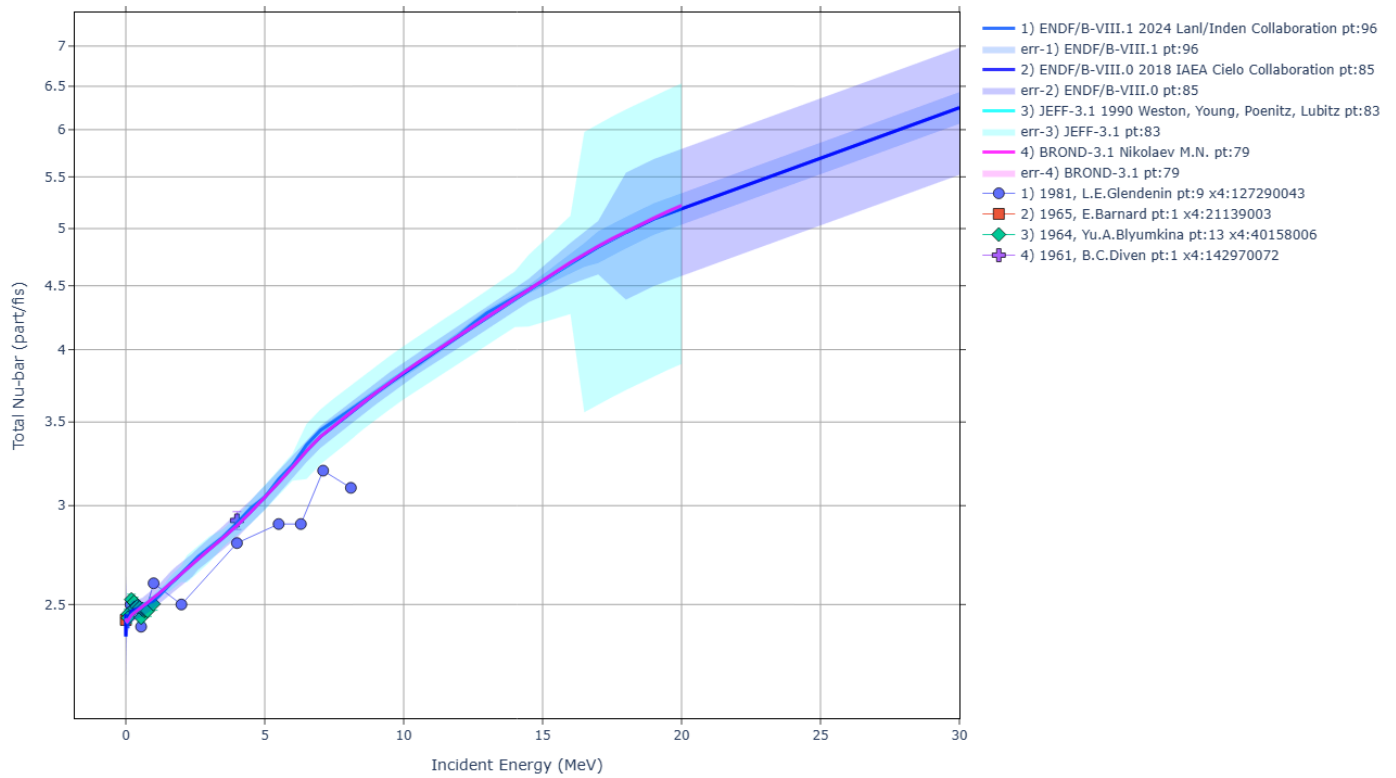
20) part2-7-fye/out00/u235-nf-kr85m-cumFY.html.png



part2-9-nubar/ NUBAR - average number of neutrons per fission

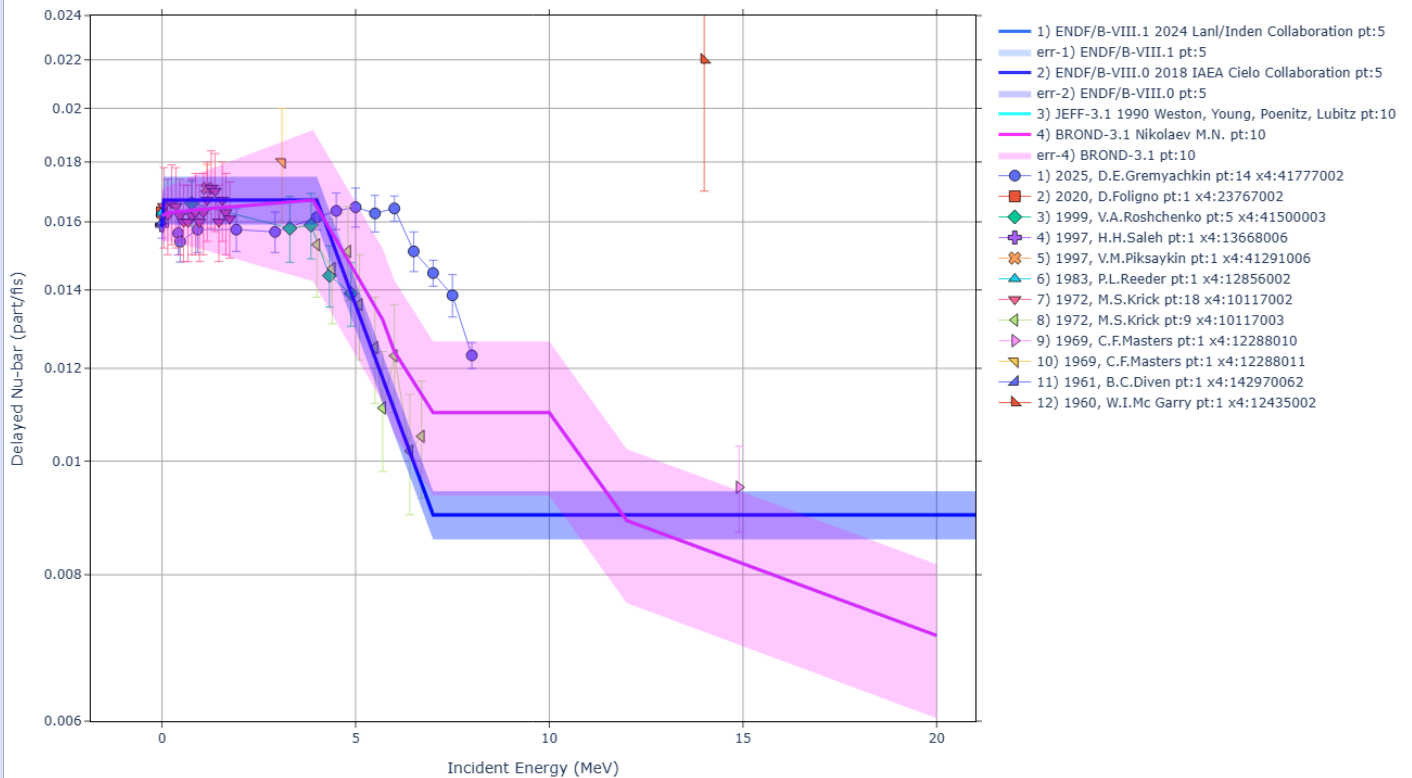
21) part2-9-nubar/out00/U235nf_-nu.html.png

EXFOR-ENDF NUBAR: 92-U-235(N,F),,NU Exp.Datasets:4 Points:24
 X4Pro, by V.Zerkin, 2026, ver.2026-08-24 //running:2026-10-02 13:53:54



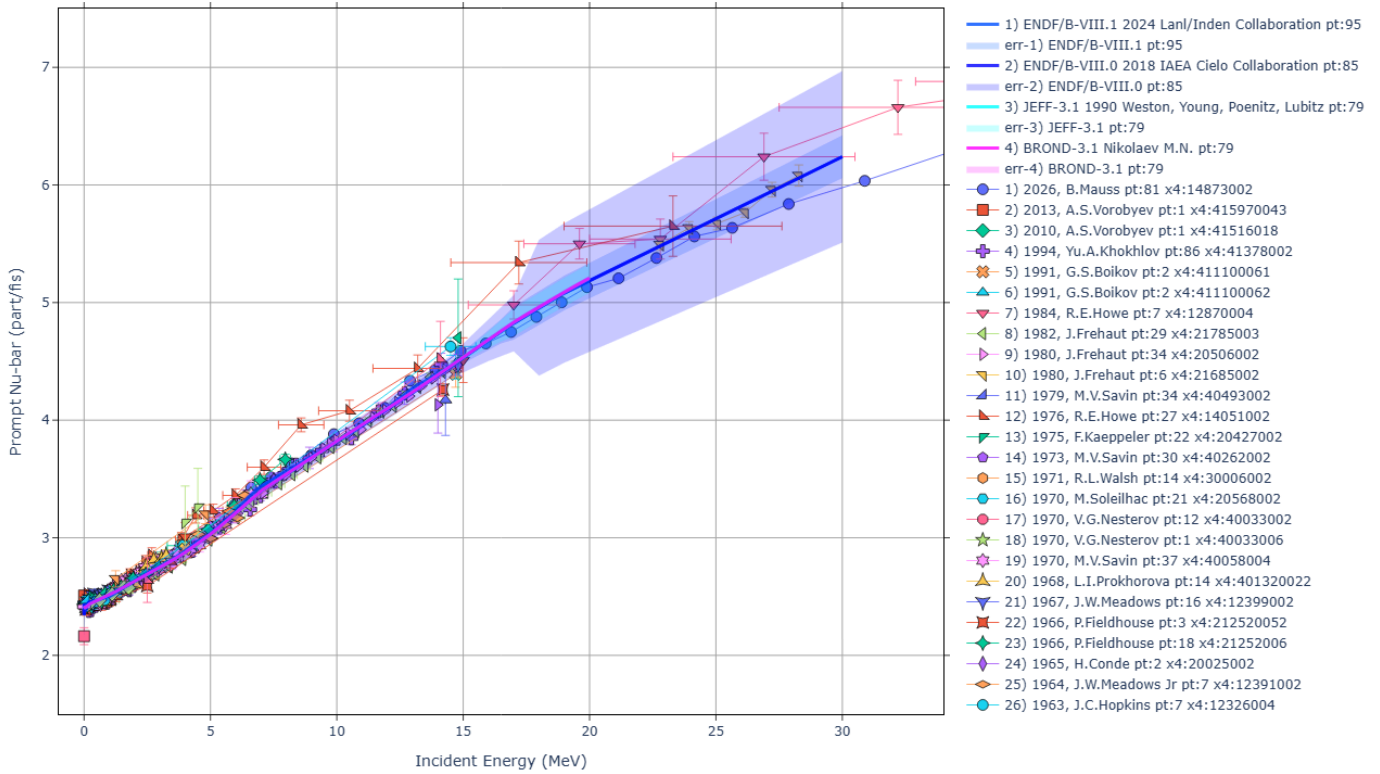
22) part2-9-nubar/out00/U235nf_dl-nu.html.png

EXFOR-ENDF NUBAR: 92-U-235(N,F),DL,NU Exp.Datasets:12 Points:54
 X4Pro, by V.Zerkin, 2026, ver.2026-08-24 //running:2026-10-02 13:53:51



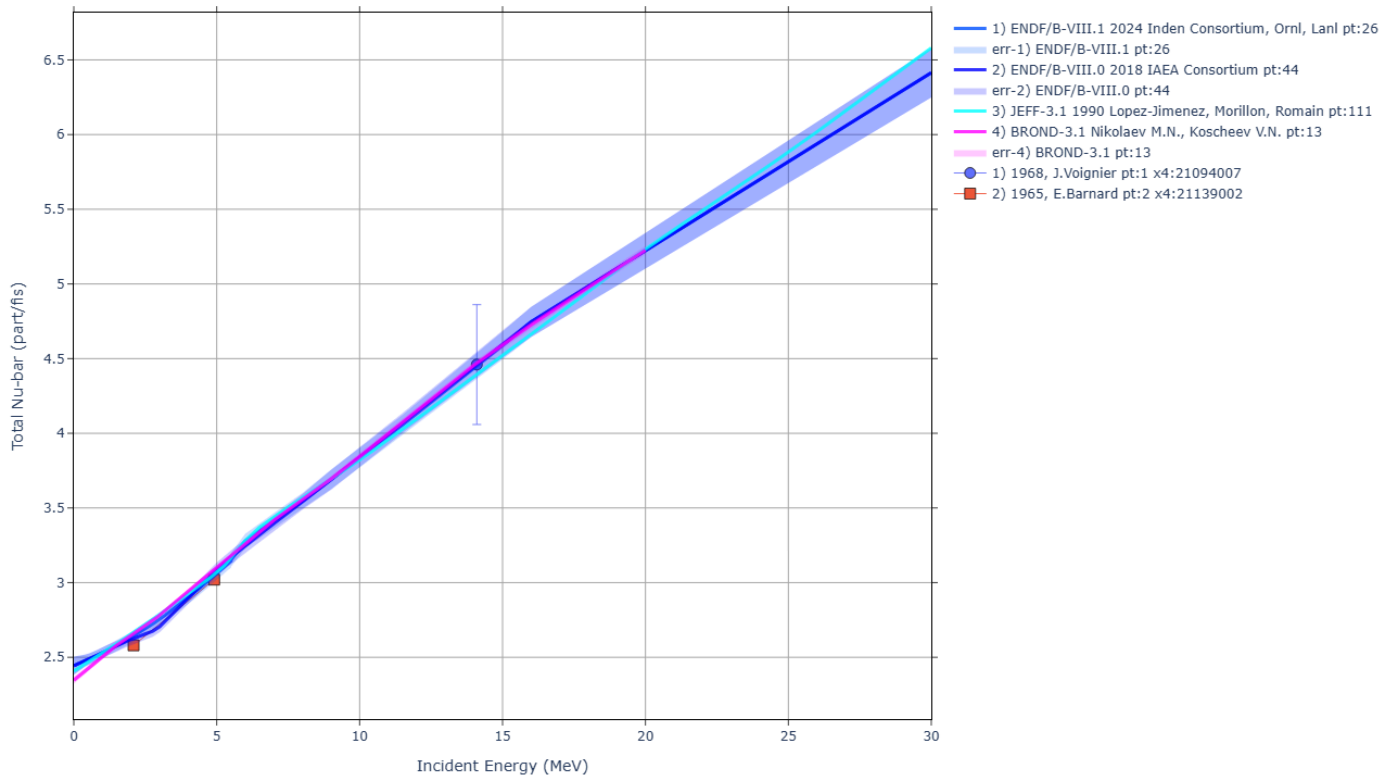
23) part2-9-nubar/out00/U235nf_pr-nu.html.png

EXFOR-ENDF NUBAR: 92-U-235(N,F),PR,NU Exp.Datasets:38 Points:535
X4Pro, by V.Zerkin, 2026, ver.2026-08-24 //running:2026-10-02 13:53:48

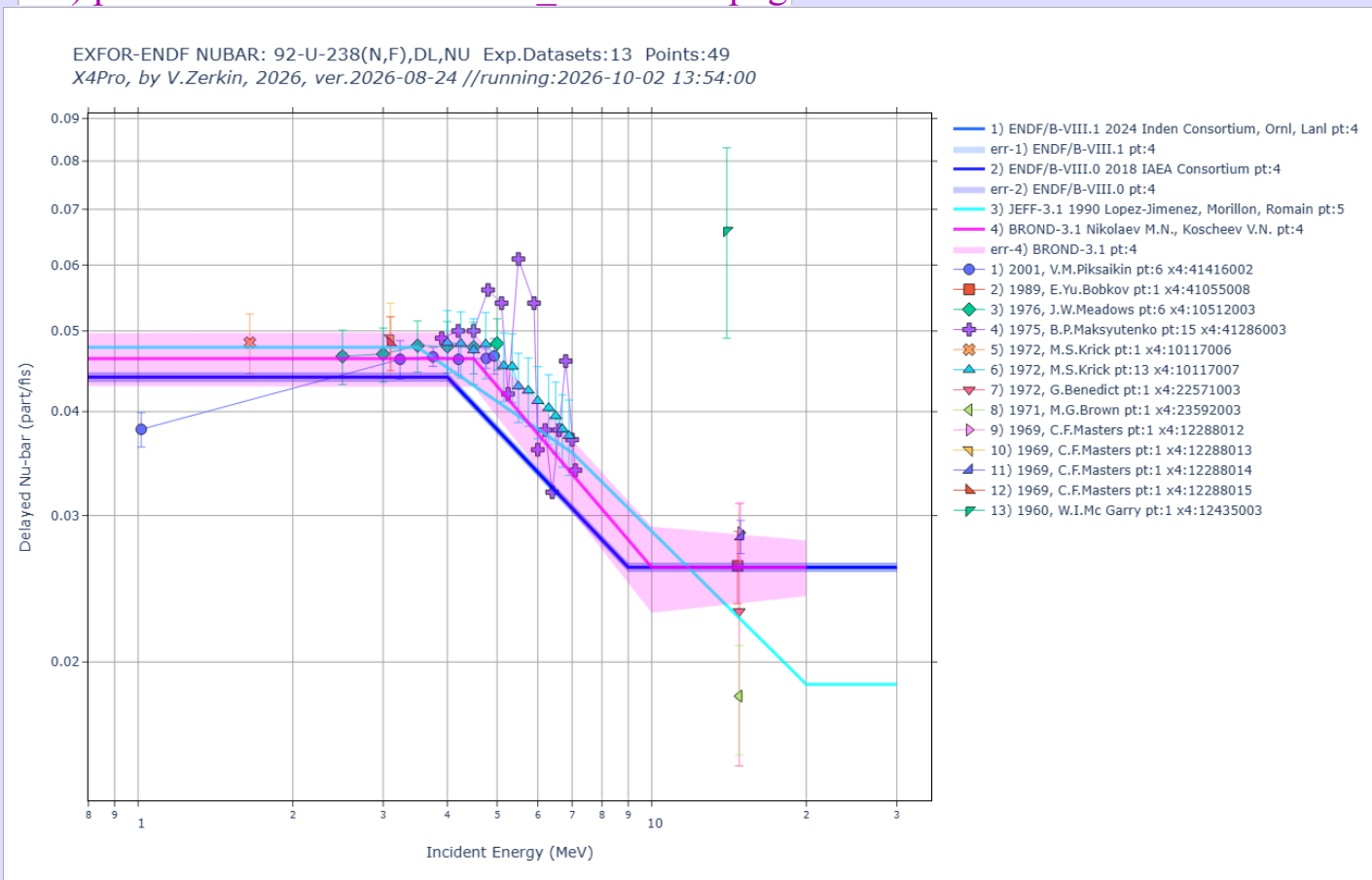


24) part2-9-nubar/out00/U238nf_-nu.html.png

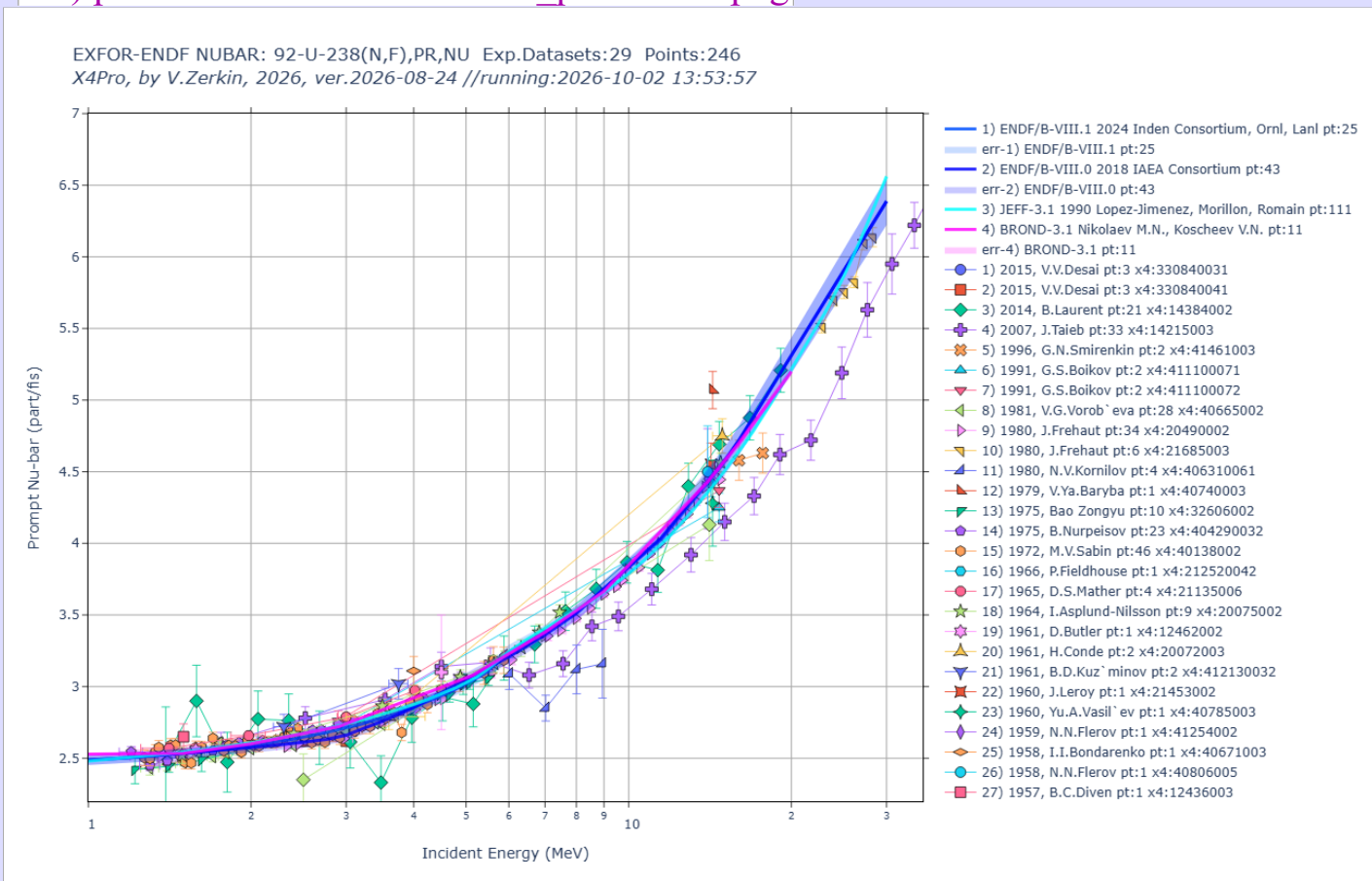
EXFOR-ENDF NUBAR: 92-U-238(N,F),,NU Exp.Datasets:2 Points:3
X4Pro, by V.Zerkin, 2026, ver.2026-08-24 //running:2026-10-02 13:54:03



25) part2-9-nubar/out00/U238nf_dl-nu.html.png

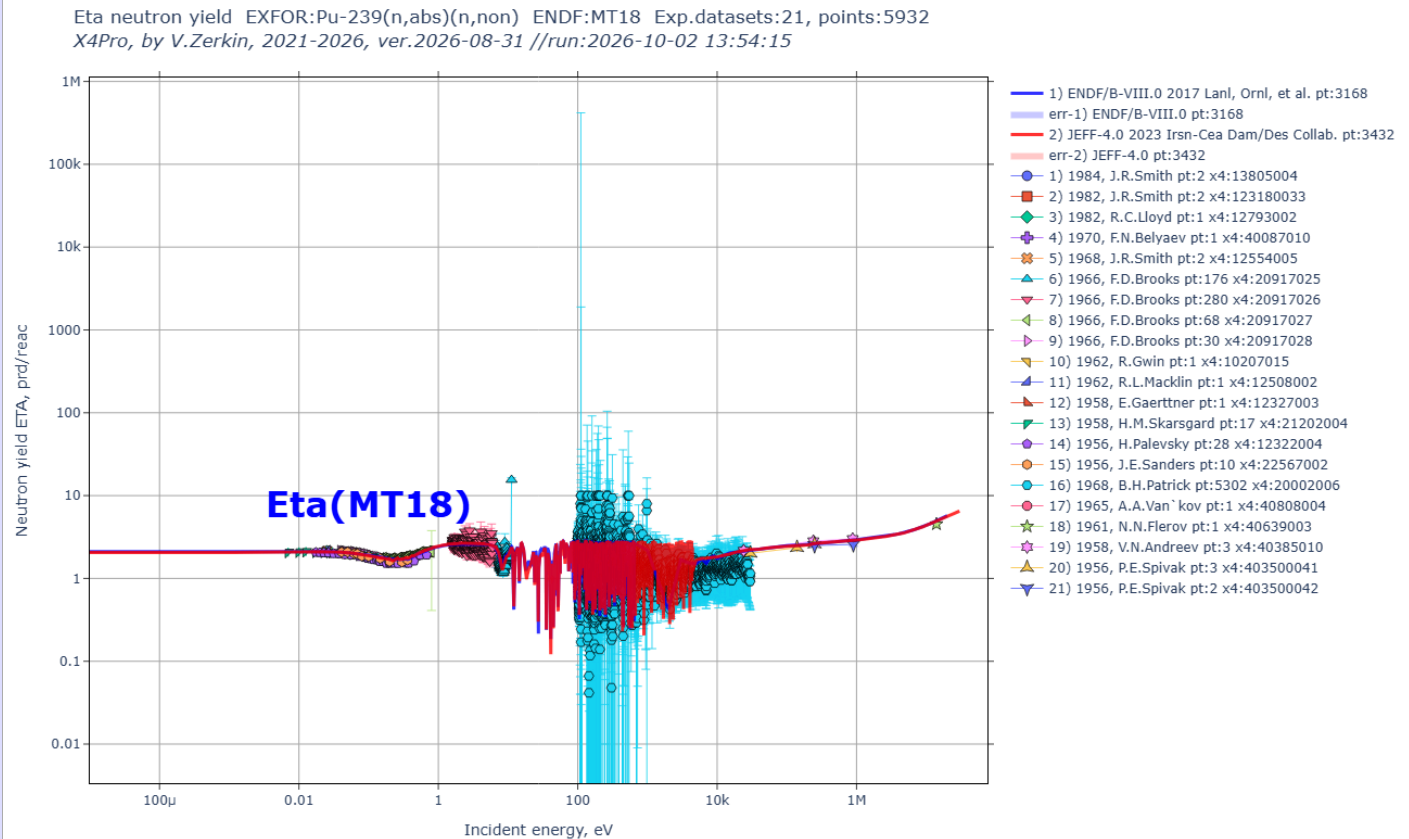


26) part2-9-nubar/out00/U238nf_pr-nu.html.png

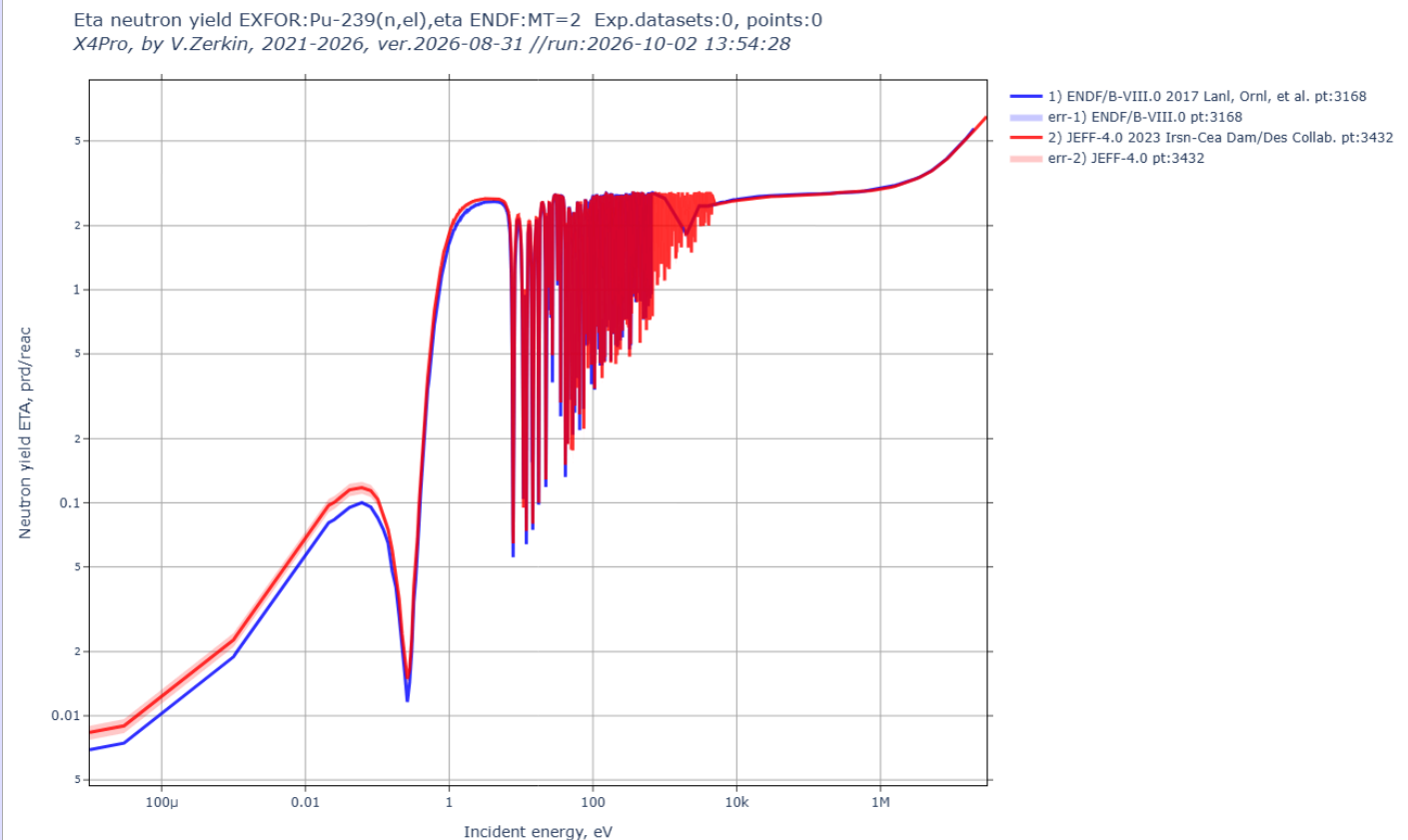


part2-A-eta/ Eta - average neutron yield per nonelastic event

27) part2-A-eta/out00/Pu239-eta.html.png

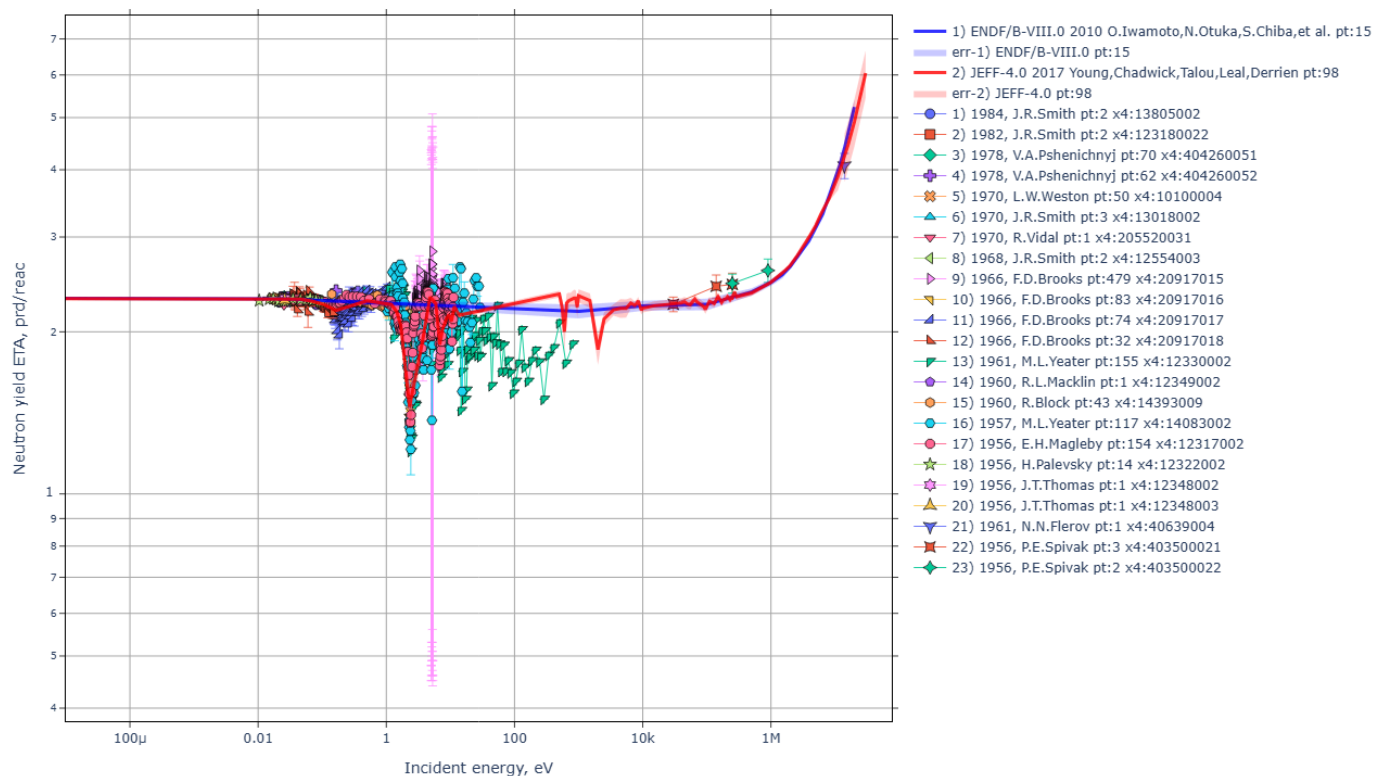


28) part2-A-eta/out00/Pu239nel-eta.html.png



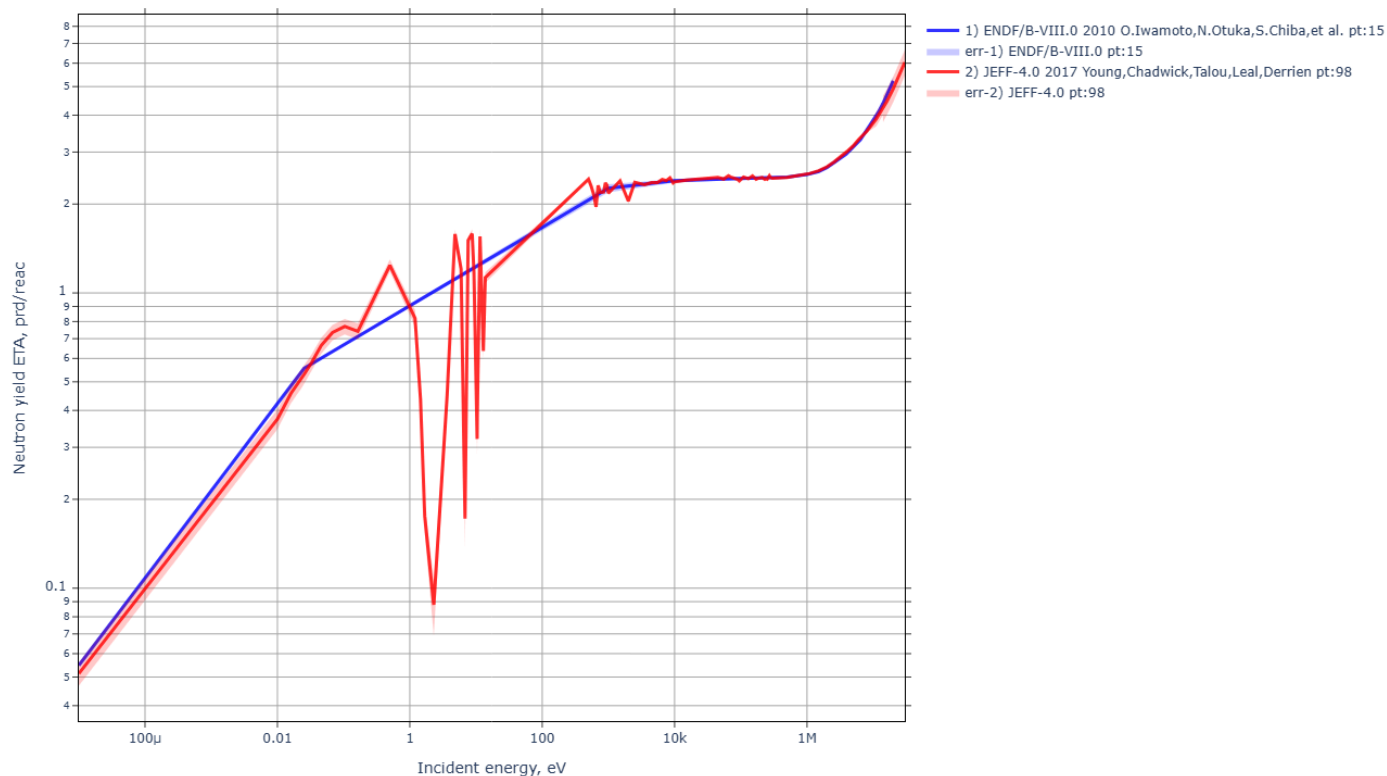
29) part2-A-eta/out00/U233-eta.html.png

Eta neutron yield EXFOR:U-233(n,abs)(n,non) ENDF:MT18 Exp.datasets:23, points:1352
X4Pro, by V.Zerkin, 2021-2026, ver.2026-08-31 //run:2026-10-02 13:54:12



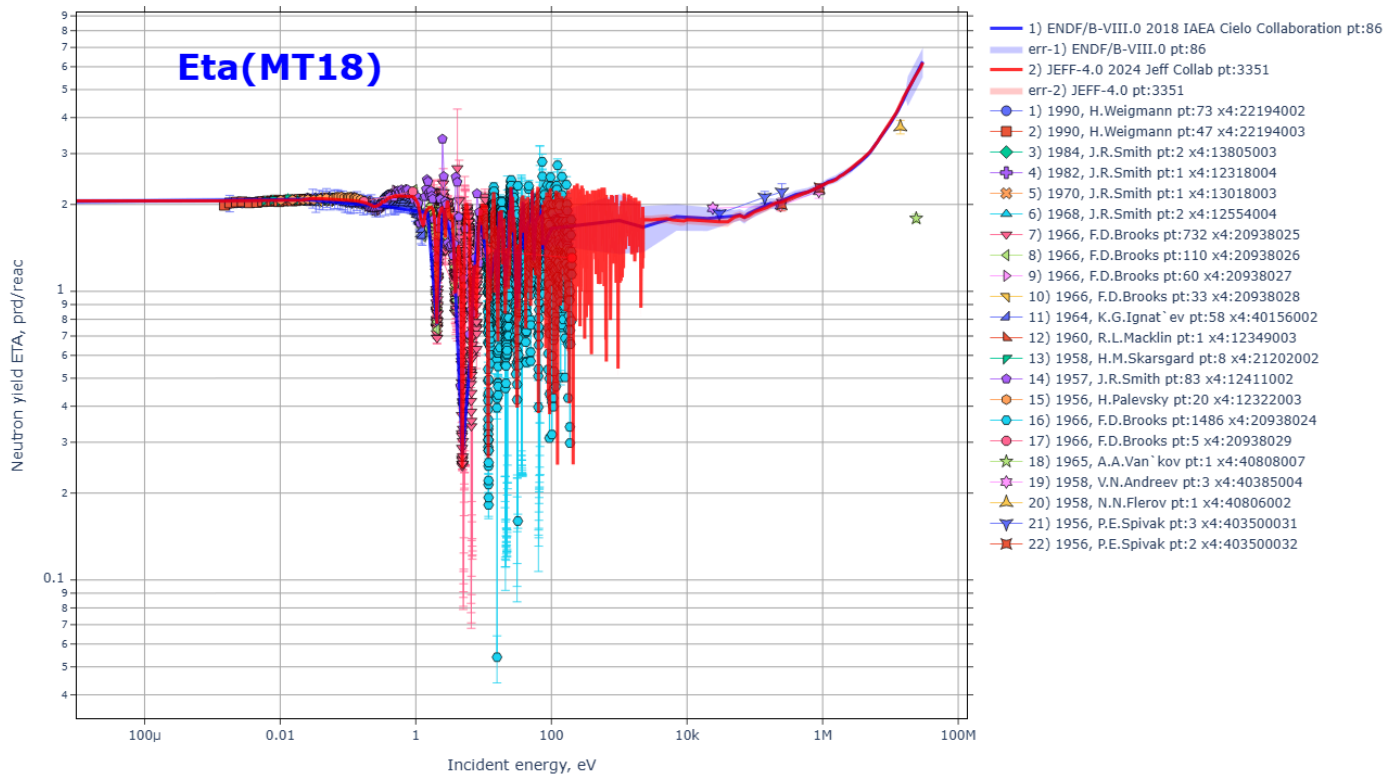
30) part2-A-eta/out00/U233nel-eta.html.png

Eta neutron yield EXFOR:U-233(n,el),eta ENDF:MT=2 Exp.datasets:0, points:0
X4Pro, by V.Zerkin, 2021-2026, ver.2026-08-31 //run:2026-10-02 13:54:25



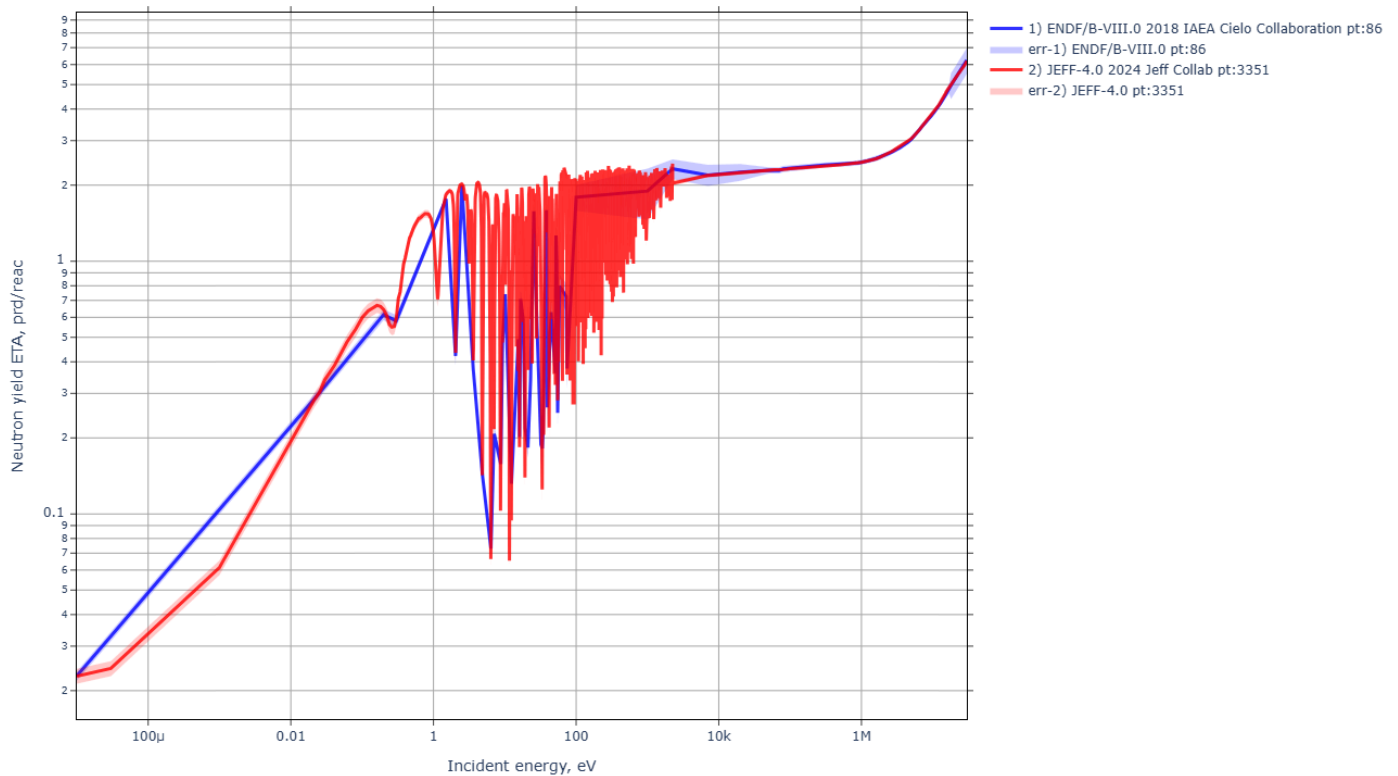
31) part2-A-eta/out00/U235-eta.html.png

Eta neutron yield EXFOR:U-235(n,abs)(n,non) ENDF:MT18 Exp.datasets:22, points:2732
X4Pro, by V.Zerkin, 2021-2026, ver.2026-08-31 //run:2026-10-02 13:54:06

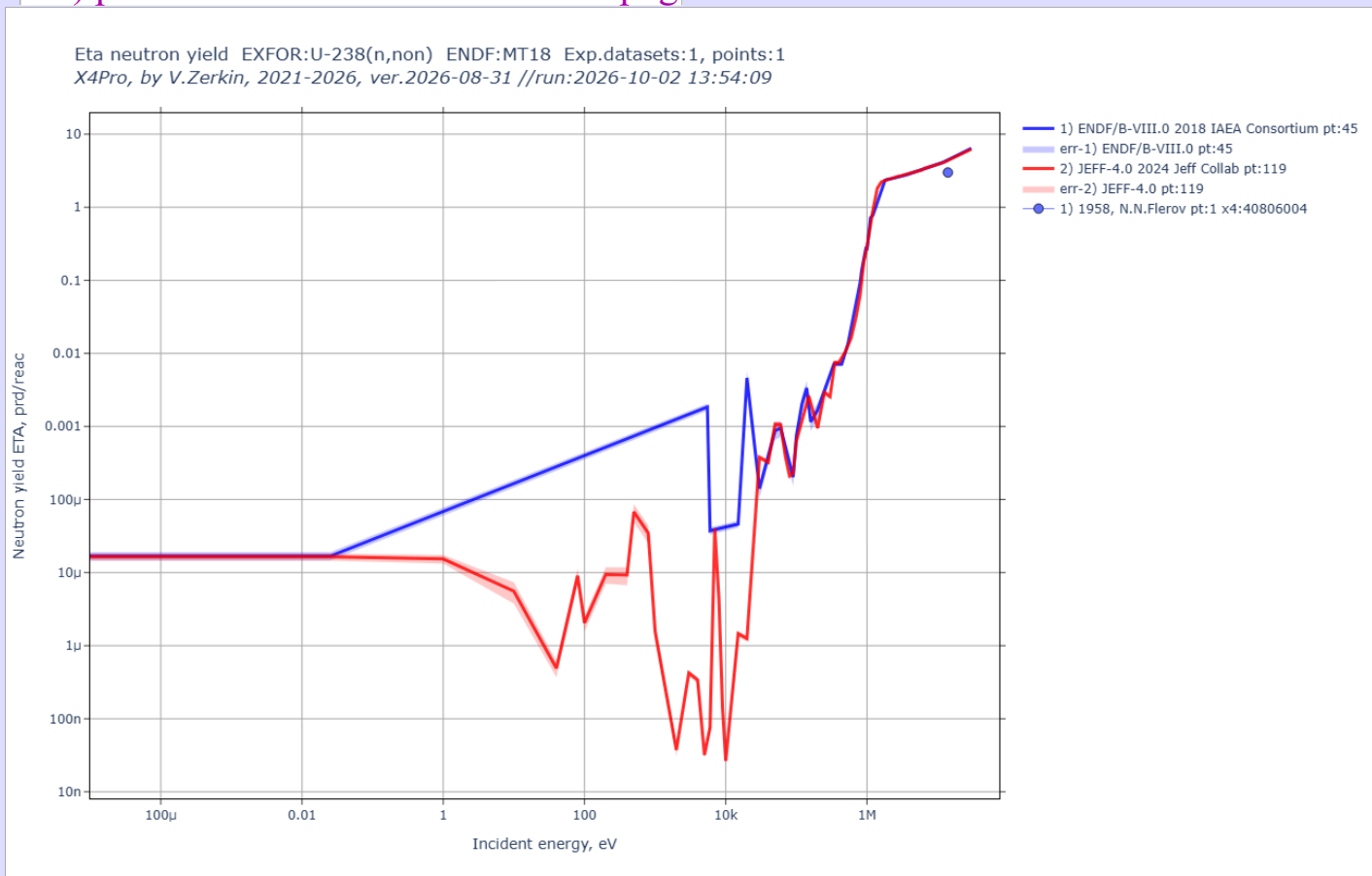


32) part2-A-eta/out00/U235nel-eta.html.png

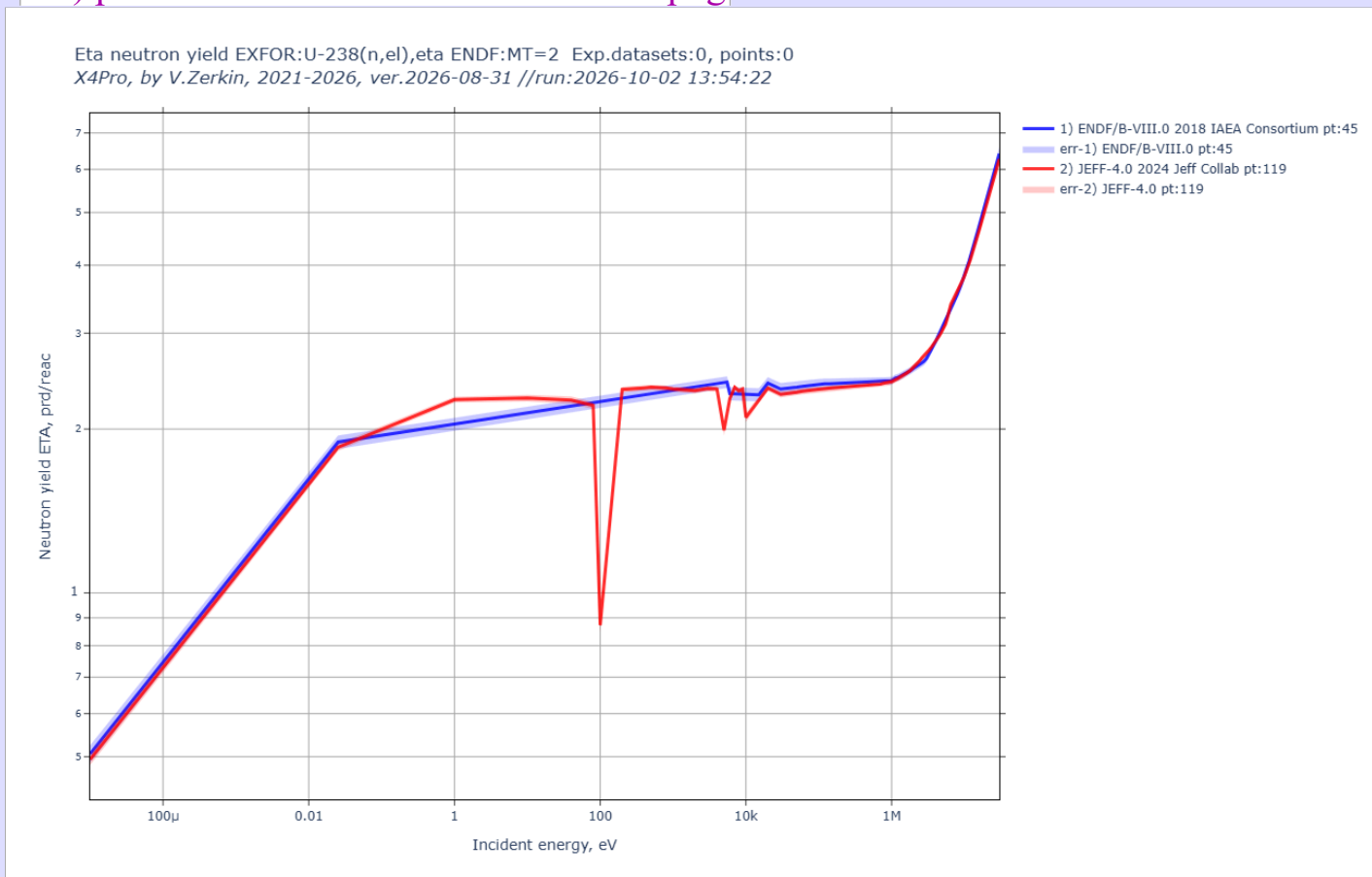
Eta neutron yield EXFOR:U-235(n,el),eta ENDF:MT=2 Exp.datasets:0, points:0
X4Pro, by V.Zerkin, 2021-2026, ver.2026-08-31 //run:2026-10-02 13:54:19



33) part2-A-eta/out00/U238-eta.html.png



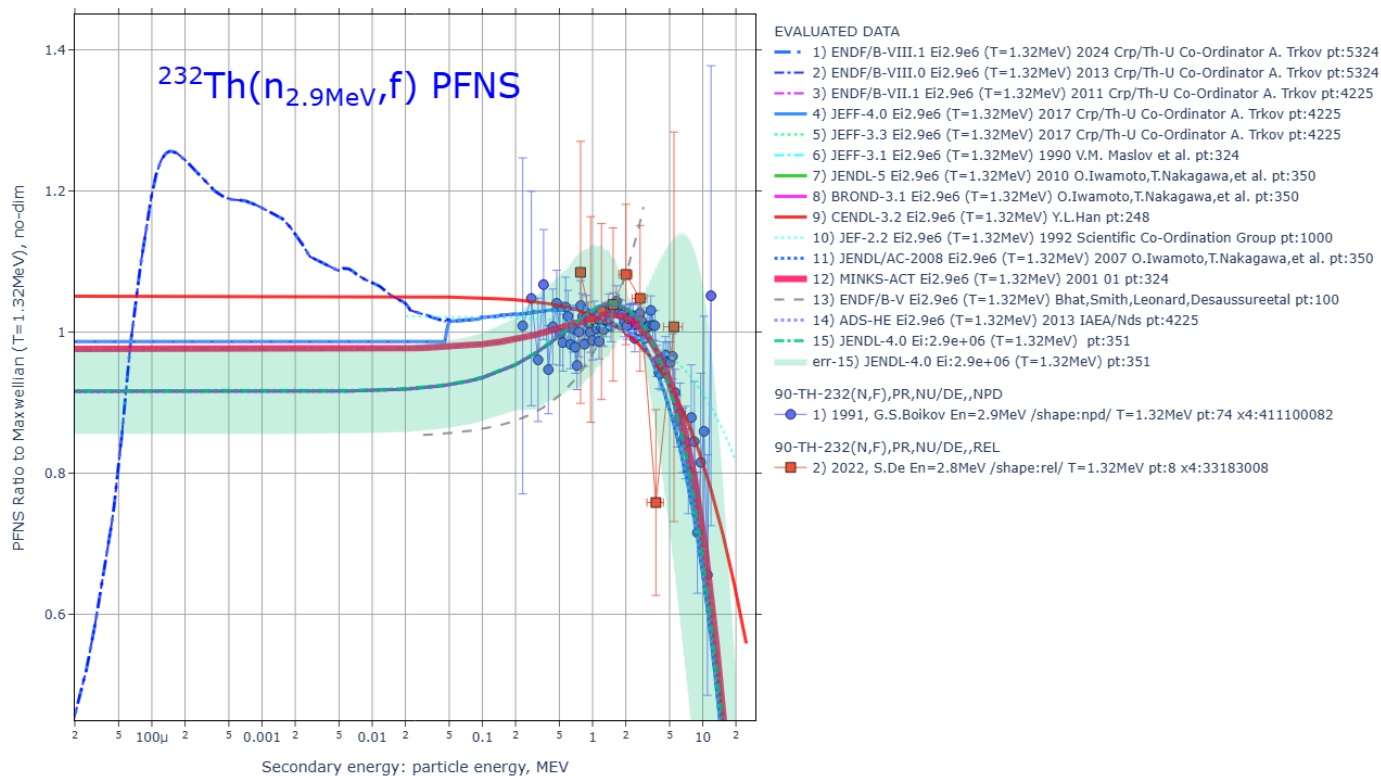
34) part2-A-eta/out00/U238nel-eta.html.png



part2-B-pfns/ PFNS - energy spectrum of prompt fission neutrons

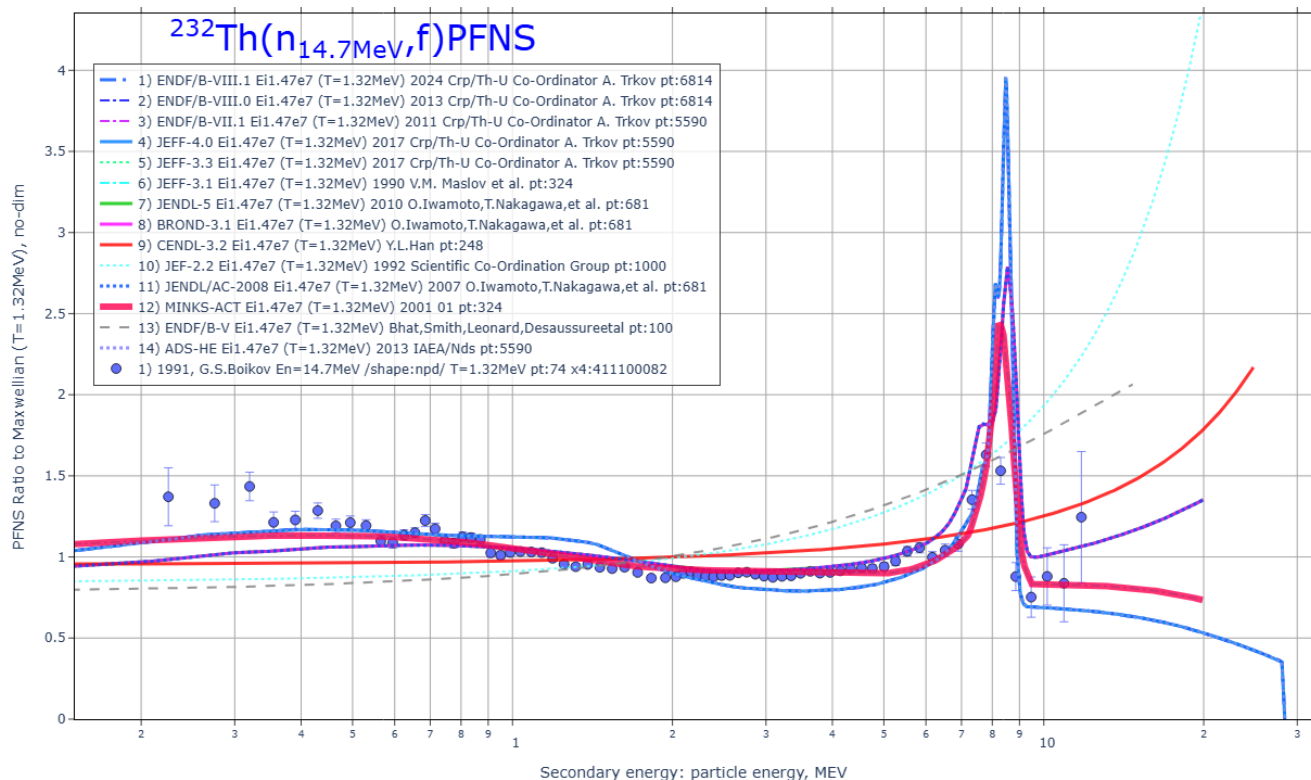
35) part2-B-pfns/out00/pfns232th02_9.html.png

Reaction:90-TH-232(N,F),PR,NU/DE,,NPD Quantity:MFQ:y=Data(En,E2) Datasets:2 Points:82
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-10-01 //running:2026-10-02 13:55:54

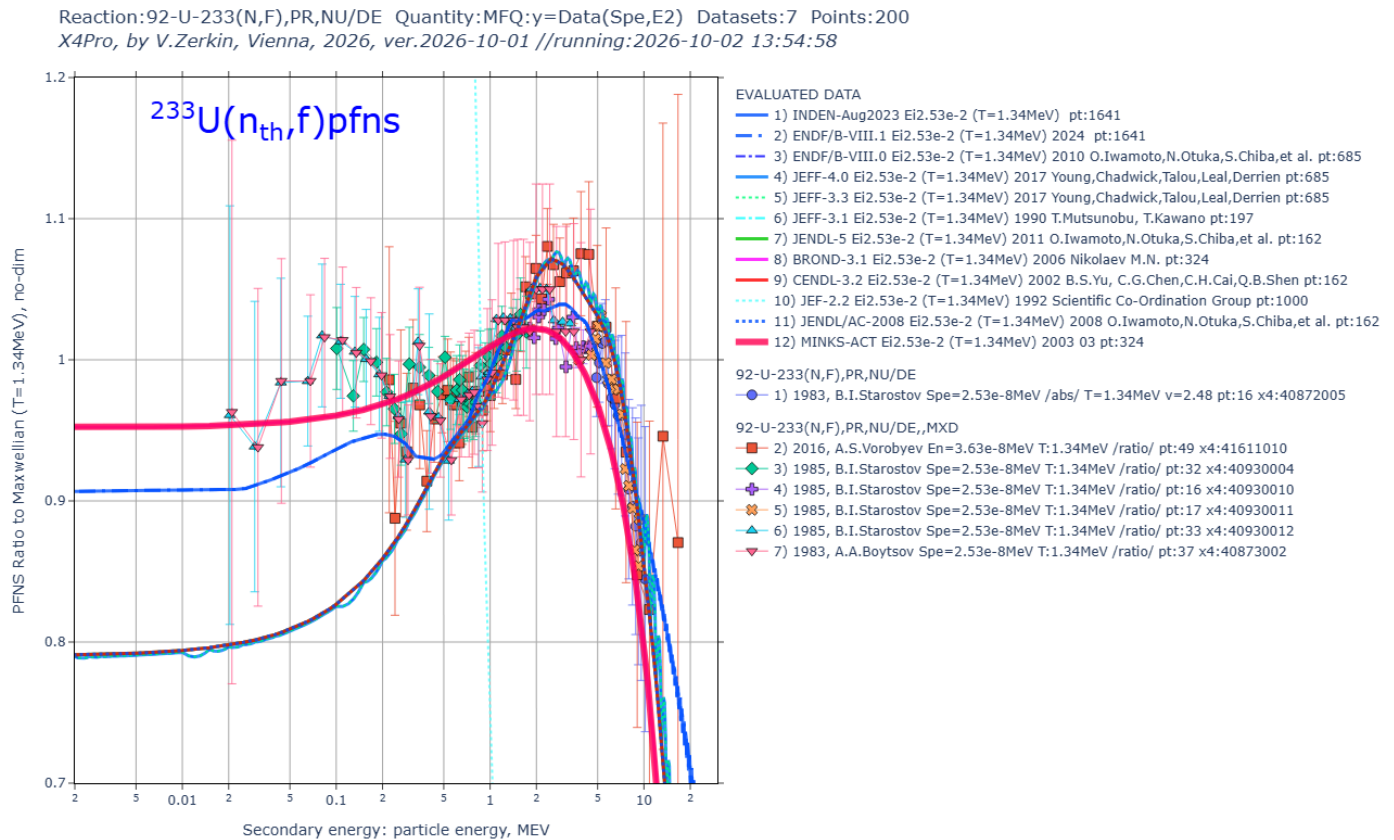


36) part2-B-pfns/out00/pfns232th14_7.html.png

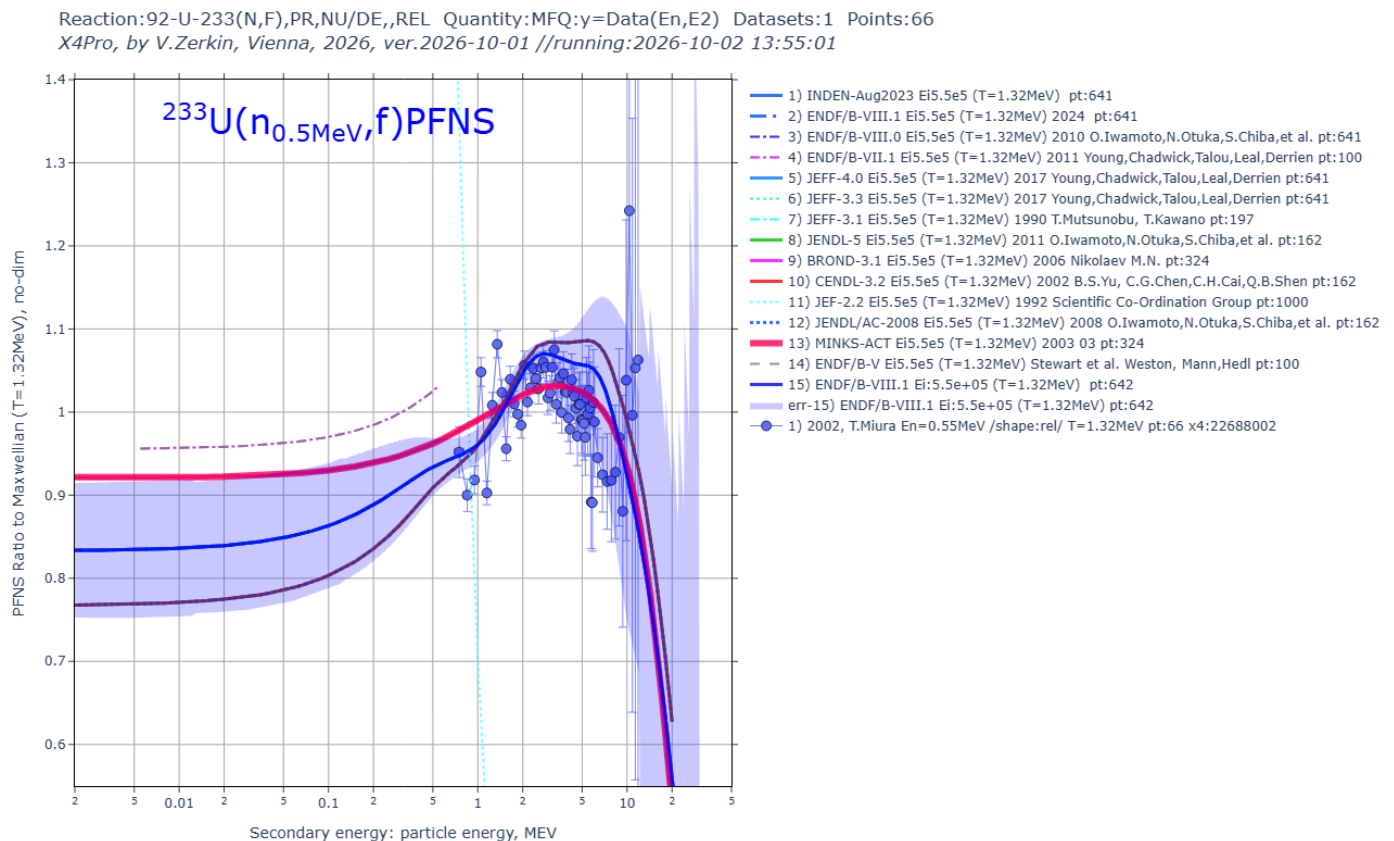
Reaction:90-TH-232(N,F),PR,NU/DE,,NPD Quantity:MFQ:y=Data(En,E2) Datasets:1 Points:74
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-10-01 //running:2026-10-02 13:55:58



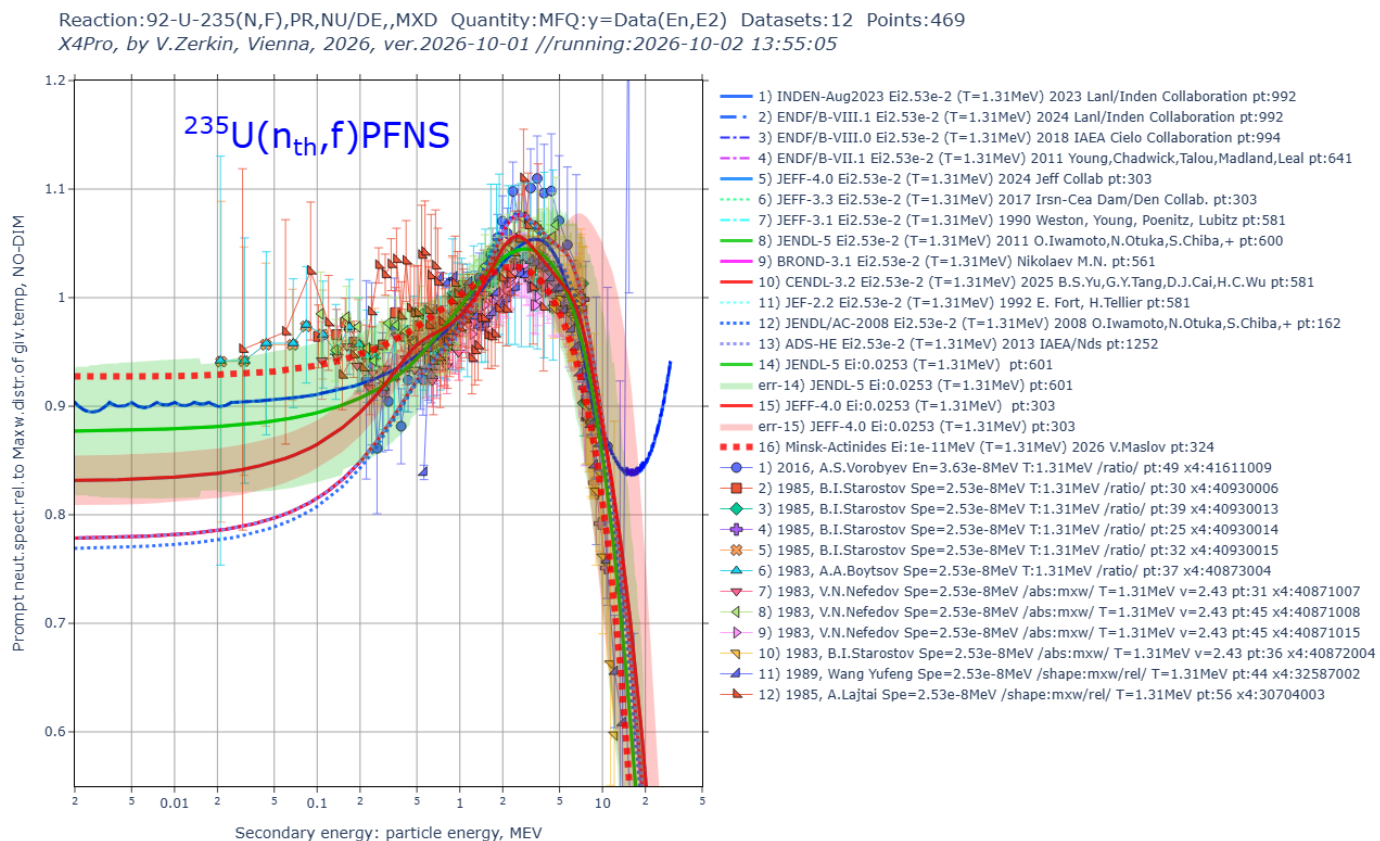
37) part2-B-pfns/out00/pfns233u00_0th.html.png



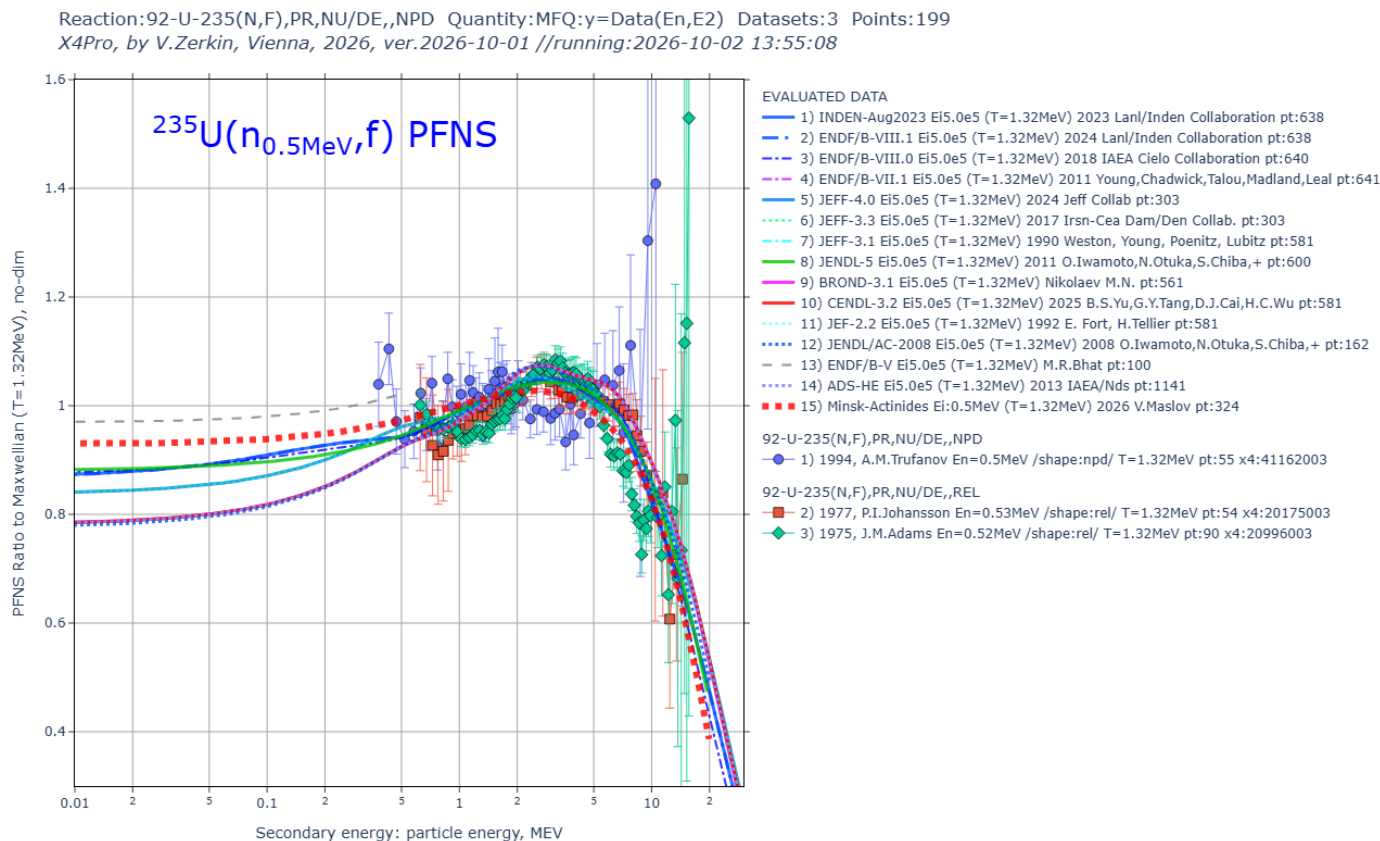
38) part2-B-pfns/out00/pfns233u00_5.html.png



39) part2-B-pfns/out00/pfns235u00_0th.html.png

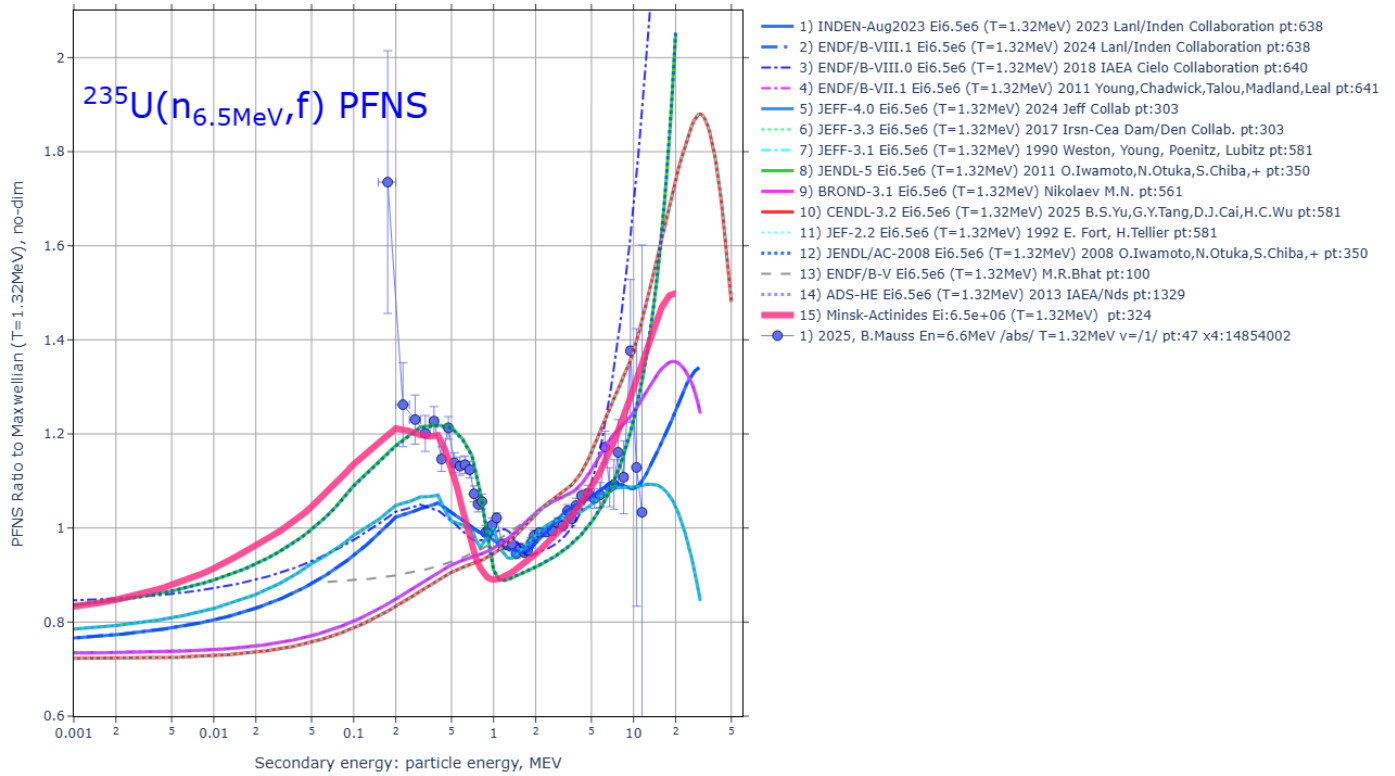


40) part2-B-pfns/out00/pfns235u00_5.html.png



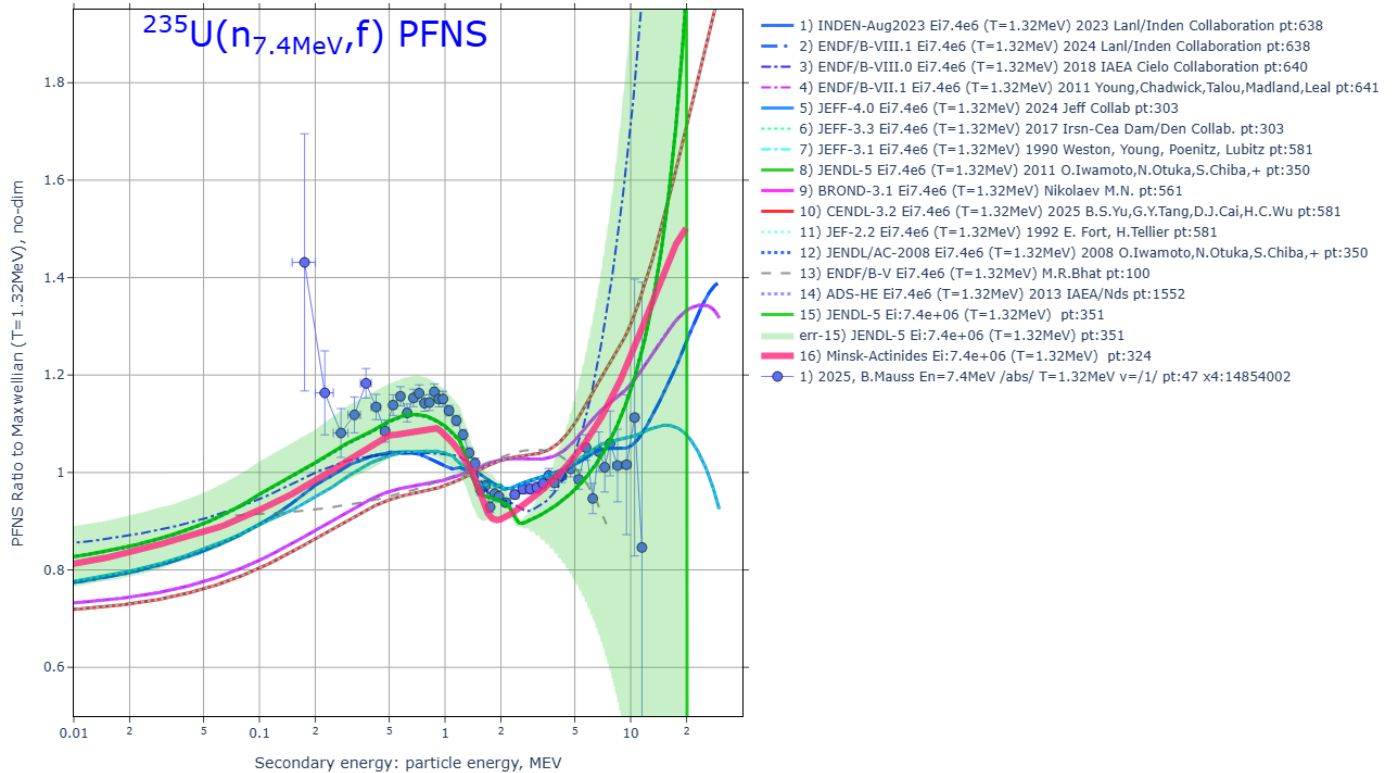
41) part2-B-pfns/out00/pfns235u06_6.html.png

Reaction: 92-U-235(N,F),PR,NU/DE Quantity: MFQ: y=Data(En,E2) Datasets: 1 Points: 47
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-10-01 //running: 2026-10-02 13:55:15



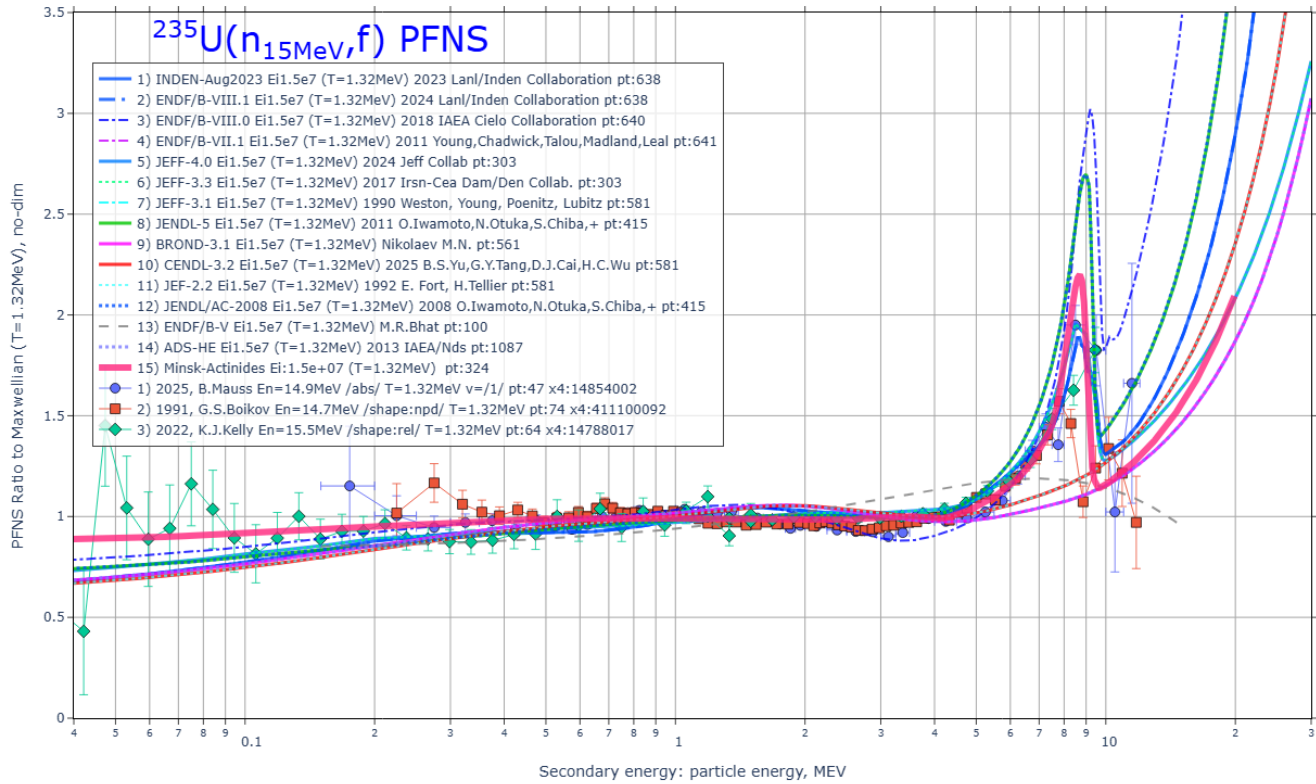
42) part2-B-pfns/out00/pfns235u07_4.html.png

Reaction: 92-U-235(N,F),PR,NU/DE Quantity: MFQ: y=Data(En,E2) Datasets: 1 Points: 47
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-10-01 //running: 2026-10-02 13:55:11



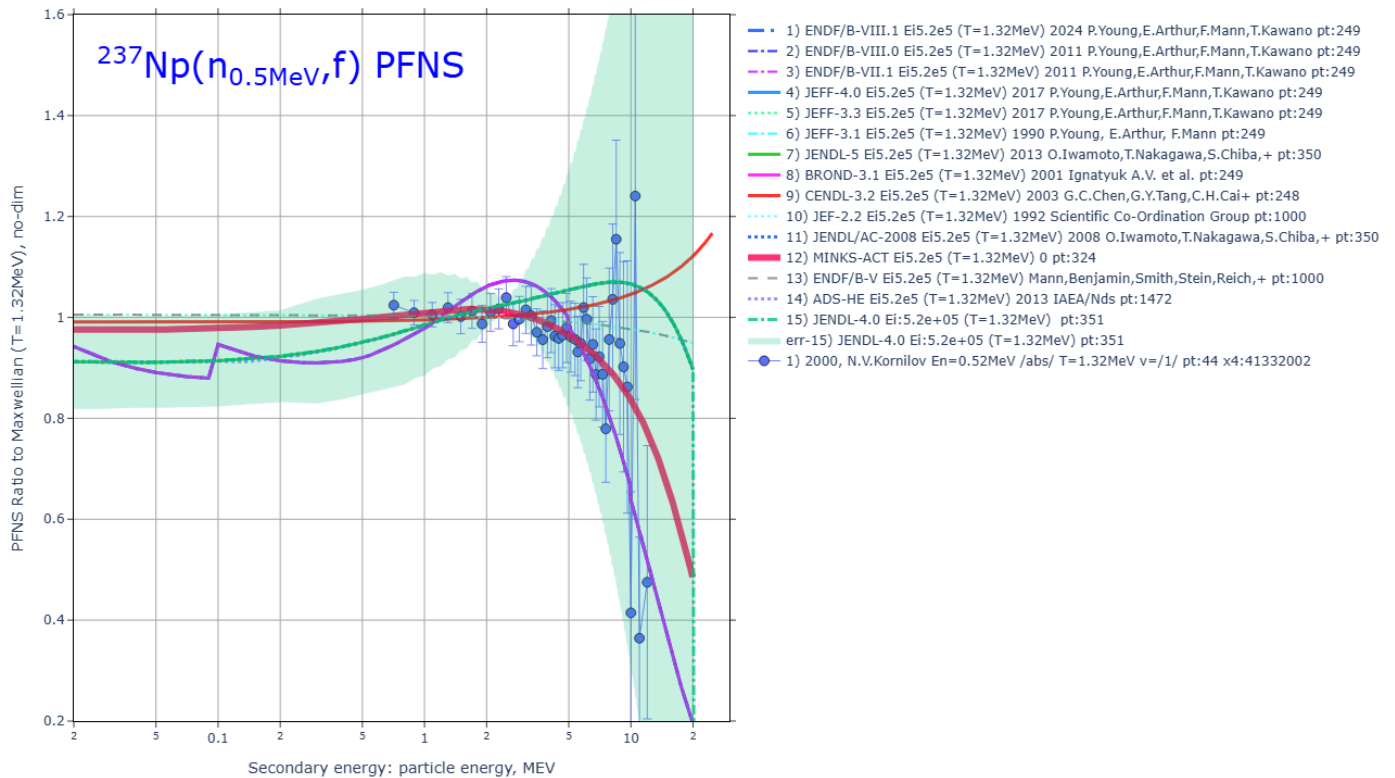
43) part2-B-pfns/out00/pfns235u15.html.png

Reaction: 92-U-235(N,F),PR,NU/DE Quantity: MFQ: y=Data(En,E2) Datasets: 3 Points: 185
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-10-01 //running: 2026-10-02 13:55:20



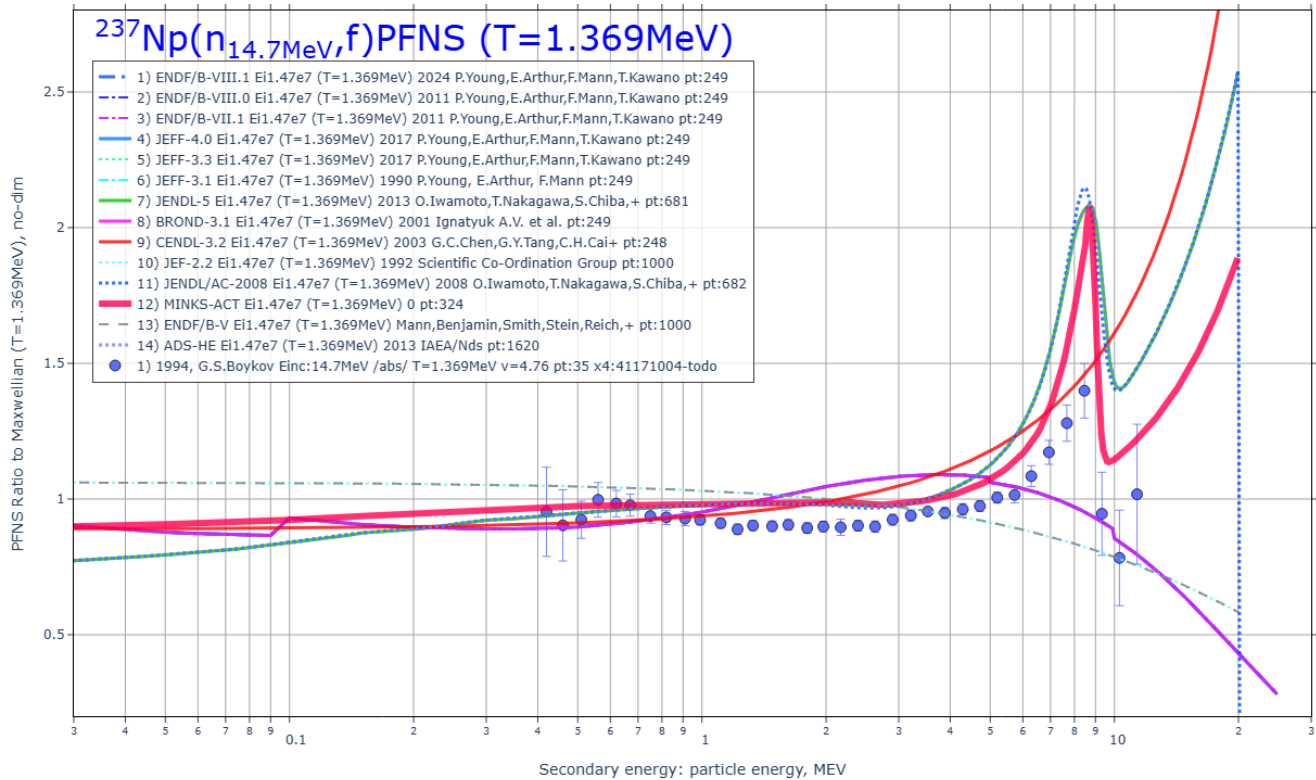
44) part2-B-pfns/out00/pfns237np00_5.html.png

Reaction: 93-NP-237(N,F),PR,NU/DE Quantity: MFQ: y=Data(En,E2) Datasets: 1 Points: 44
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-10-01 //running: 2026-10-02 13:56:01



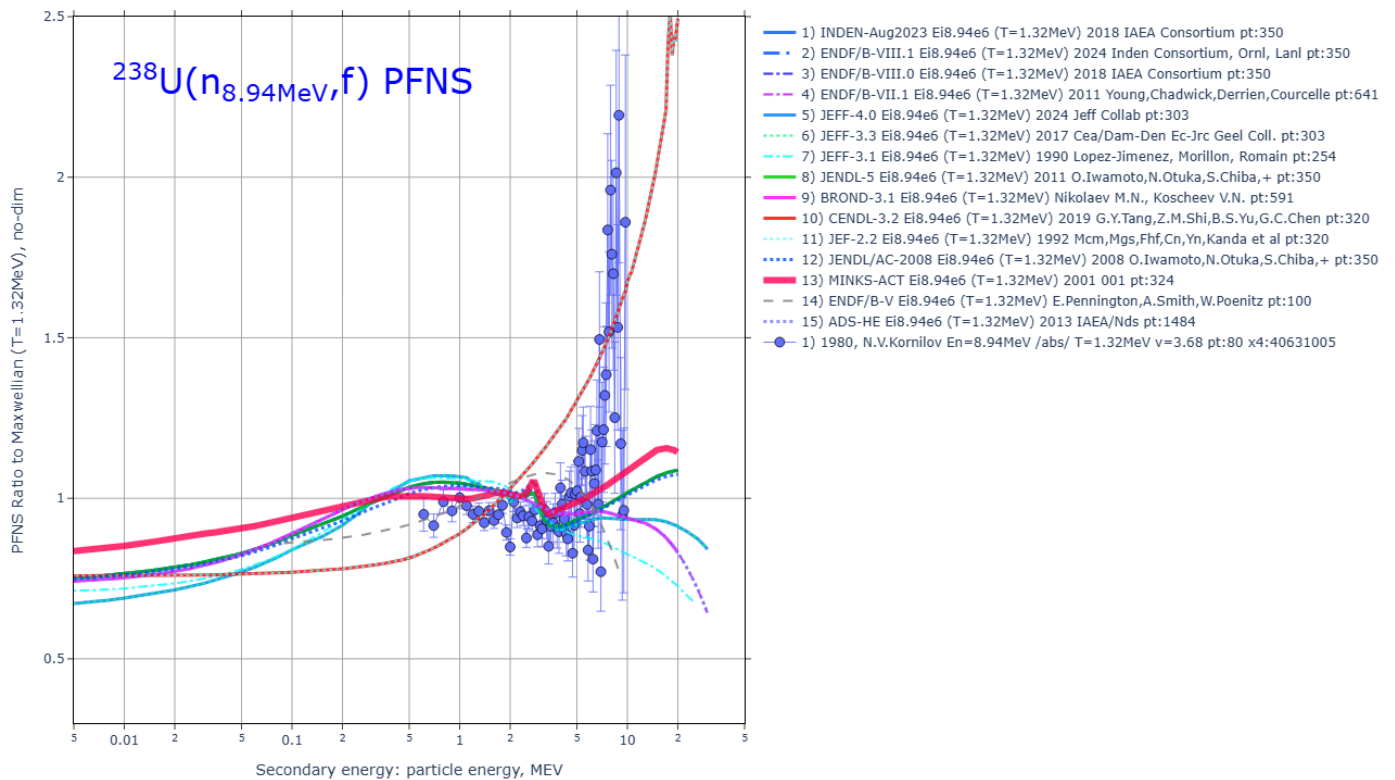
45) part2-B-pfns/out00/pfns237np14_7.html.png

Reaction: 93-NP-237(N,F),PR,NU/DE Quantity: MFQ: y=Data(En,E2) Datasets: 1/0 Points: 35/0
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-10-01 //running: 2026-10-02 13:56:04



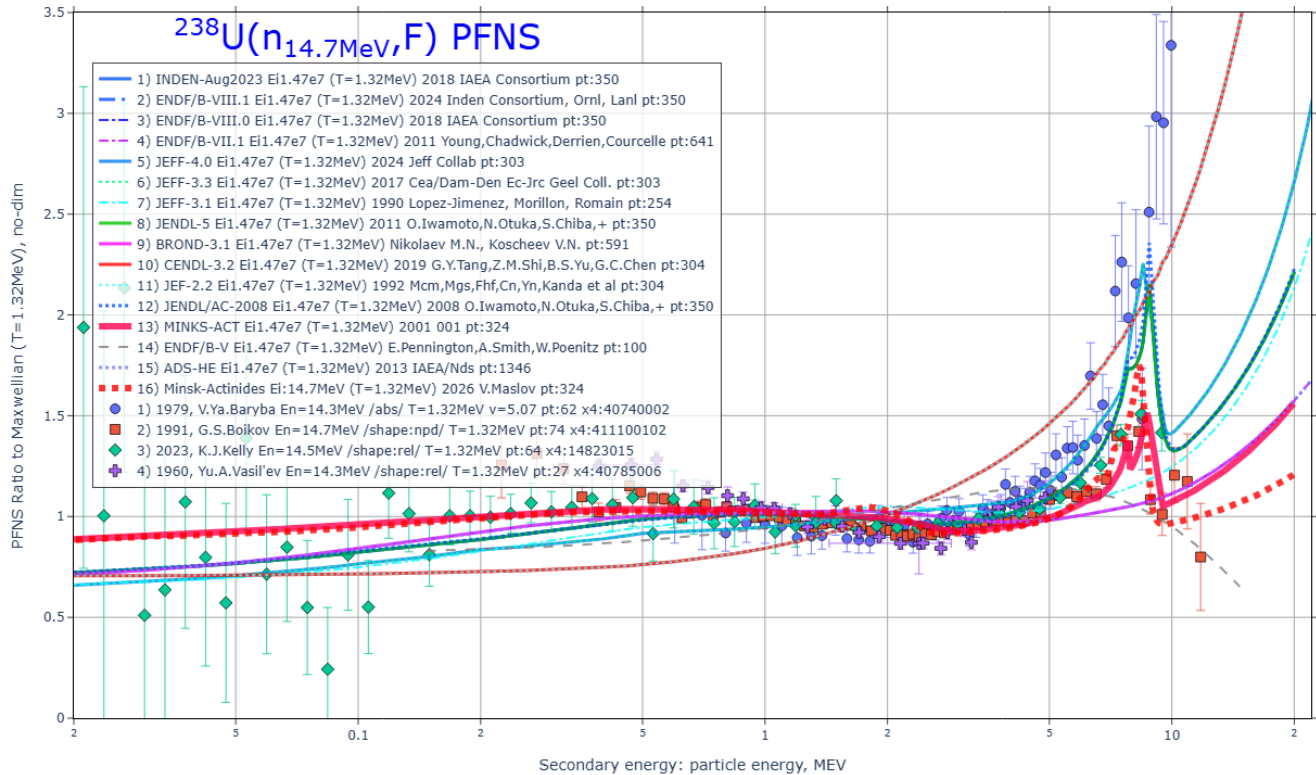
46) part2-B-pfns/out00/pfns238u08_94.html.png

Reaction: 92-U-238(N,F),PR,NU/DE Quantity: MFQ: y=Data(En,E2) Datasets: 1 Points: 80
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-10-01 //running: 2026-10-02 13:55:24



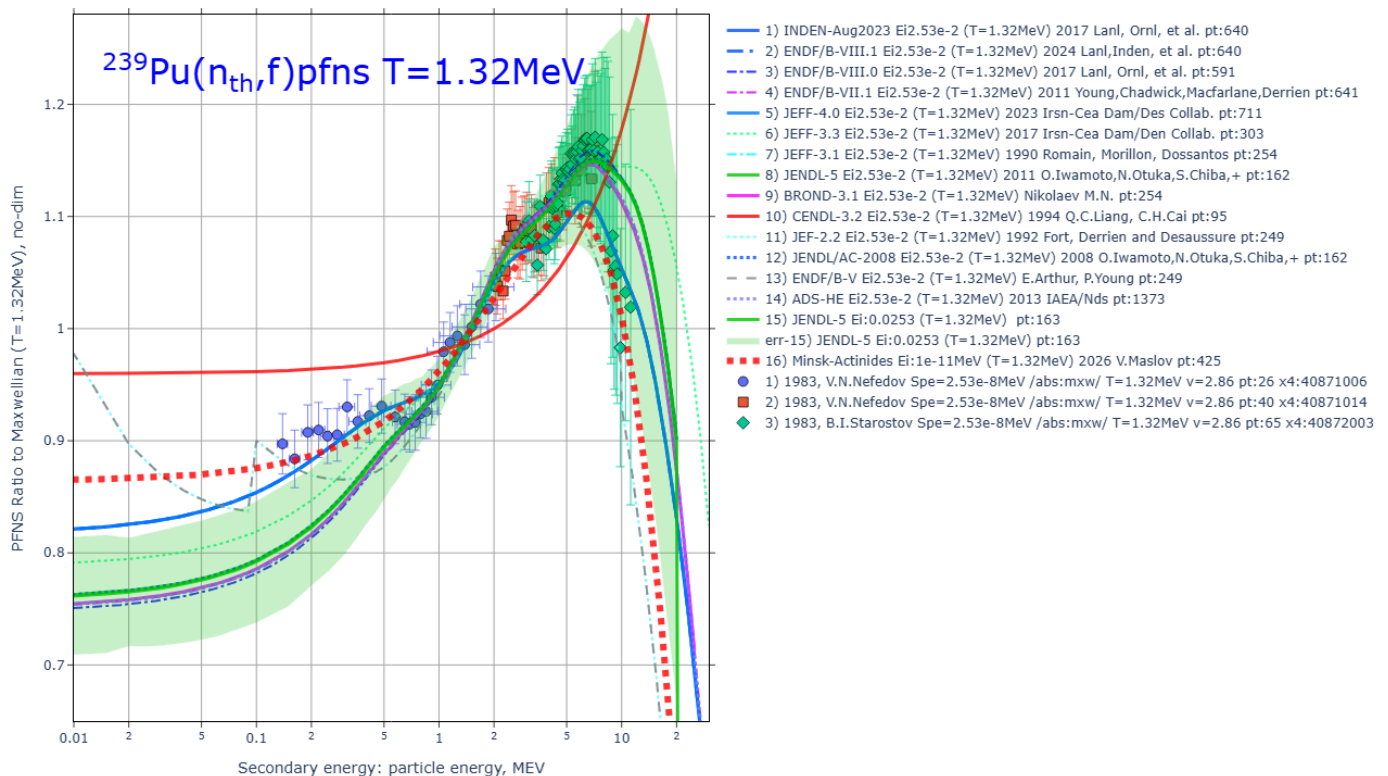
47) part2-B-pfns/out00/pfns238u14_7.html.png

Reaction: 92-U-238(N,F),PR,NU/DE Quantity: MFQ:y=Data(En,E2) Datasets: 4 Points: 227
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-10-01 //running:2026-10-02 13:55:27



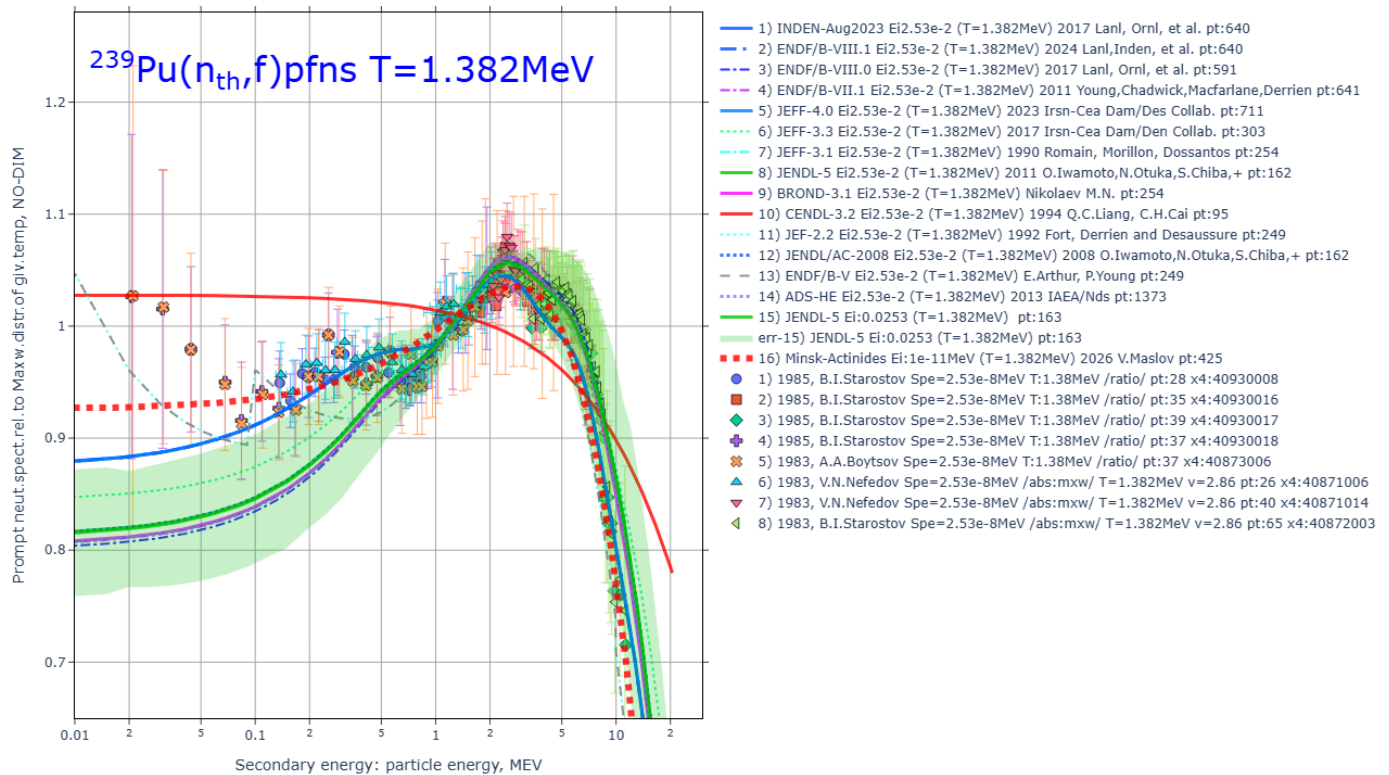
48) part2-B-pfns/out00/pfns239pu00_0th.html.png

Reaction: 94-PU-239(N,F),PR,NU/DE,,MXW Quantity: MFQ:y=Data(Spe,E2) Datasets: 3 Points: 131
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-10-01 //running:2026-10-02 13:55:30



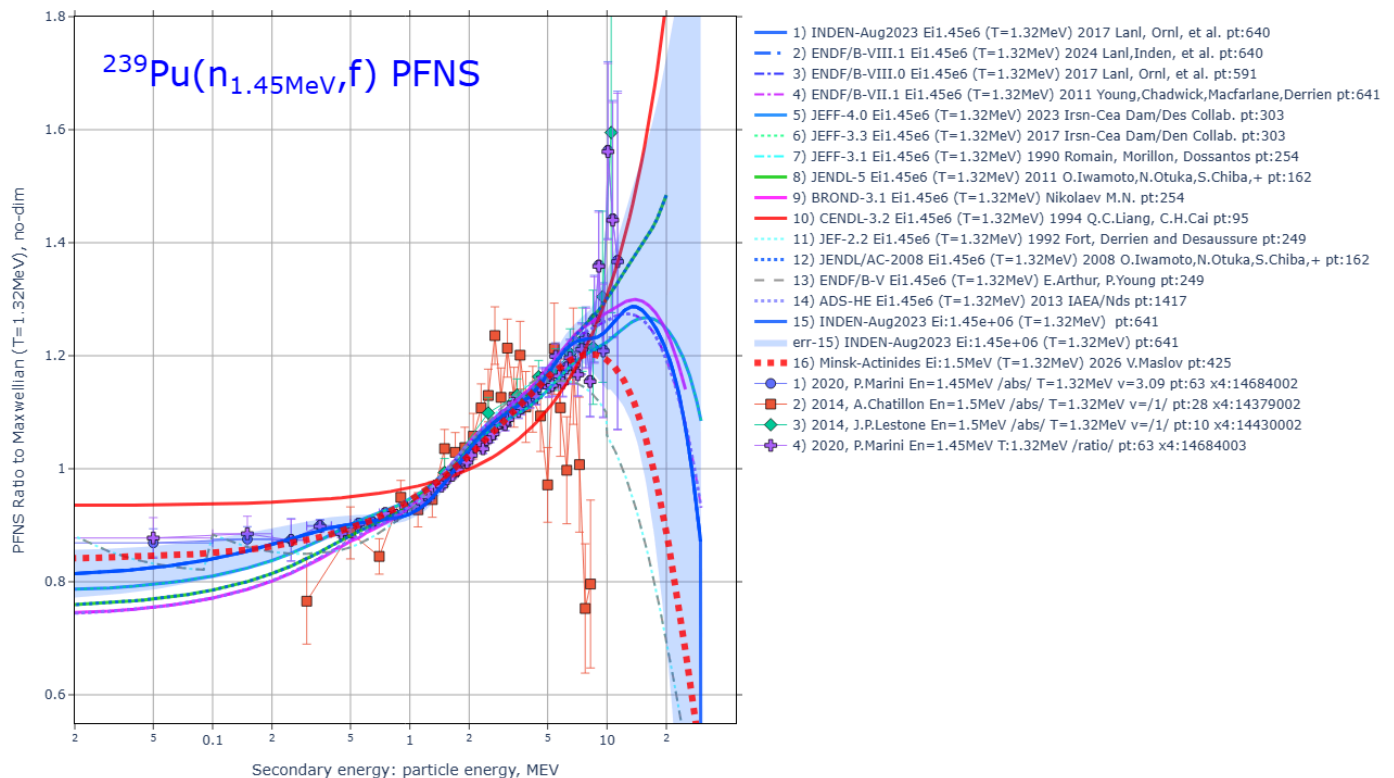
49) part2-B-pfns/out00/pfns239pu00_0th138.html.png

Reaction:94-PU-239(N,F),PR,NU/DE,,MXD Quantity:MFQ:y=Data(Spe,E2) Datasets:8 Points:307
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-10-01 //running:2026-10-02 13:55:34



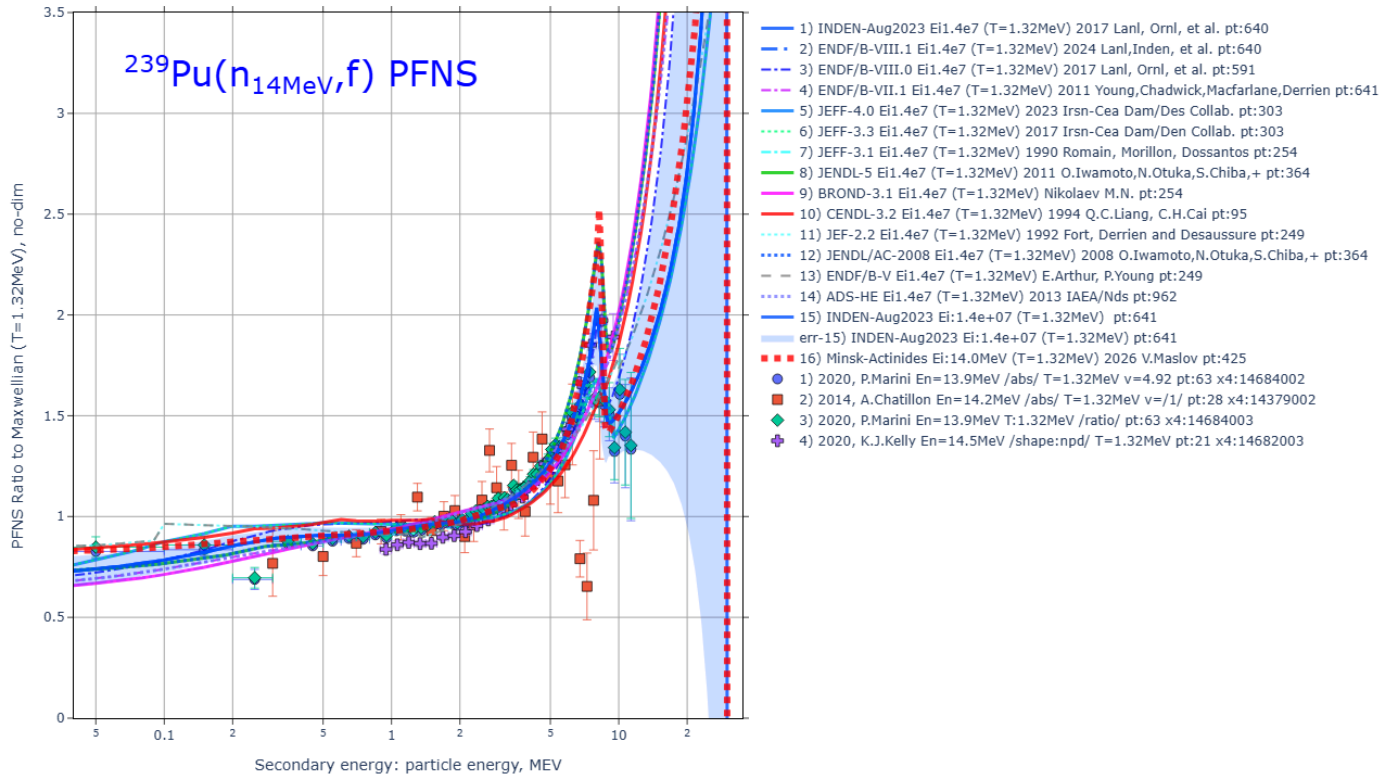
50) part2-B-pfns/out00/pfns239pu01_45.html.png

Reaction:94-PU-239(N,F),PR,NU/DE Quantity:MFQ:y=Data(En,E2) Datasets:4 Points:164/165
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-10-01 //running:2026-10-02 13:55:37



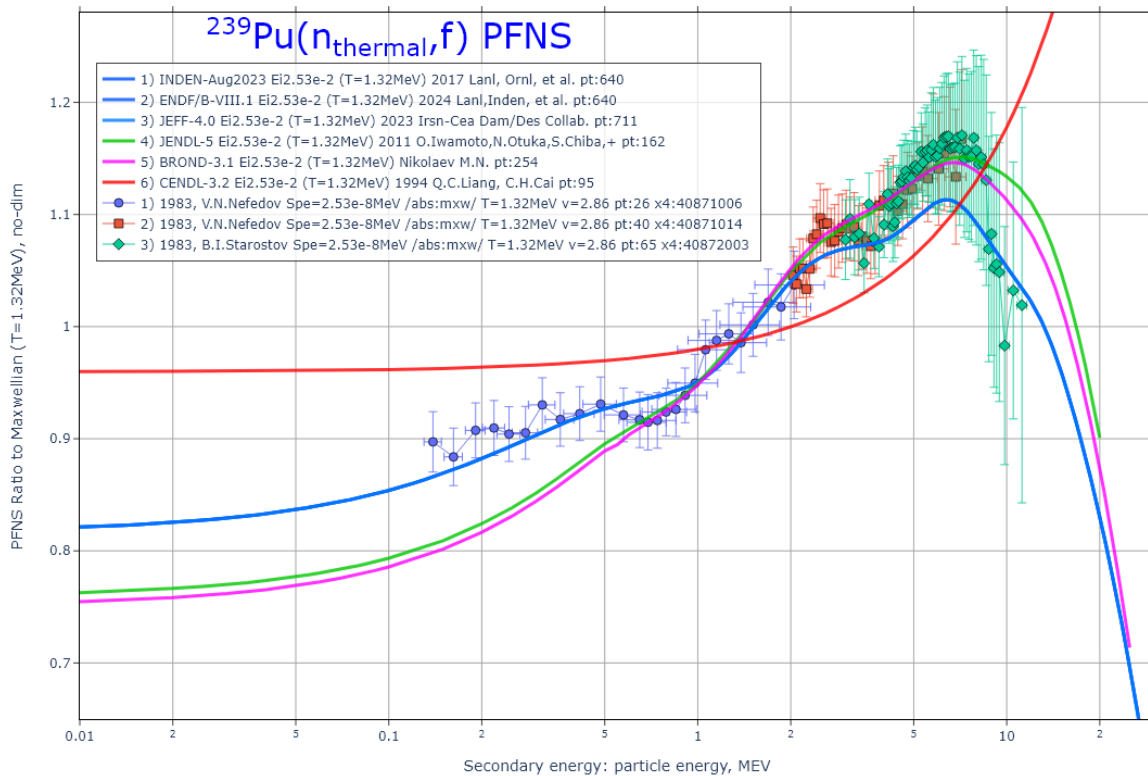
51) part2-B-pfns/out00/pfns239pu14.html.png

Reaction: 94-PU-239(N,F),PR,NU/DE Quantity: MFQ: y=Data(En,E2) Datasets: 4/5 Points: 175/215
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-10-01 //running: 2026-10-02 13:55:45



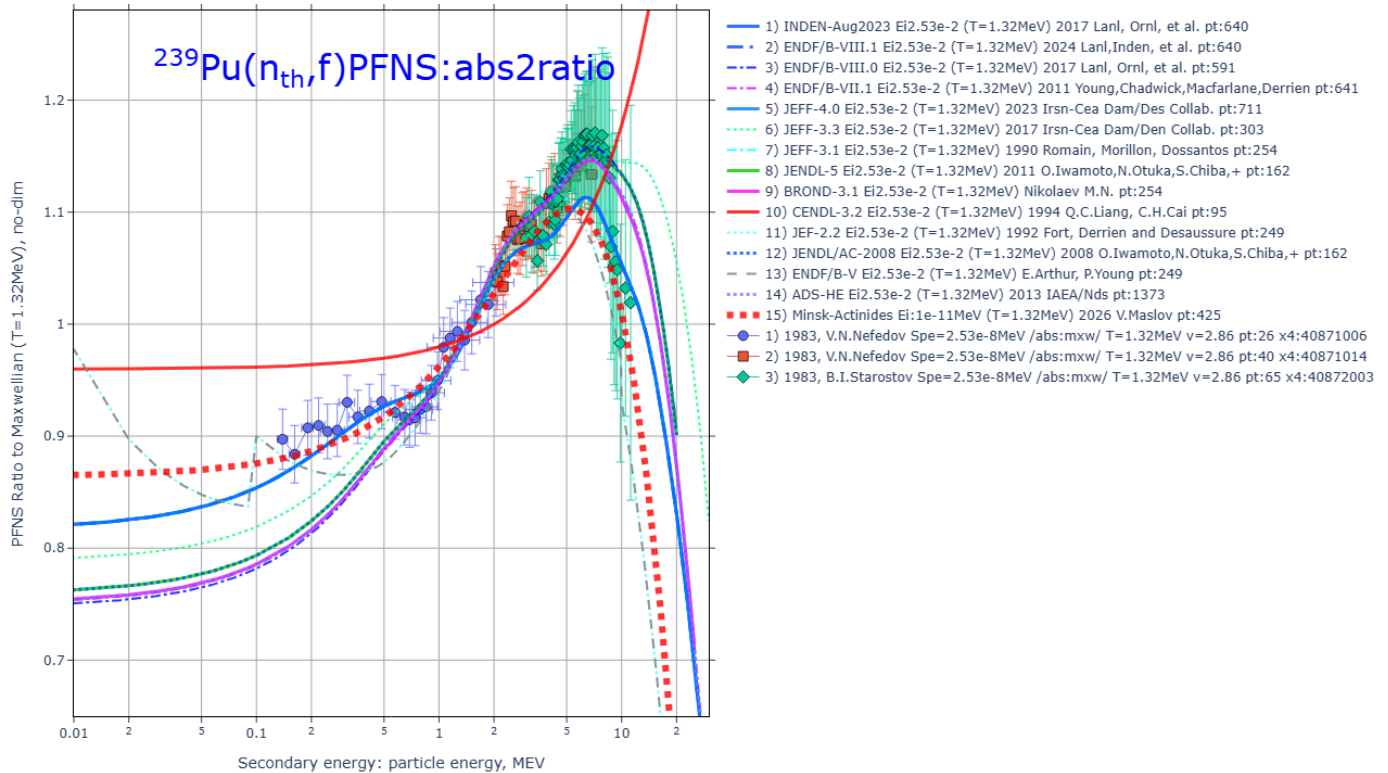
52) part2-B-pfns/out00/pfns239pu_t2.html.png

Reaction: 94-PU-239(N,F),PR,NU/DE,,MXW Quantity: MFQ: y=Data(Spe,E2) Datasets: 3 Points: 131
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-20 //running: 2026-09-22 19:02:32



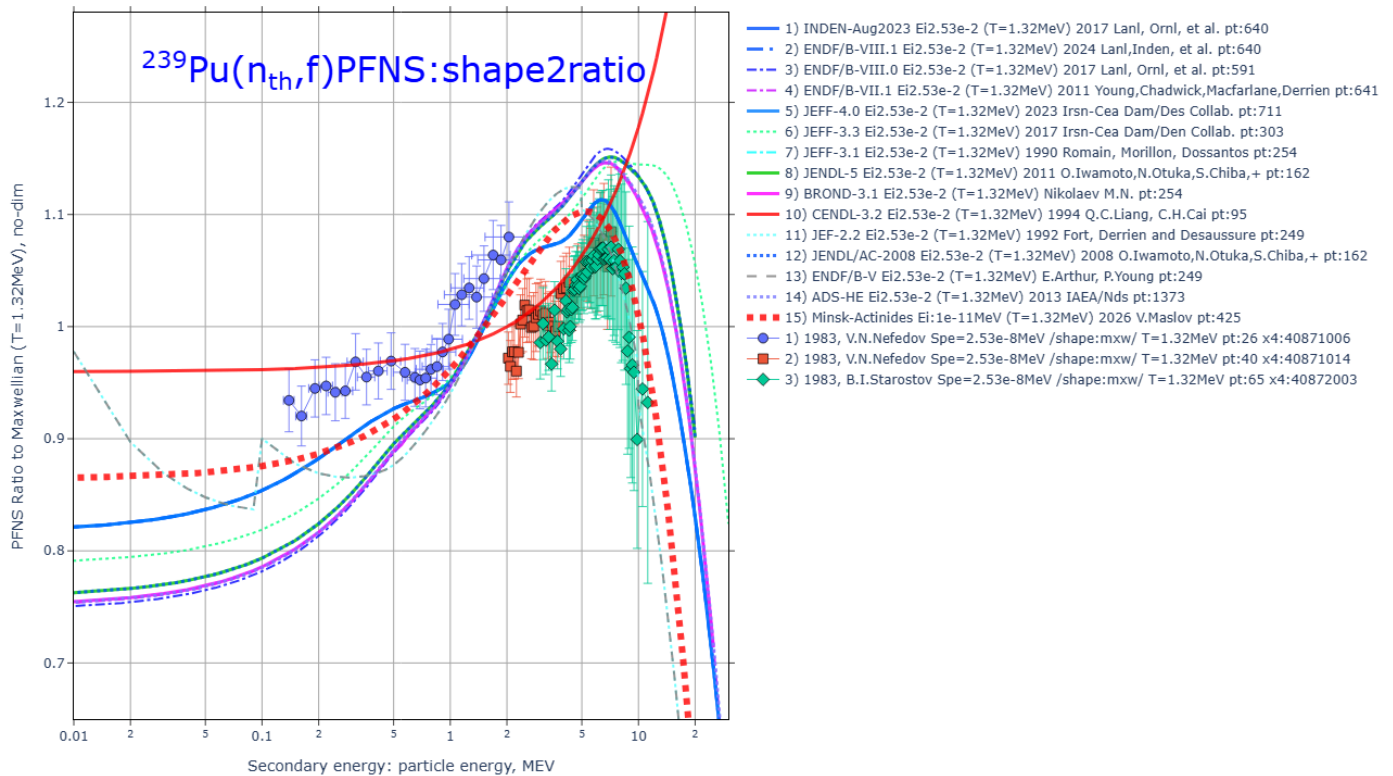
53) part2-B-pfns/out00/pfns239pu_t_abs.html.png

Reaction:94-PU-239(N,F),PR,NU/DE,,MXW Quantity:MFQ:y=Data(Spe,E2) Datasets:3 Points:131
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-10-01 //running:2026-10-02 13:56:16



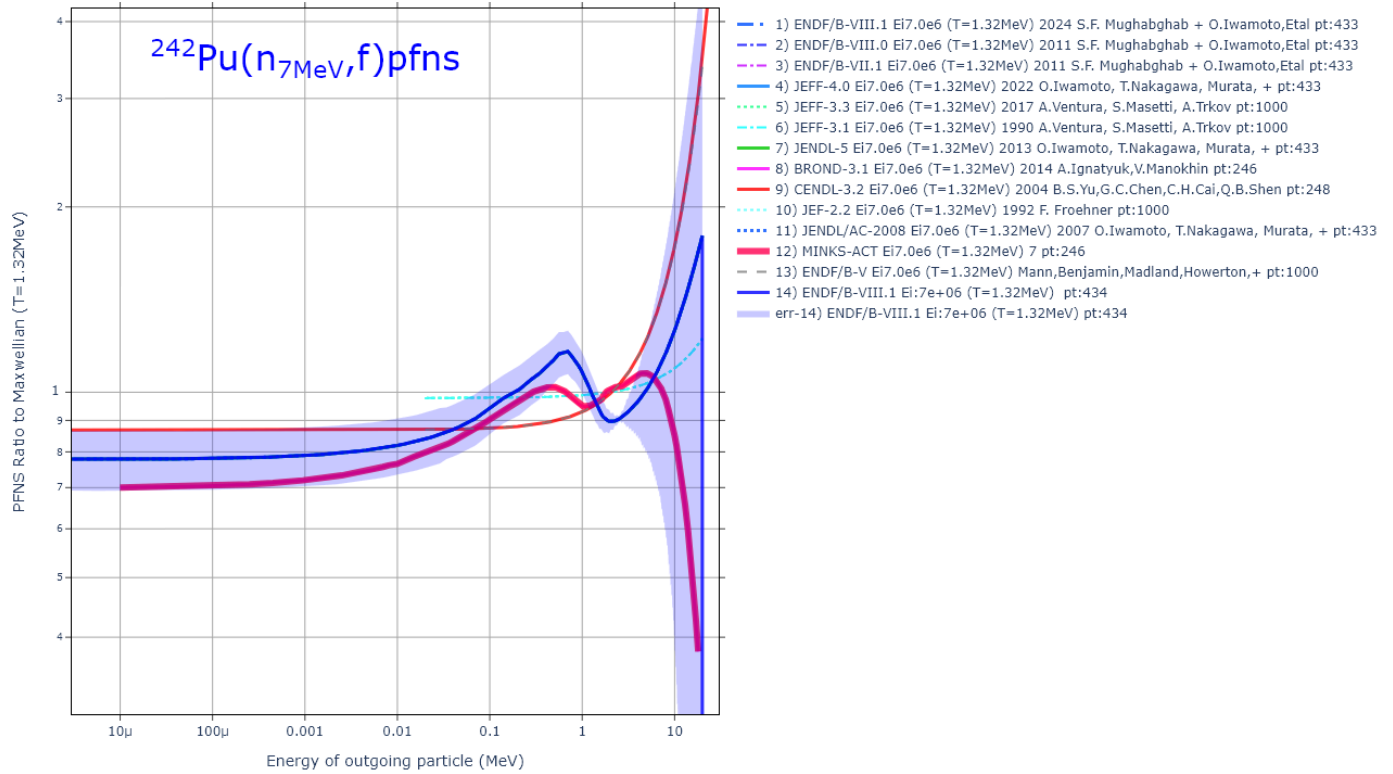
54) part2-B-pfns/out00/pfns239pu_t_shp.html.png

Reaction:94-PU-239(N,F),PR,NU/DE,,MXW Quantity:MFQ:y=Data(Spe,E2) Datasets:3 Points:131
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-10-01 //running:2026-10-02 13:56:20



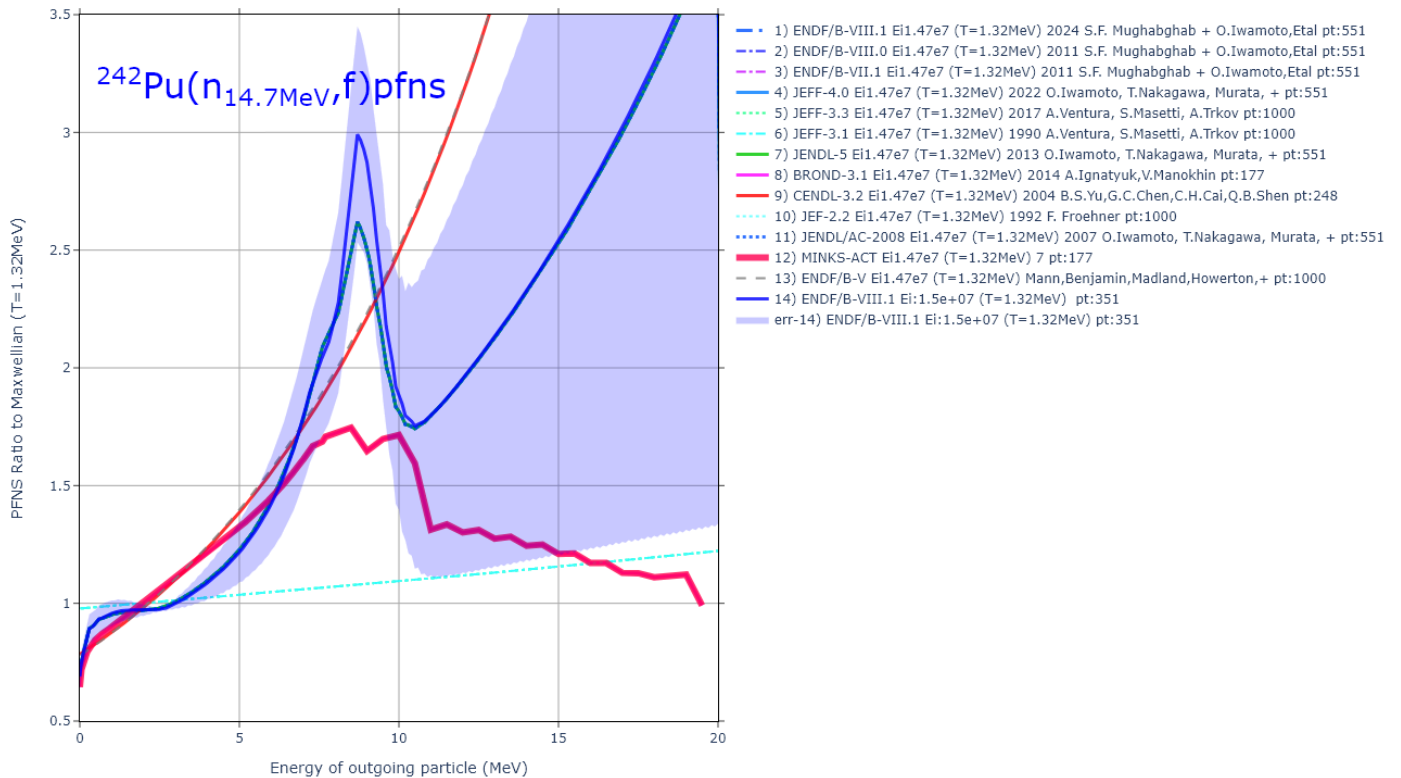
55) part2-B-pfns/out00/pfns242pu07.html.png

Reaction: Pu-242(n,f),pfns ENDF: MF=5, MT=18, Einc=7MeV, T=1.32MeV
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-10-01 //running:2026-10-02 17:04:34

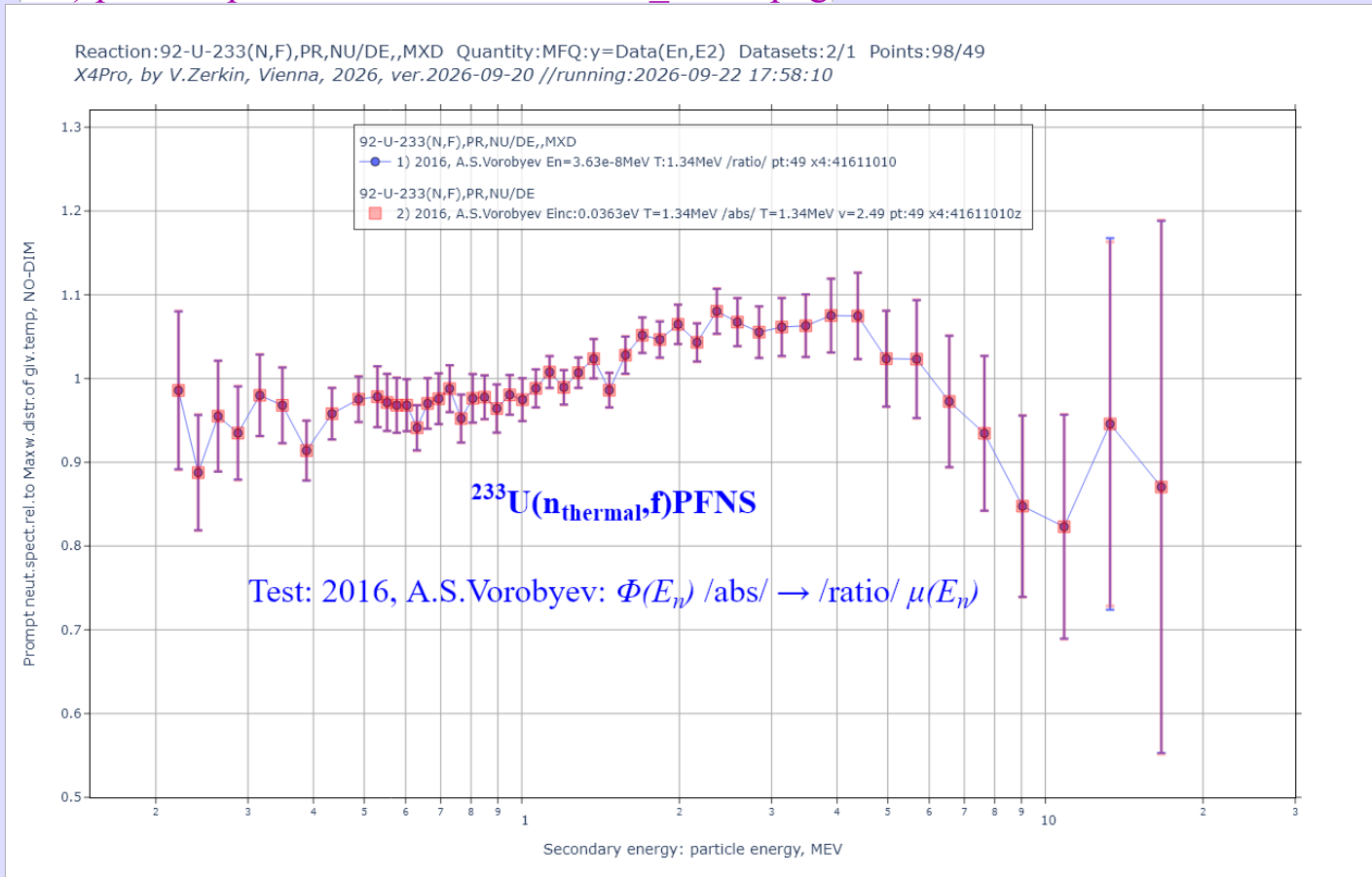


56) part2-B-pfns/out00/pfns242pu14_7.html.png

Reaction: Pu-242(n,f),pfns ENDF: MF=5, MT=18, Einc=14.7MeV, T=1.32MeV
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-10-01 //running:2026-10-02 17:04:36

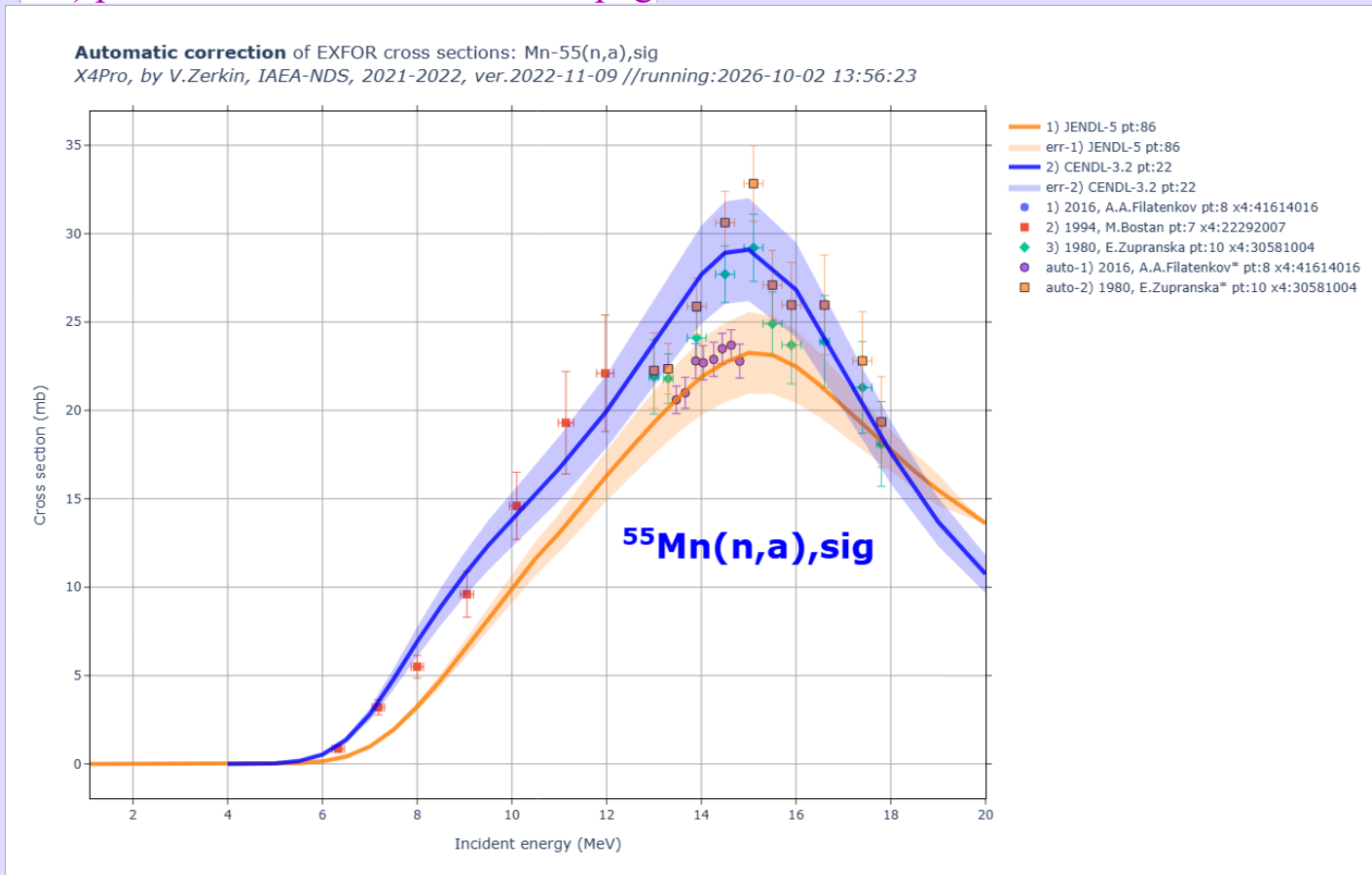


57) part2-B-pfns/out00/zabs2mxw233u_t.html.png



part3-1-auto1/ Automatically renormalize EXFOR cross sections using old monitor and new standards data

58) part3-1-auto1/out00/auto1.html.png

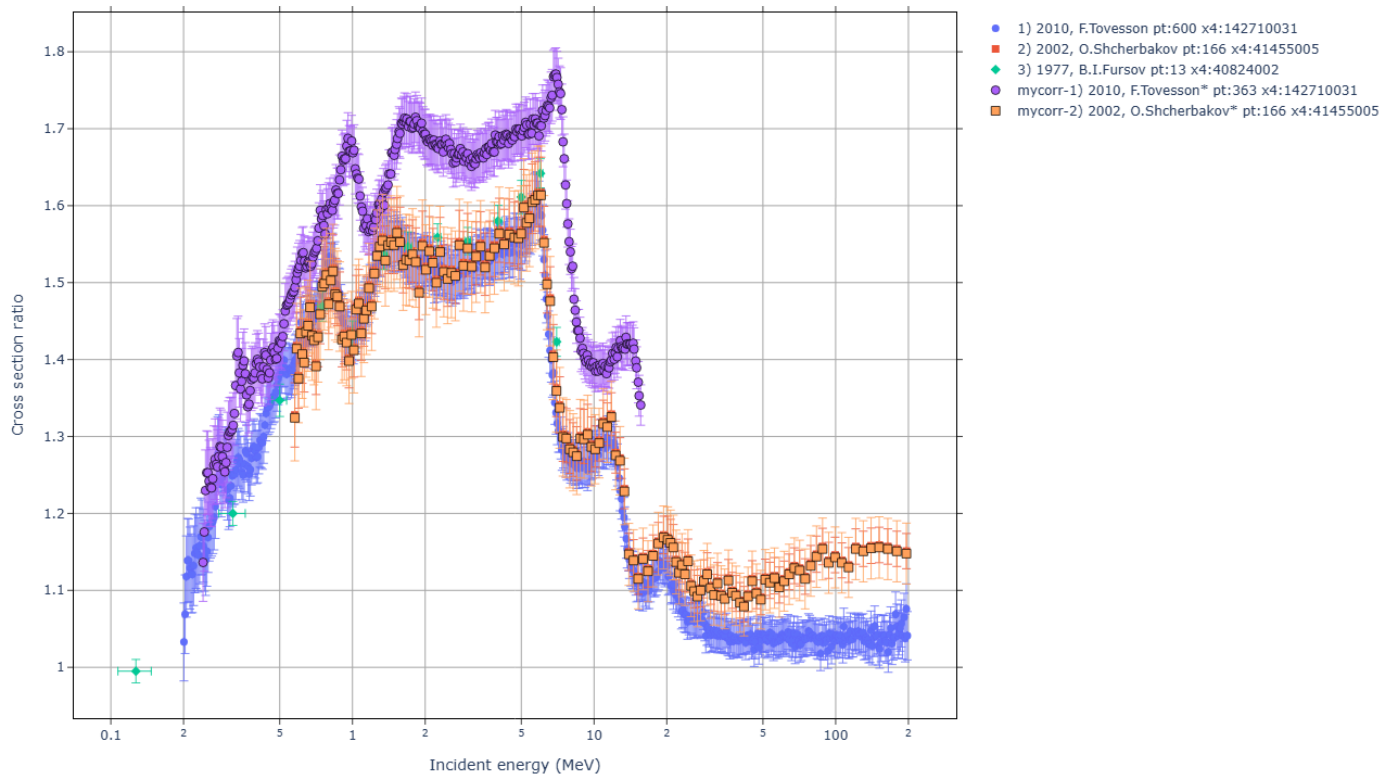


part3-2-user1/ *Correct data from EXFOR by user's program*

59) part3-2-user1/out00/user1.html.png

Local user's corrections of EXFOR cross sections ratios: Pu-239/U-235(n,f)CS

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-10-02 13:56:25

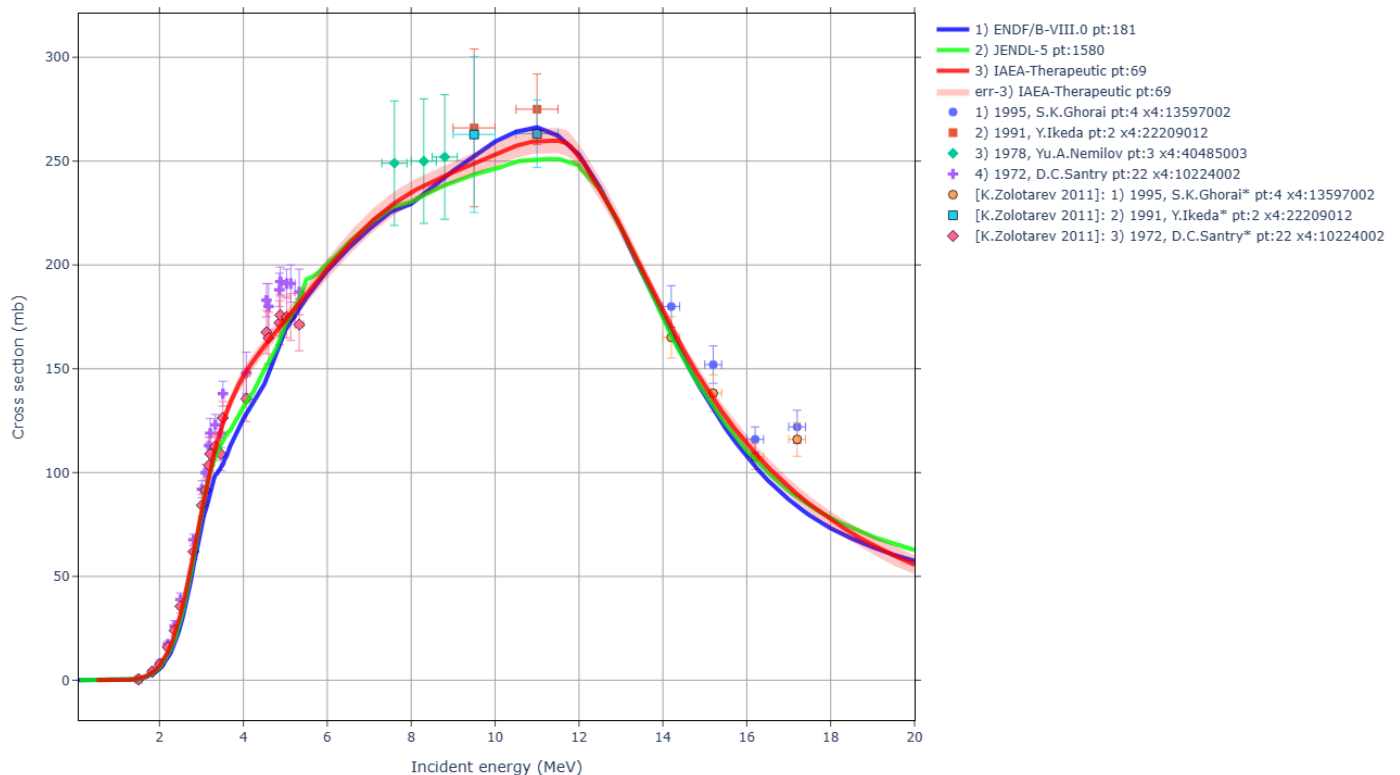


part3-3-expert1/ *Apply experts corrections stored in X4Pro database*

60) part3-3-expert1/out00/expert1.html.png

Apply experts corrections from database to EXFOR data: Zn-64(n,p),sig

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-10-02 13:56:28

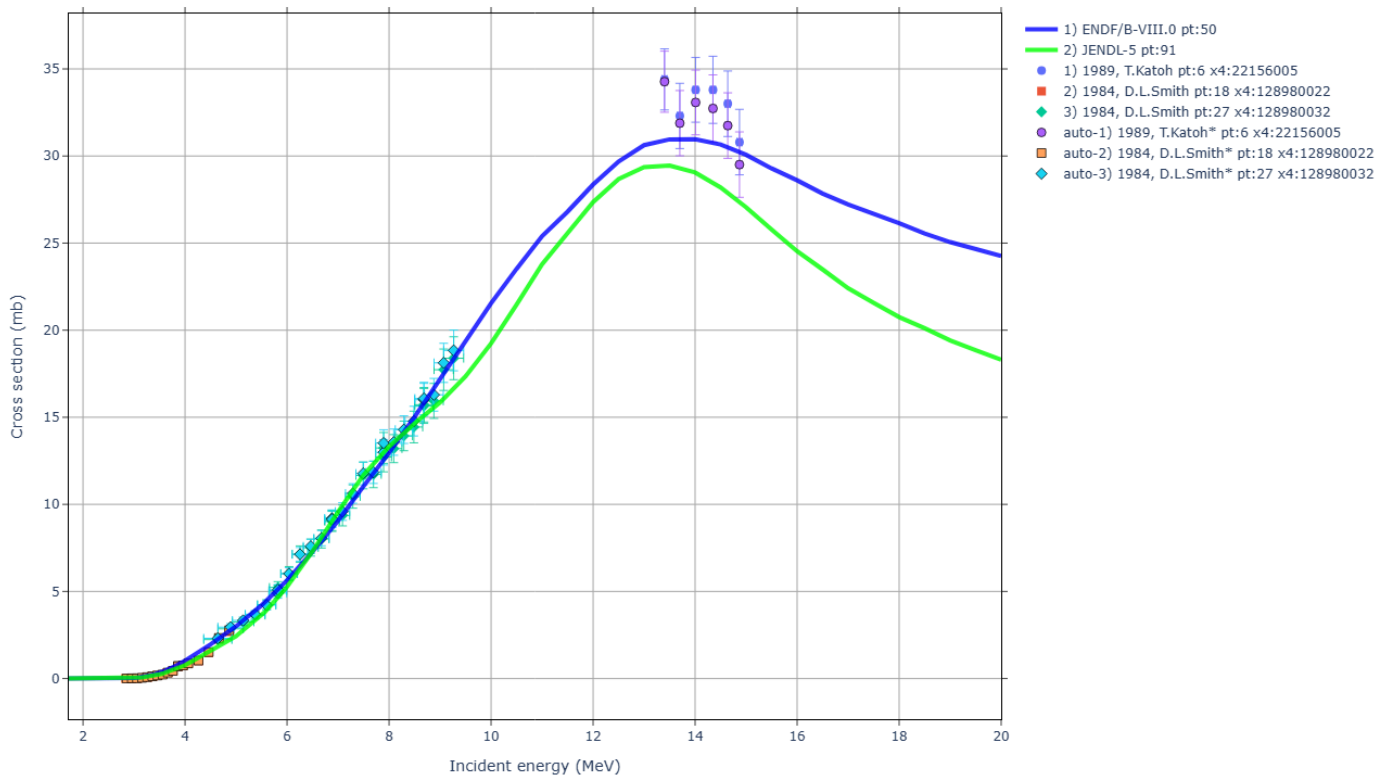


part3-4-auto2/ Automatically renormalize EXFOR cross sections using old monitor and new standards, new decay data

61) part3-4-auto2/out00/auto2.html.png

Automatic correction of EXFOR cross sections: V-51(n,p)CS

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-10-02 13:56:32

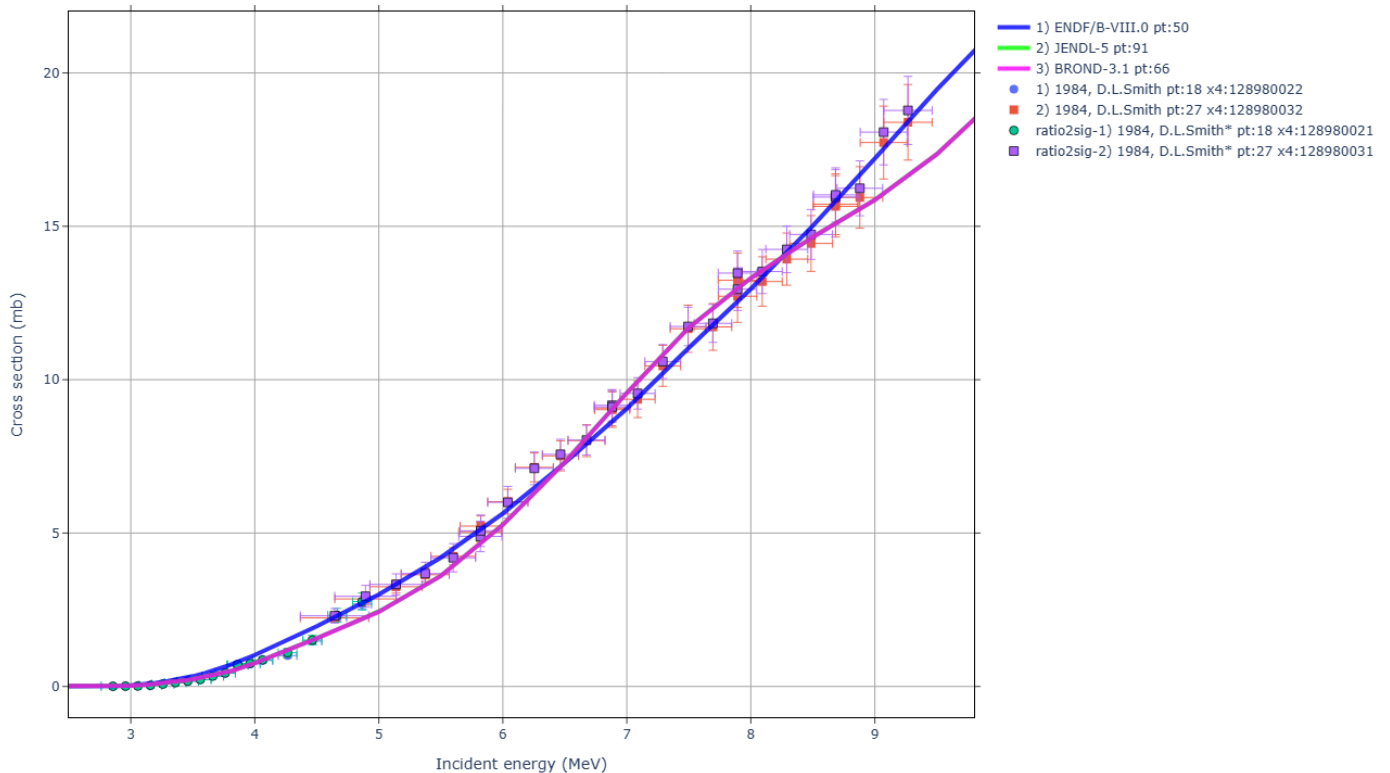


part3-5-ratio2sig/ Retrieve ratios, calculate cross sections and uncertainties using latest standards

62) part3-5-ratio2sig/out00/ratio2cs.html.png

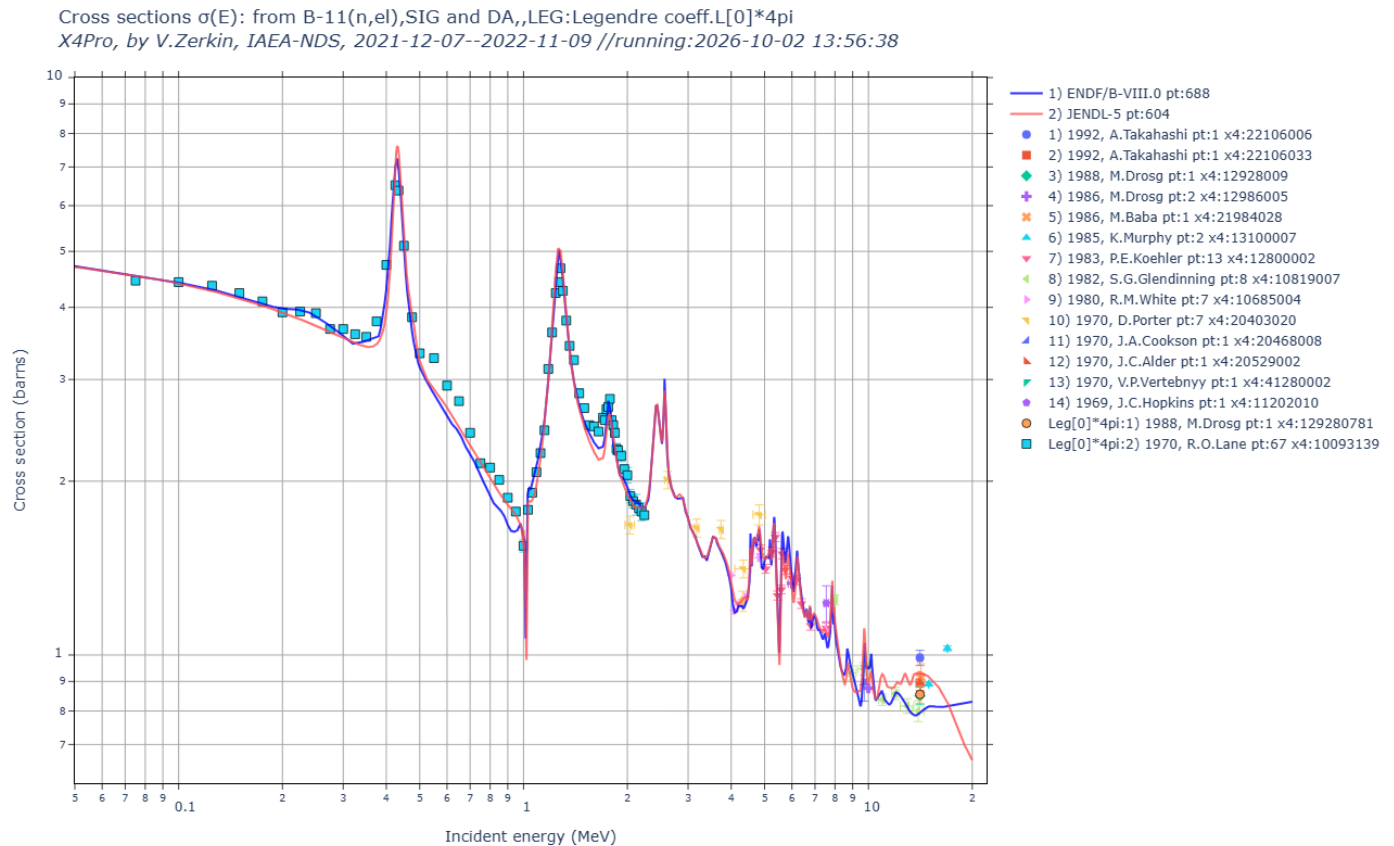
Ratio to cross section of EXFOR data: V-51(n,p)CS

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-10-02 13:56:35



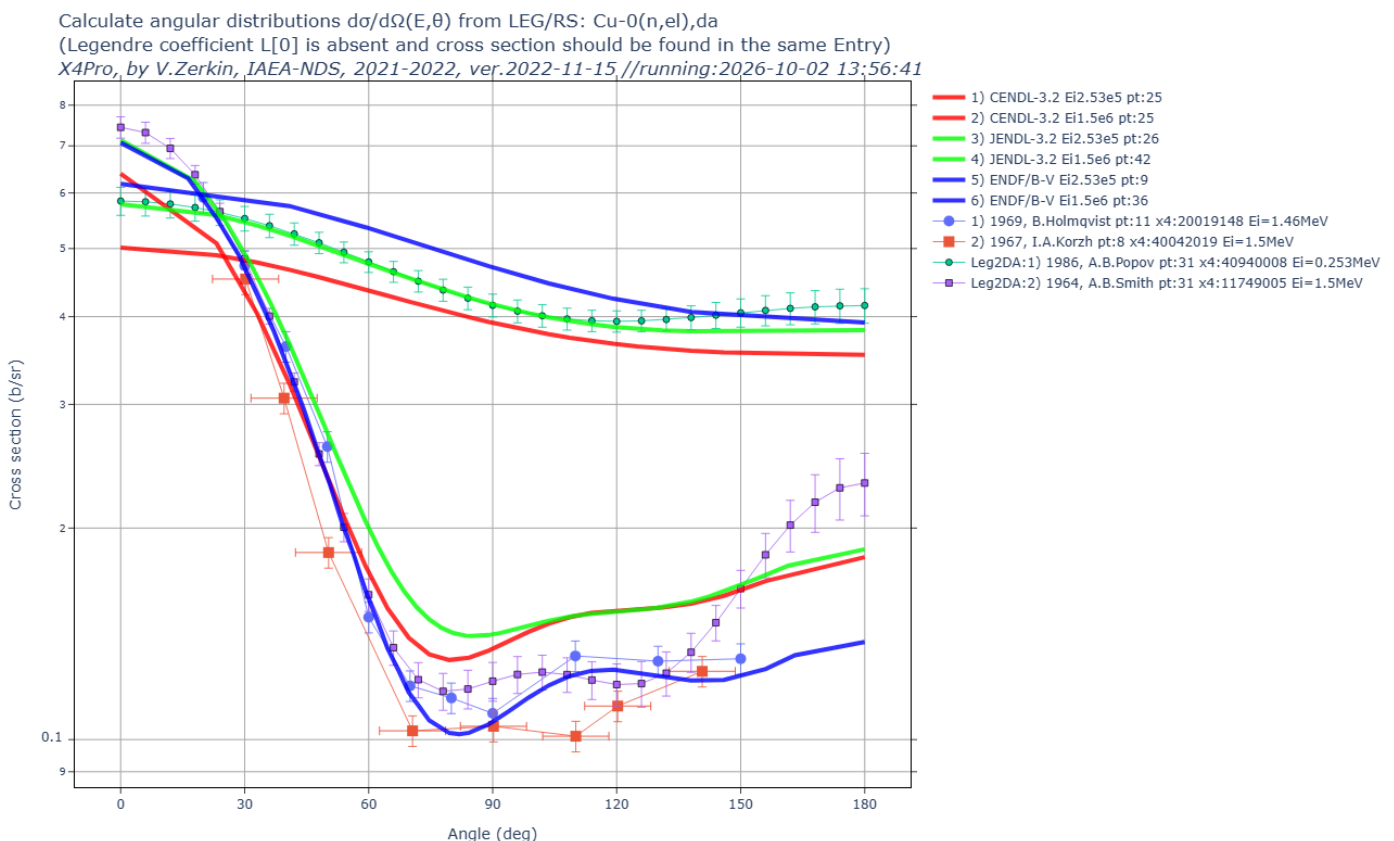
part3-6-daleg2sig/ Retrieve Legendre coefficient $L[0]$ from EXFOR, calculate cross sections and errors

63) part3-6-daleg2sig/out00/B11nel.html.png



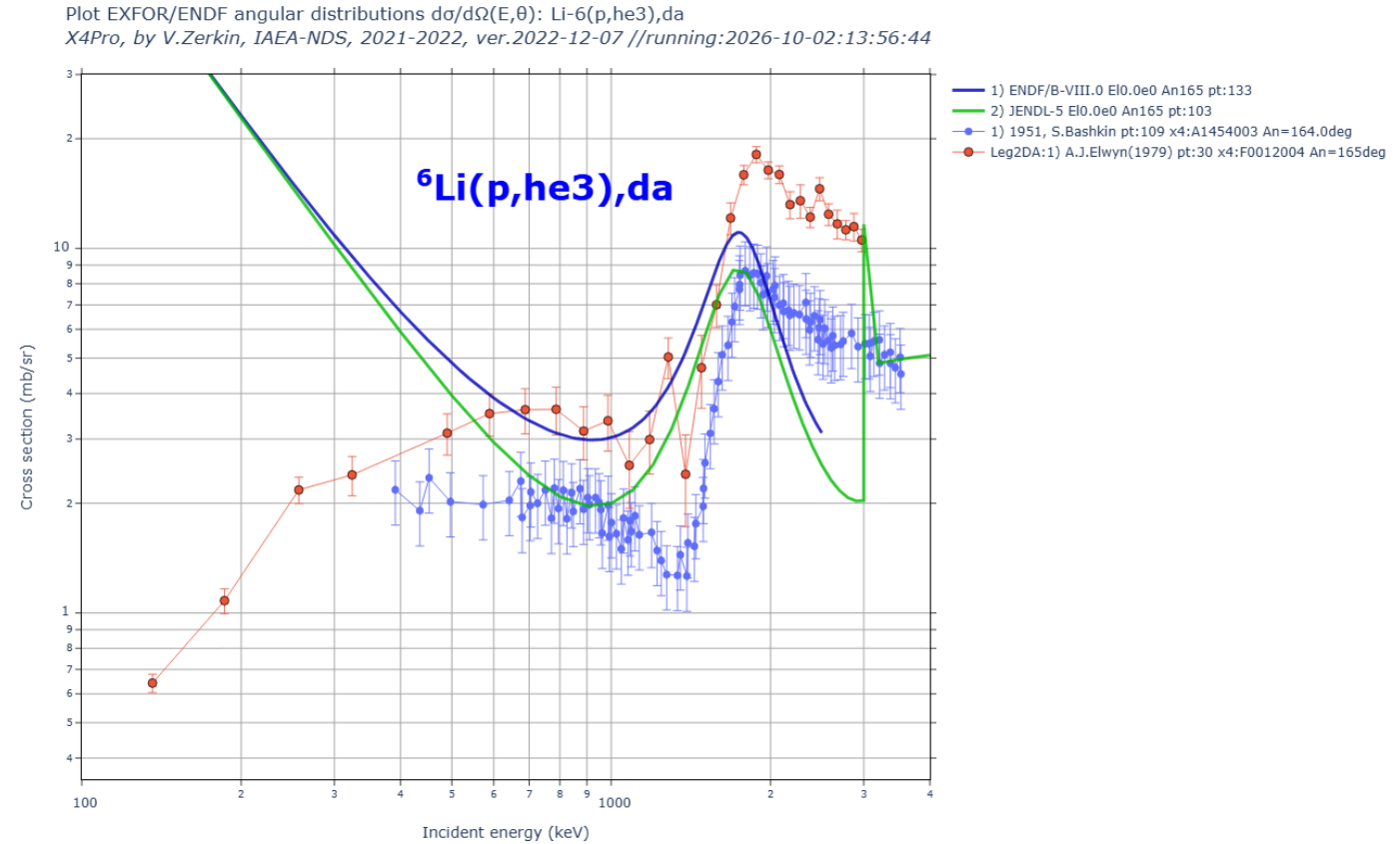
part3-7-legrs2da/ EXFOR LEG/RS and SIG from the same Entry; calculate missing Legendre coefficient $L[0]$; calculate angular distributions

64) part3-7-legrs2da/out00/Cu0nel_da.html.png



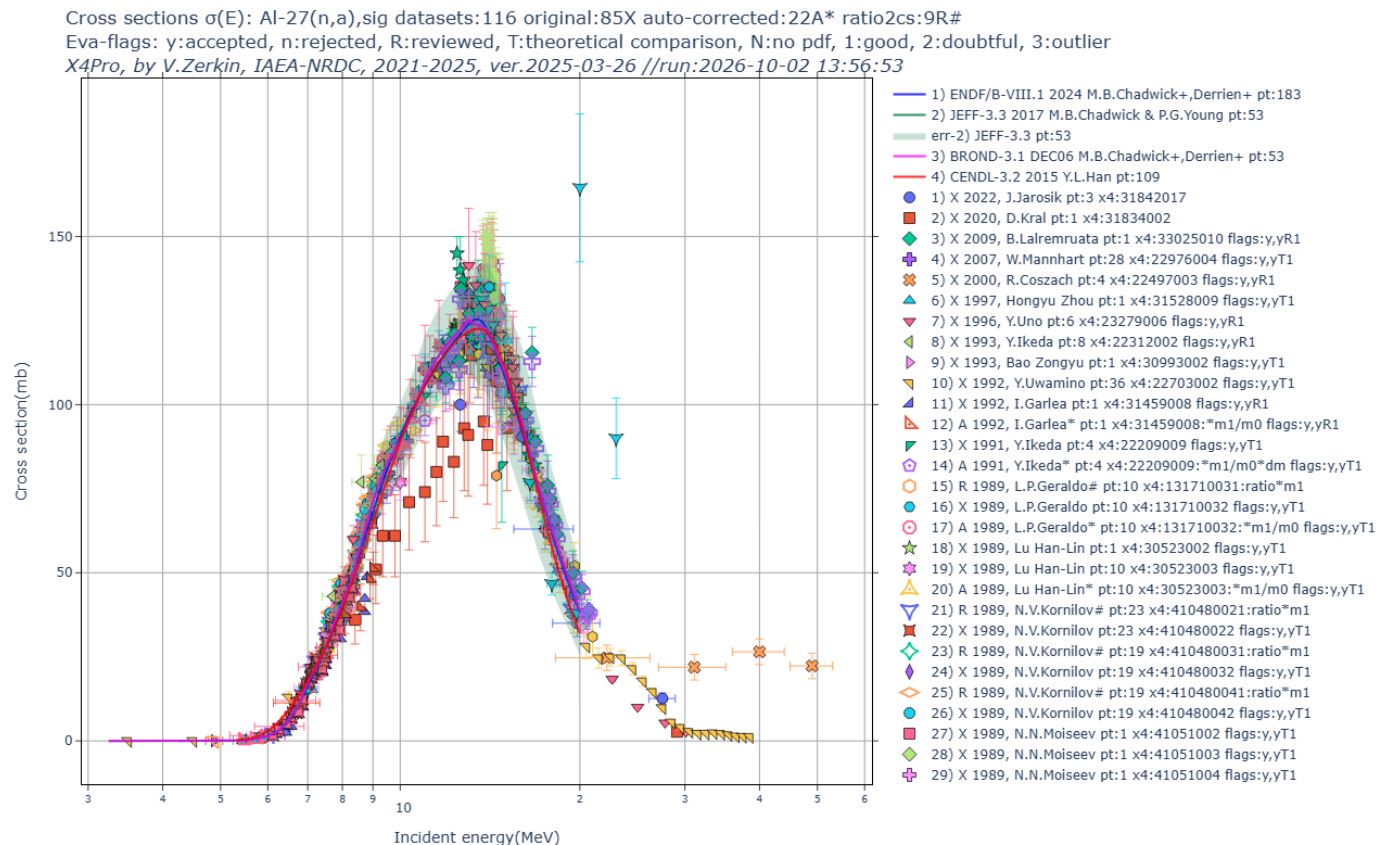
part3-8-leg2r33/ Retrieve Legendre coefficient calculate angular distributions store R33dat (draft of R33)

65) part3-8-leg2r33/out00/dal_ei_leg2da.html.png

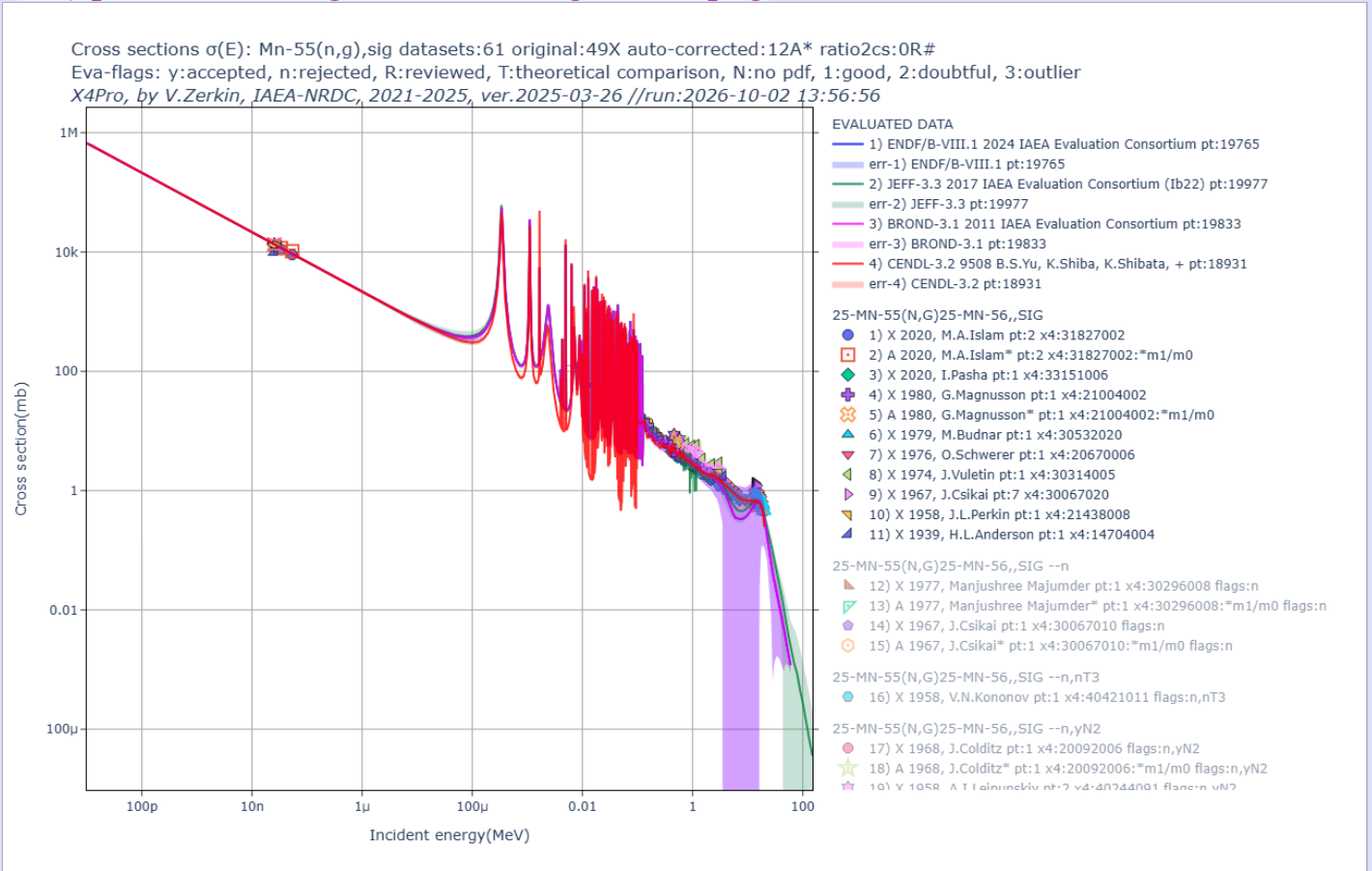


part6-1-eval2sig/ Retrieve CS, ratios, monit-data; renormalize to new standards and decay data; convert ratios to CS

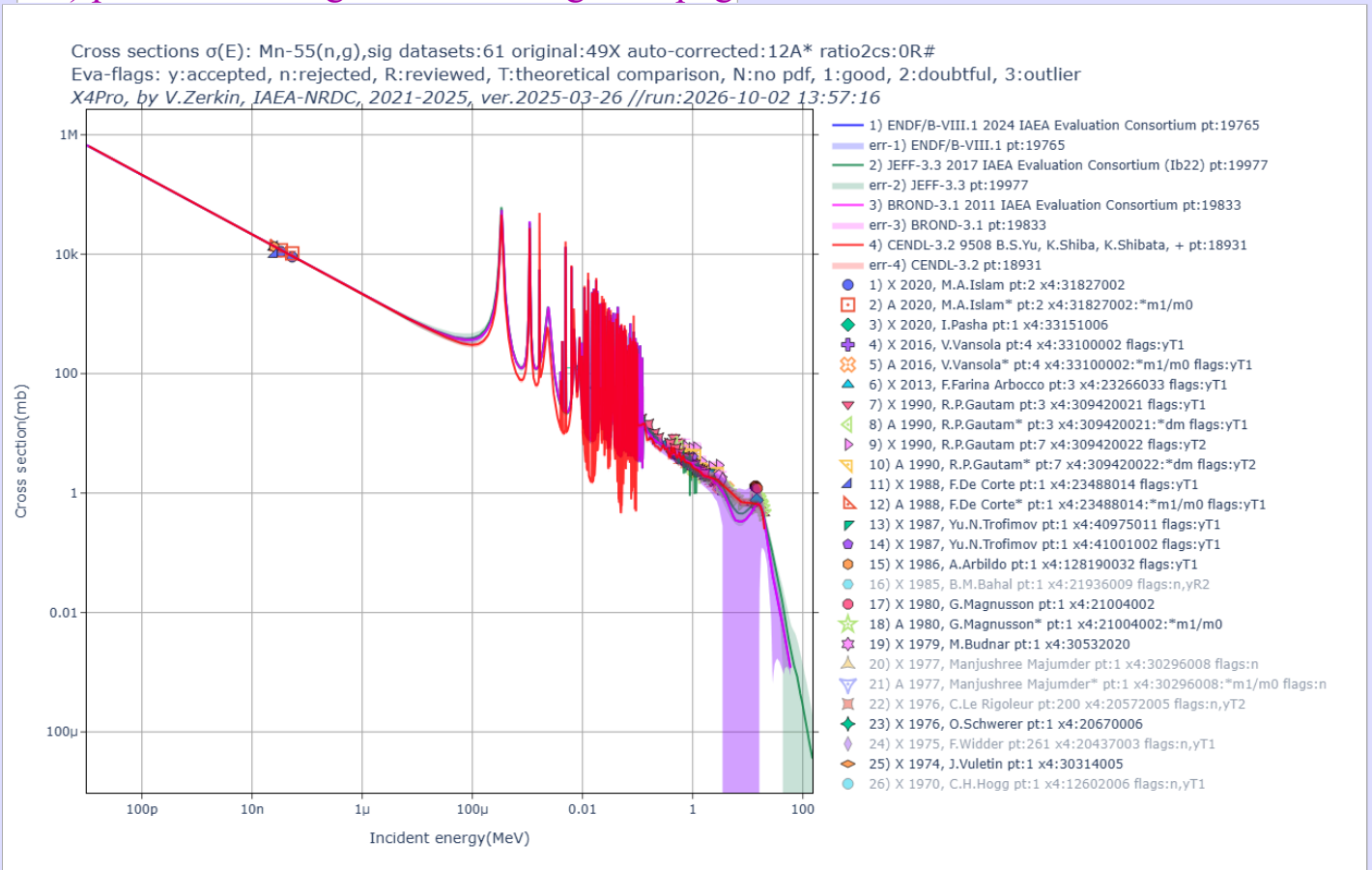
66) part6-1-eval2sig/out00/Al27na.html.png



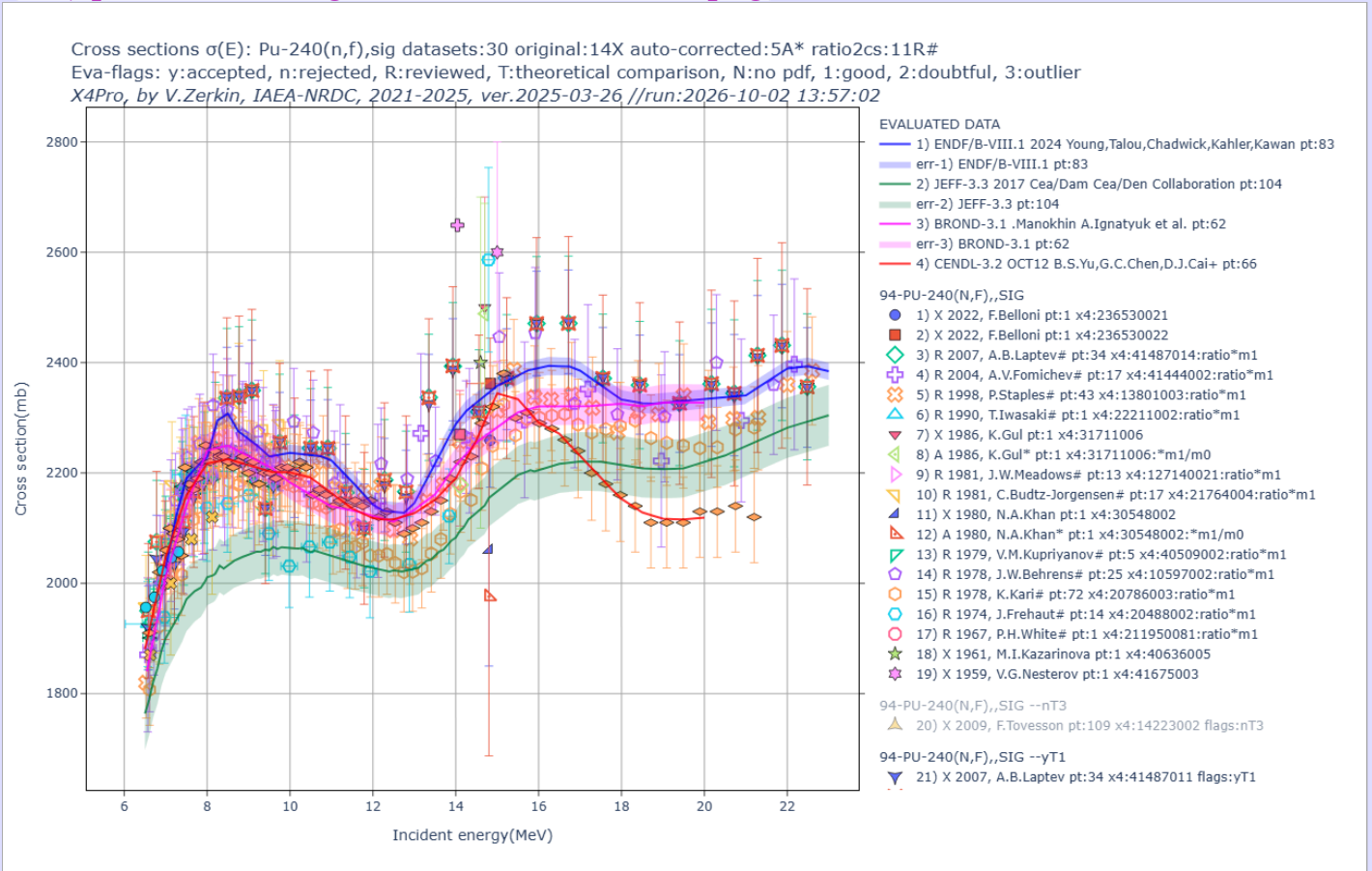
67) part6-1-eval2sig/out00/Mn55ng-n.html.png



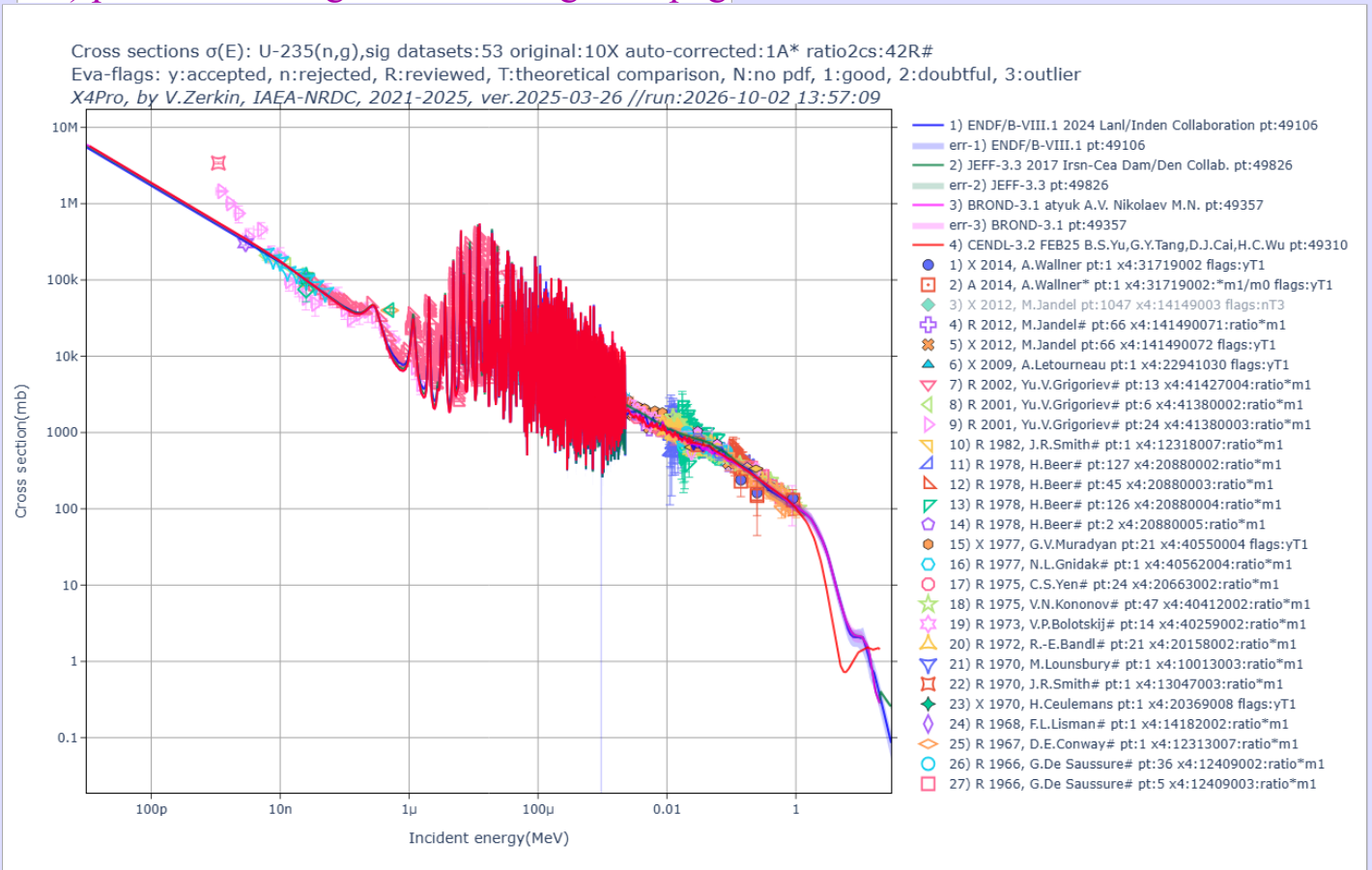
68) part6-1-eval2sig/out00/Mn55ng.html.png



69) part6-1-eval2sig/out00/Pu240nf-n.html.png



70) part6-1-eval2sig/out00/U235ng.html.png

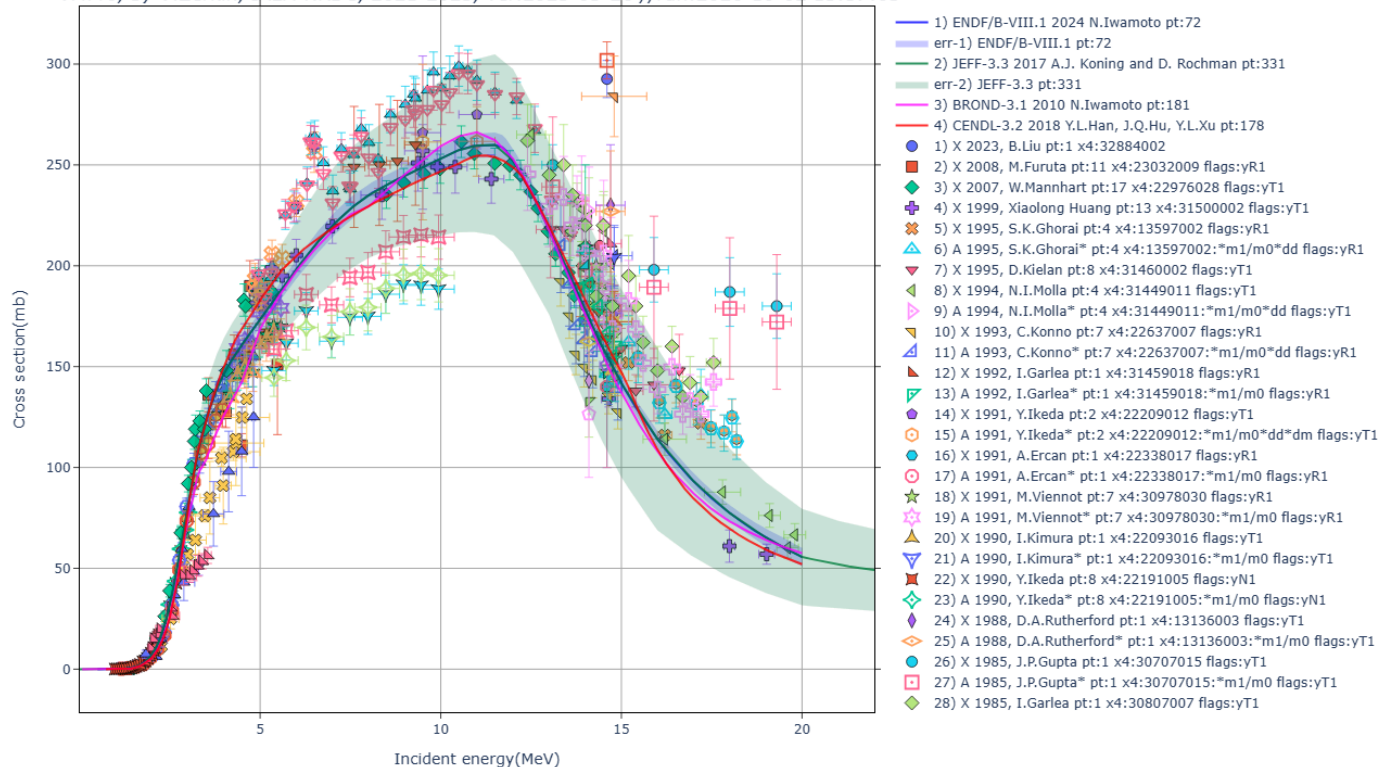


71) part6-1-eval2sig/out00/Zn64np.html.png

Cross sections $\sigma(E)$: Zn-64(n,p),sig datasets:87 original:51X auto-corrected:30A* ratio2cs:6R#

Eva-flags: y:accepted, n:rejected, R:reviewed, T:theoretical comparison, N:no pdf, 1:good, 2:doubtful, 3:outlier

X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-03-26 //run:2026-10-02 13:57:05



72) myplot1.png

Interactive plot: EXFOR:ETA + ENDF:MT452,18,102,MF1,3,31,33

X4Pro - advanced plot

by V.Zerkin, 2025-2026

